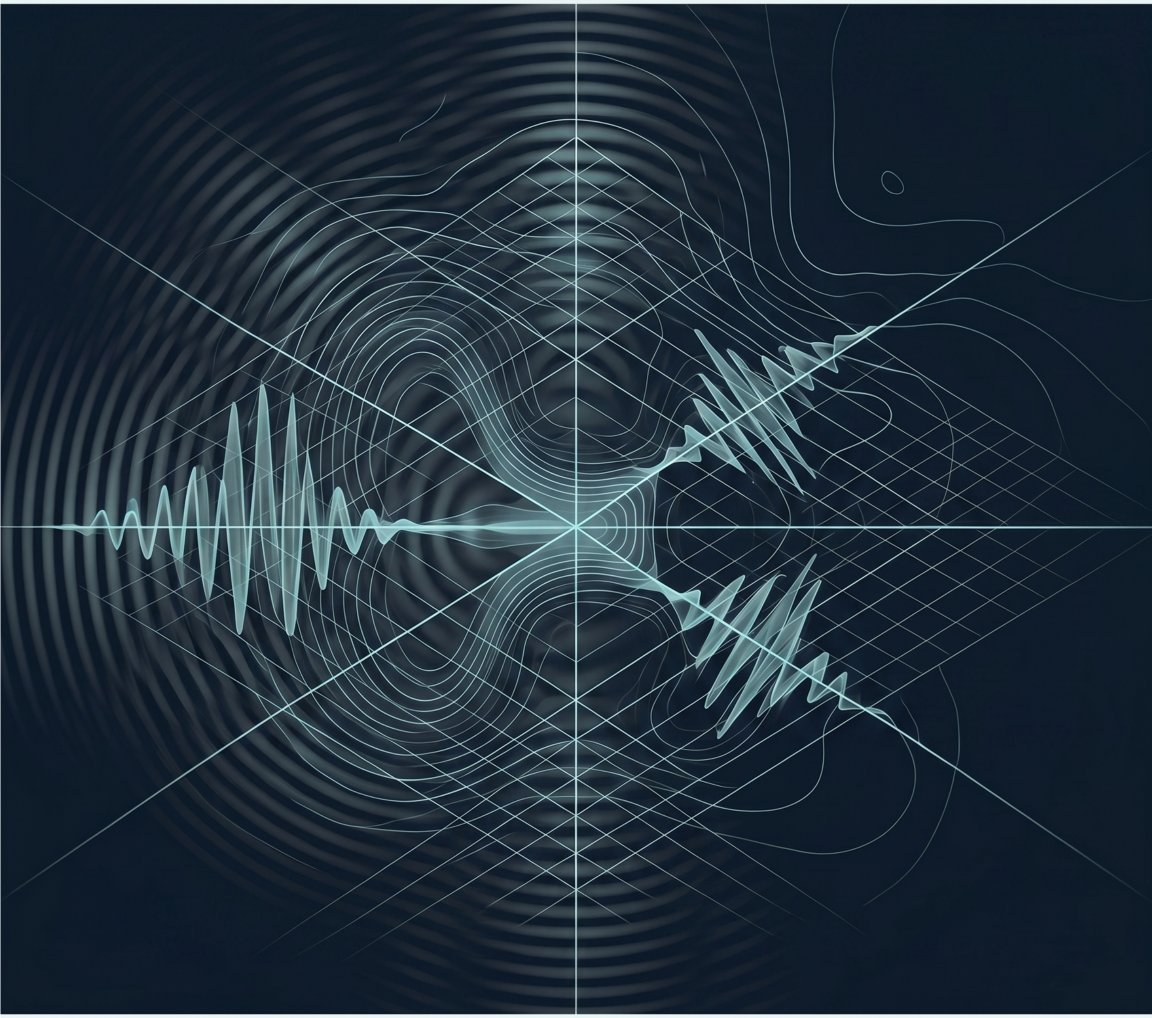
Quantum Mechanics

Core track, 2012 Winter Modern Physics Stanford



Original lectures by Leonard Susskind

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ABOUT THESE NOTES

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CHAPTER 1

QUANTUM LOGIC, MEASUREMENT, AND VECTOR SPACES

Quantum mechanics is not classical mechanics with a little random noise added. Its unpredictability coexists with exact conservation and reversible evolution, yet the questions we may ask of a system cannot be separated from the physical means by which we ask them. The purpose of this first lecture is to make that difference visible before introducing the formal machinery. We shall move from the two-slit experiment and reversibility to Heisenberg's microscope, and only then ask what kind of mathematical object a quantum state must be.

1.1 The Contract of the Course

People who encounter these lectures on the Internet see the lecturer but not the audience, and the class can therefore look puzzling. It is neither a standard undergraduate course nor a graduate course. It belongs to Stanford's continuing-education program: people from the surrounding community, along with some people connected to the university, return to theoretical physics because they want the subject itself.

The age distribution makes the purpose unusually clear. Only a handful of the roughly one hundred people in the room are under forty; many are over seventy, a few over eighty, and earlier classes have even included people over ninety. The audience wants to reach the basic ideas quickly. That does not mean replacing physics by metaphors. We shall use equations, sometimes hard equations, but only as much machinery as is needed to get the ideas right. The aim is a theoretical minimum in the literal sense: efficient, straightforward, and exact enough to support the next step.

This lecture also belongs to a larger sequence of approximately six ten-lecture courses. Classical mechanics came first because it supplies the language of motion, energy, momentum, and time evolution. A course on quantum entanglement provided an earlier entrance to quantum ideas. The present course is nevertheless designed to be substantially self-contained. We begin with the deepest difference between the two subjects: classical and quantum physics do not organize possibilities according to the same logic.

1.2 Classical Randomness Is the Wrong Model

Quantum mechanics contains unpredictability, but of a very special kind. Einstein objected that the laws of nature should not amount to a throw of dice; Bohr's reply was that Einstein should not prescribe what nature may do. The familiar exchange can be misleading if it suggests that quantum theory simply inserts a random-number generator into Newton's laws. Let us construct that classical model first and see why it fails.

Imagine that the moon ordinarily obeys Newton's equation

$$\mathbf{F} = m\mathbf{a} \tag{1.1}$$

together with the law of gravity, but that every tenth of a second an independent throw of the dice gives it a small extra kick. One outcome pushes the moon slightly forward, another slightly backward, and most outcomes may do nothing. Its future orbit would become unpredictable even if its initial position and velocity were known exactly.

This randomness would leave a characteristic scar. Each kick would change the energy by a small positive or negative amount. The average change might vanish, but the accumulated energy would execute a random walk. Exact energy conservation would be lost.

Quantum evolution does not work that way. Prepare a closed system with a sharp energy and let it evolve under a time-independent Hamiltonian. A later energy measurement returns the same energy exactly, even though other measurement outcomes may be unpredictable.[1] Quantum uncertainty affects some questions while leaving the relevant conservation law exact. It cannot be represented by occasional violations of an otherwise classical equation.

[1]Editorial clarification: The qualification about a closed system and a time-independent Hamiltonian states the standard condition implicit in the lecture's conservation argument.

1.3 Two Slits and the Failure of Classical Alternatives

The two-slit experiment gives the cleanest first view of the new logic. Photons, electrons, or neutrons may be used; even a bowling ball has a quantum wavelength in principle, but the corresponding effect is far too small to observe. Imagine a source so weak that particles leave one at a time, perhaps one every five minutes. Each particle passes a barrier and eventually produces one localized flash or one darkened spot on a screen.

Open only the upper slit. Repeated particles build a broad arrival profile, which we denote by $P_1(y)$. Close that slit and open only the lower one. The second experiment produces another broad profile $P_2(y)$, displaced on the screen. If the particles merely received independent classical kicks while passing the barrier, opening both slits would produce the pointwise sum

$$P_{\mathrm{cl}}(y) = P_1(y) + P_2(y). \tag{1.2}$$

Every particle would use one route or the other, and two available routes could not make an already accessible point inaccessible.

The experiment gives an interference pattern instead. There are points y_0 for which

$$P_{12}(y_0) = 0, \qquad P_1(y_0) > 0, \qquad P_2(y_0) > 0. \tag{1.3}$$

Either slit by itself supplies arrivals at y_0; opening both removes them. The particles may be separated

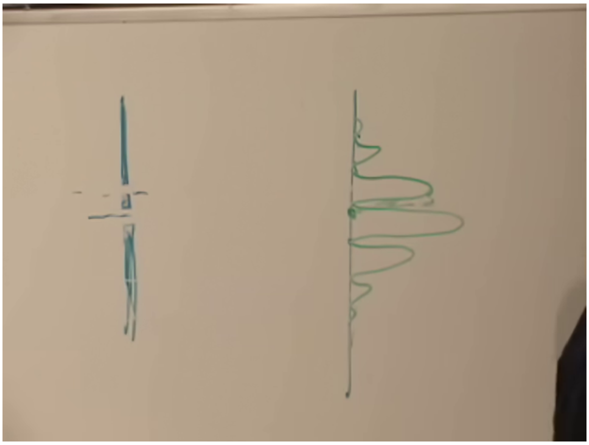

Figure 1.1: The completed two-slit interference profile on the board. The horizontal lobes represent arrival probability along the screen, including the nodes at which no particles arrive.

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by minutes or years, so the pattern cannot be attributed to collisions among successive particles.

One can invent an elaborate memory hidden in the barrier or screen, but interference is far too general to be explained by a special mechanism built for this apparatus. The broad conclusion is unavoidable: the classical rule for mutually exclusive alternatives is not the fundamental rule. Later we shall express the replacement in terms of probability amplitudes, which may cancel before probabilities are formed. At this stage the observed nonadditivity in Eq. (1.3) is enough.

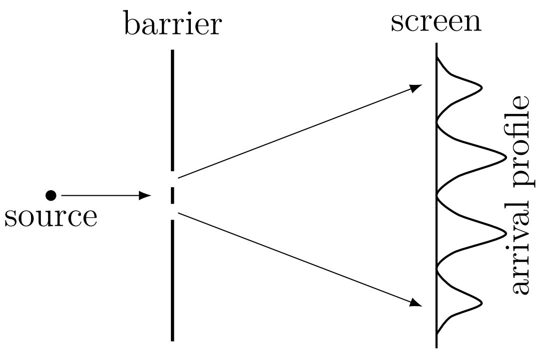

Figure 1.2: A clean reconstruction of the board geometry. The measured probability is built one detection at a time, yet it contains alternating maxima and nodes.

1.4 Reversibility and Conservation of Information

A second route to the same conclusion begins with deterministic motion in a finite state space. For a two-state coin, one reversible law may leave heads and tails unchanged,

$$H \mapsto H, \qquad T \mapsto T, \tag{1.4}$$

while another may interchange them,

$$H \mapsto T, \qquad T \mapsto H. \tag{1.5}$$

For a fanciful three-sided coin with heads, tails, and feet, consider

$$H \mapsto T, \qquad T \mapsto F, \qquad F \mapsto H. \tag{1.6}$$

Starting from any state, the system cycles forever.

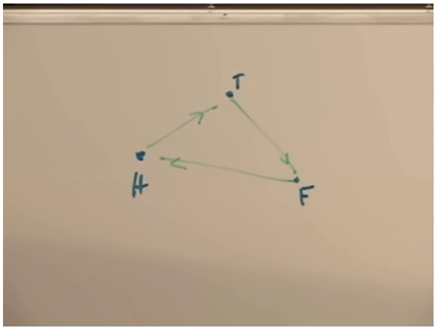

Figure 1.3: The three-state deterministic cycle. Reversing every arrow gives a unique route back from each state.

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Let the system run for a million time steps, reverse every arrow, and run for another million. It returns to its starting point. This is an operational meaning of reversibility and of information conservation: the present state still contains everything required to reconstruct the past. We need not even know the law in detail; we need only a switch that replaces the evolution by its inverse.

Now add a classical accident. With probability one in a million, suppose the coin stays put when the rule says to move. A short reversal test may succeed, but after ten million steps one or more accidents are likely. The reverse dynamics does not know when the random pauses occurred. In the three-state example the final reverse run is then no more likely to find the original state than either of the other two. Classical randomness has destroyed information.

1.5 An Electron Forward and Backward

Apply the reversal test to a quantum particle. One slit is enough. Send one electron through it, remove the screen, and do not determine where the electron goes. After a time t, reverse the dynamics and let the entire undisturbed system evolve for another time t. In the ideal experiment the electron returns along the reversed evolution and arrives back at the gun. Quantum spreading has not acted like a sequence of independent classical kicks.

Insert an intermediate detector and the result changes. If the detector records where the electron emerged, then reversing the electron's dynamics alone does not erase that record. The exact return test fails. In modern language, the electron has become correlated with degrees of freedom in the detector; the closed combined system may still evolve reversibly, but the electron can no longer be treated as though nothing else carries its history.[2]

The classical contrast is deliberately mundane. Looking at a coin—the lecture uses a Chilean peso—need not turn heads into tails. If sufficiently energetic light is used, it can of course knock the coin over, but classical physics allows the illumination to be made arbitrarily gentle while still revealing the state. An intermediate classical observation therefore need not spoil the reversal experiment. Quantum measurement admits no analogous limit in

[2] Editorial clarification: This closed-system formulation sharpens the lecture's contrast without adding a new formalism.

which a perfectly sharp observation has an arbitrarily small effect.

The two-slit experiment tells the same story. Interference survives only if nothing records which slit the electron used. The recorder need not be a human observer. A gas molecule, another electron, or any other physical degree of freedom may retain the path information. If the upper path changes one molecule and the lower path changes another, then the alternatives leave distinguishable records. The interference disappears and the observed probabilities reduce to the classical sum in Eq. (1.2).

Question from the audience. If a particle passes through one slit and is deflected, does momentum conservation not give the plate an opposite recoil, thereby recording the path?[a]

Response. The recoil is real, but a physical kick is not automatically a readable record. A plate localized well enough to define the slits has an intrinsically broad momentum distribution. The recoil shifts that distribution. If the shift is small compared with its width, the post-collision momentum does not reveal whether the kick occurred.

[a]Lecture timestamp: 00:44:30–00:44:47.

The uncertainty principle closes the apparent loophole. Making the plate's momentum sharp enough to resolve a tiny kick makes its position correspondingly uncertain, so the slit geometry itself becomes blurred. The apparatus cannot simultaneously define

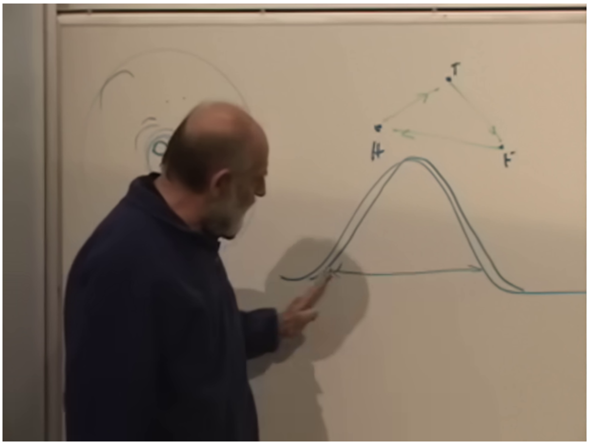

Figure 1.4: The plate's broad momentum profile and its small recoil-induced shift. The two nearly coincident curves are the part of the board relevant to the argument.

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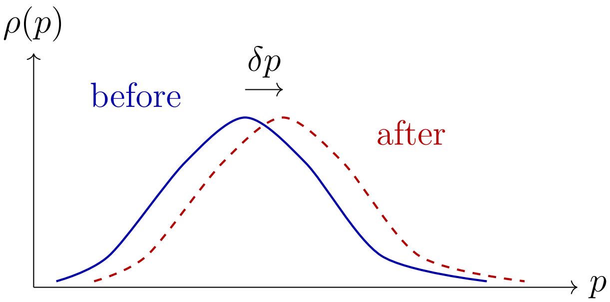

Figure 1.5: The recoil becomes a path record only when the shift δp is distinguishable relative to the initial width.

the paths sharply and record an arbitrarily small path-dependent recoil.

1.6 Why Classical Uncertainty Is Not Fundamental

Before deriving the microscope argument, distinguish quantum uncertainty from an ordinary statistical cloud. A Brownian particle is knocked by surrounding molecules, and an ensemble of such particles spreads in position and velocity. In classical physics this uncertainty reflects incomplete observation. With a better microscope, a more accurate detector, and gentler illumination, one imagines determining the position and velocity of each particle as accurately as desired.

Quantum mechanics contains a logical obstruction to that refinement. It is not merely that our apparatus is coarse or expensive. A procedure that makes position sharper necessarily changes what can be said about momentum. Heisenberg first encountered this through abstract mathematics, where the operations corresponding to position and momentum do not commute. The microscope thought experiment supplies the physical illustration.

1.7 Heisenberg's Microscope

To locate a particle optically, scatter light from it and focus the scattered wave into an image. The required chain begins with the energy of one photon,

$$E = hf = \hbar\omega, \qquad \omega = 2\pi f, \qquad \hbar = \frac{h}{2\pi}. \tag{1.7}$$

Visible light has a frequency of order 10^{15} Hz. Light also carries momentum: an absorbed beam heats a door because it carries energy and can push the door because it carries momentum.

For a nonrelativistic particle,

$$E_{\text{kin}} = \frac{p^2}{2m} = \frac{1}{2}pv, \tag{1.8}$$

whereas for light relativity gives

$$E = cp, \qquad p = \frac{E}{c}. \tag{1.9}$$

One cycle has period $T = 1/f$, and in that time a light wave advances by one wavelength. Hence

$$c = \frac{\lambda}{T} = \lambda f, \qquad f = \frac{c}{\lambda}. \tag{1.10}$$

Combining this with Eqs. (1.7) and (1.9) gives

$$p_\gamma = \frac{hf}{c} = \frac{h}{\lambda}. \tag{1.11}$$

Short wavelength means large photon momentum.

An image cannot resolve detail much smaller than its wavelength. To locate a particle on a scale Δx, the probe must therefore satisfy, at the order-of-magnitude level,

$$\lambda \lesssim \Delta x. \tag{1.12}$$

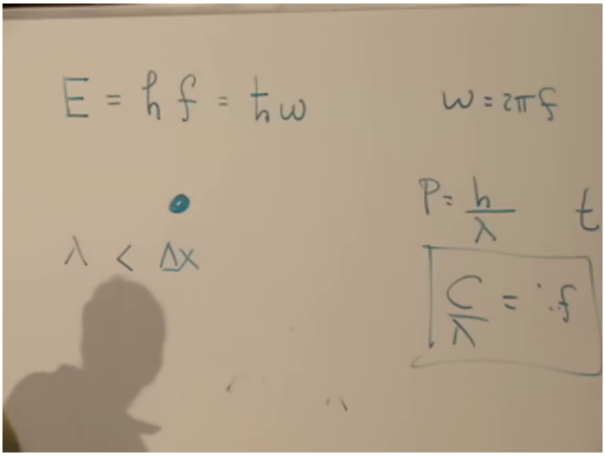

Figure 1.6: The board at the decisive step: $E = hf = \hbar\omega$, $p = h/\lambda$, $c/\lambda = f$, and the imaging condition $\lambda < \Delta x$.

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But the photon used to obtain that resolution carries momentum h/λ. When it scatters from the particle, the unknown scattering direction gives the particle an uncertain kick of the same order:

$$\Delta p_{\text{kick}} \sim \frac{h}{\lambda}. \tag{1.13}$$

Equations (1.12) and (1.13) give the heuristic scale

$$\Delta x \, \Delta p_{\text{kick}} \gtrsim h \qquad \text{(order of magnitude).} \tag{1.14}$$

This is not the later sharp inequality with its convention-dependent numerical factor; it is the microscope logic that motivates it.

Why does the same argument not seem fatal classically? A classical light wave may be divided

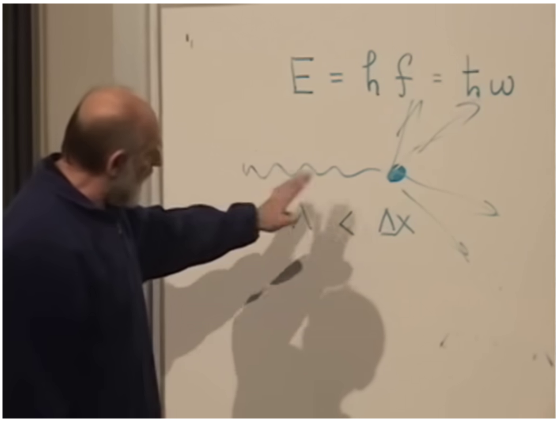

Figure 1.7: A short-wavelength photon localizes the particle sharply, then scatters in an uncontrolled direction and transfers momentum.

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into arbitrarily small amounts of energy while its wavelength is held fixed. One could imagine forming the same sharp image with an arbitrarily weak wave. Quantum light comes in indivisible packets. At a fixed wavelength the smallest available probe is one photon, and that photon already carries h/λ. A precise position measurement cannot be made arbitrarily gentle. A gentle measurement is possible only by using a long wavelength and accepting a coarse position.

1.8 Measuring Velocity Gently

After the break, a question asks how velocity can be measured at all. The answer is to use two deliberately imprecise position measurements and allow a long time between them. Suppose the particle's mass is known, so that measuring velocity also determines its nonrelativistic momentum.

Use a long-wavelength photon for the first observation. It gives only

$$x_1 = x + O(\lambda), \tag{1.15}$$

but its momentum is small enough that the particle receives only a gentle kick. Wait for a time t, then make a second long-wavelength observation:

$$x_2 = x + vt + O(\lambda). \tag{1.16}$$

The inferred displacement and velocity are

$$d = x_2 - x_1 = vt + O(\lambda), \tag{1.17}$$

$$v_{\text{meas}} = \frac{d}{t} = v + O\left(\frac{\lambda}{t}\right). \tag{1.18}$$

No matter how large λ is, the measurement contribution λ/t can be made small by waiting long enough.

The tradeoff has not disappeared. Velocity is determined accurately only while position remains uncertain on the large scale λ. The method does not assign both a sharp position and a sharp momentum at one instant.

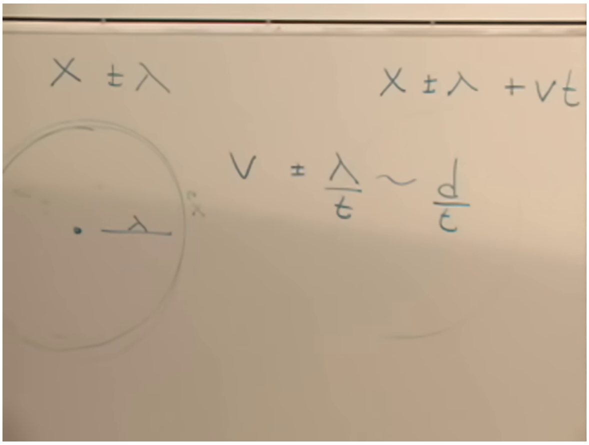

Figure 1.8: The two coarse position readings and the resulting velocity estimate $v = d/t$ with uncertainty of order λ/t.

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Question from the audience. Does the term λ/t include every source of uncertainty in this velocity experiment?[a]

Response. No. It describes the uncertainty contributed by the two coarse position readings. The original preparation supplies another uncertainty, and the complete accounting requires keeping both. A fixed reference object such as a wall can have its mean position determined by averaging many gentle photon measurements over a long time. The detailed follow-up is deferred rather than answered by pretending that λ/t is the whole uncertainty principle.

[a] Lecture timestamp: 01:14:07–01:16:16.

1.9 Classical States Are Points in Phase Space

The experiments point to a failure of classical logic, and the failure appears at the most elementary question: what is a state? For a classical coin the alternatives H and T are points in a set. For a die the set has six points. For one classical particle, a state is a point in phase space,

$$(x, p), \tag{1.19}$$

and for many particles it is the corresponding list of all positions and momenta. Dynamics is a flow carrying one point of this set into another.

If the apparatus is coarse, we may describe our knowledge by a probability density on phase space,

$$\rho(x, p) \geq 0, \qquad \int dx\, dp\, \rho(x, p) = 1. \tag{1.20}$$

The distribution may be concentrated in a small region and fall away from its center. Classical logic treats this as incomplete knowledge, not as the maximal possible specification of the state. More delicate, less disturbing measurements can in principle concentrate ρ more closely around one phase-space point.

That conclusion is exactly what quantum mechanics refuses. A quantum state requires more structure than a point-set description supplies. Of course any collection of states can be regarded as a set in the formal mathematical sense; the claim is that set membership and classical probability are not the natural operations that organize quantum alternatives. The state space must support linear combinations.

1.10 Complex Vector Spaces and Ket Vectors

A vector space is a collection of objects called vectors together with two operations. Vectors may be added, and they may be multiplied by scalars. For the spaces used in quantum mechanics, the scalars are complex numbers. Thus if $|a\rangle, |b\rangle \in V$ and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha |a\rangle + \beta |b\rangle \in V. \tag{1.21}$$

This closure under linear combinations is the first mathematical departure from a classical list of alternatives. Multiplying heads by $2 + 4i$ has no meaning in the classical two-point state space; multiplying a ket vector by a complex scalar is a basic operation.

Dirac denoted these vectors by kets such as $|a\rangle$, half of a bracket. The complementary half will be a bra, and a bra next to a ket makes a bracket. That structure comes later; for the moment the notation only marks an abstract vector.

The word *vector* must not be confused with an arrow in ordinary three-dimensional space. To keep the distinction visible, call the familiar arrow a *pointer*. A pointer has a length and a spatial direction. It can be multiplied by a real number and added to another pointer by the parallelogram rule, so pointers form a vector space over $\mathbb{R}$. Quantum state vectors are

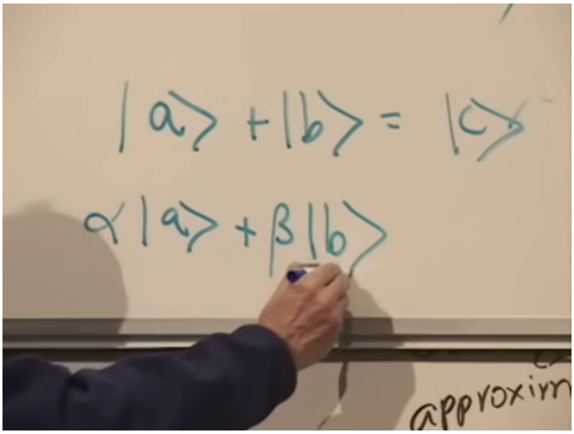

Figure 1.9: Ket-vector addition and a general complex linear combination. The result is another ket vector in the same vector space.

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more general and form spaces over $\mathbb{C}$.

The lecture occasionally uses *Hilbert space* while introducing this linear structure. Strictly, a complex vector space becomes a Hilbert space only after an inner product and an appropriate completeness condition are supplied. Those additions have not yet been introduced; at this stage we need only the two linear operations.

1.11 Functions as Vectors

Consider complex-valued functions of one real variable,

$$\psi(x) = \psi_R(x) + i\psi_I(x). \tag{1.22}$$

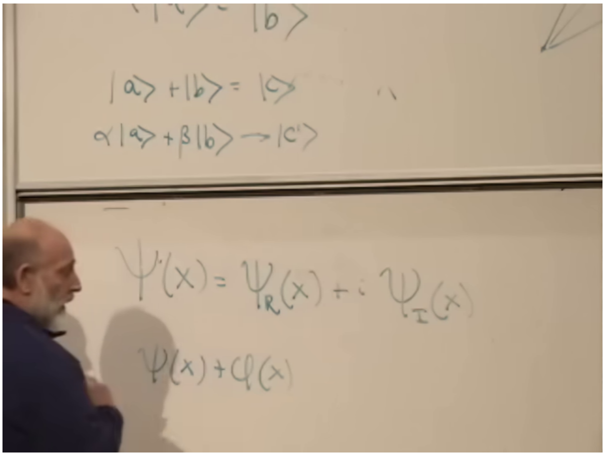

Figure 1.10: A complex function resolved into real and imaginary parts, beneath the ket-vector linear-combination rule.

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The argument x is real; the value of the function may be complex. Given two such functions and a complex number α, define

$$(\psi + \phi)(x) = \psi(x) + \phi(x), \tag{1.23}$$

$$(\alpha\psi)(x) = \alpha\psi(x). \tag{1.24}$$

Both operations produce new complex-valued functions, so any chosen class of functions closed under these operations forms a complex vector space. Real functions similarly form a vector space over $\mathbb{R}$, but not over $\mathbb{C}$, because multiplying a real function by i takes it out of the real-valued collection.

Function spaces are the examples that motivated Hilbert's work and will become central to wave mechanics. Questions of inner products, normalization, and which functions are admissible are deliberately left for later.

1.12 Columns and Dimension

A finite-dimensional example is a column of n complex numbers. For $n = 4$,

$$A = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \qquad B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}. \tag{1.25}$$

Addition and scalar multiplication are componentwise:

$$A + B = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}, \qquad \alpha A = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}. \tag{1.26}$$

No multiplication of one vector by another has been defined. Only addition of vectors and multiplication of a vector by a scalar are needed here.

Columns of a fixed length n form an n-dimensional complex vector space. Pairs, quadruples, and infinite columns each give their own spaces; throwing columns of every possible length into one collection does not give one vector space under these rules, because no addition has been defined between a two-component and a three-component column. The entries may be viewed as components, just as the entries of a spatial pointer are its components along

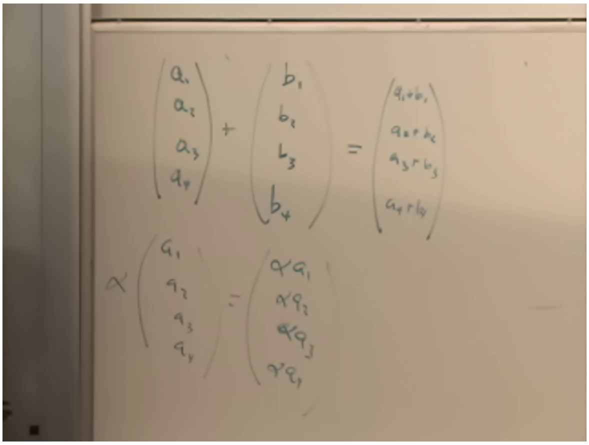

Figure 1.11: Componentwise addition and complex scalar multiplication for four-component columns.
Lecture frame: 01:42:00.

chosen axes. One may also regard a column as a function of its discrete index, but that abstraction is postponed.

We have not yet explained in what physical sense a state is represented by such a vector. The present task is only to learn the operations before applying them to quantum systems.

Question from the audience. At this stage, is the vector-space discussion supposed to sound like mumbo jumbo?[a]

Response. Its physical use has not yet been supplied, so the abstraction is expected to feel unmotivated. The definitions themselves are nevertheless precise: vectors can be added and multiplied by complex numbers. The next lectures will connect those operations to physical states.

[a]Lecture timestamp: 01:44:52–01:45:10.

The analogy with logic is worth retaining. Classical Boolean operations act on sets: *and* corresponds to intersection, while *or* corresponds to union. The lecture momentarily interchanges the two and corrects the order. Quantum logic will be organized by subspaces and linear operations rather than by ordinary unions of alternatives. That is why vector spaces are not optional mathematical decoration.

1.13 Complex Conjugation

The complex numbers themselves form a one-dimensional vector space over $\mathbb{C}$. They also carry an operation that will be indispensable when inner products are introduced. Write

$$z = x + iy. \tag{1.27}$$

Its complex conjugate is

$$z^* = x - iy. \tag{1.28}$$

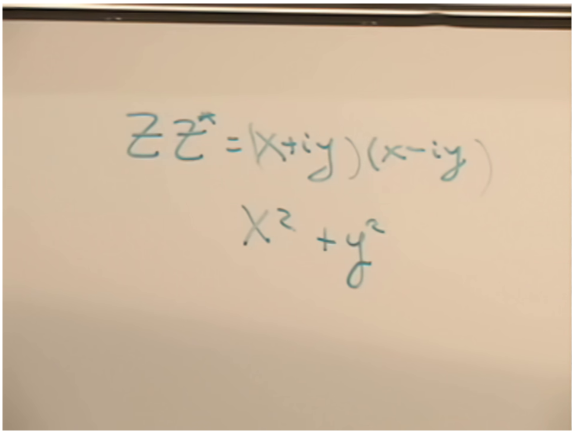

Figure 1.12: The board calculation $zz^* = (x+iy)(x-iy) = x^2 + y^2$.

Lecture frame: 01:49:30.

Geometrically, conjugation reflects the point in the complex plane across the real axis. Algebraically,

$$z^* z = (x - iy)(x + iy) \tag{1.29}$$
$$= x^2 + y^2 = |z|^2. \tag{1.30}$$

The cross terms cancel and $i(-i) = 1$, leaving the squared distance from the origin.

Conjugation is best regarded as a map from complex numbers back to complex numbers, not merely as a trick for simplifying products. The next lecture will extend the vector-space language, introduce operators, Hermitian operators, and eigenvalues, and begin applying this machinery to quantum systems.

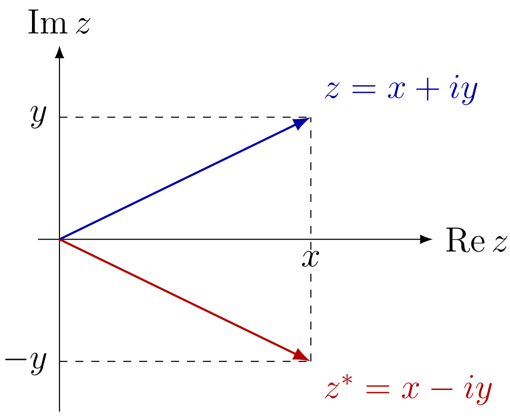

Figure 1.13: Complex conjugation is reflection across the real axis; both vectors have magnitude $\sqrt{x^2 + y^2}$.

CHAPTER 2

VECTOR SPACES, SUPERPOSITION, AND THE FIRST QUANTUM POSTULATES

We ended the previous lecture with a question about the uncertainty principle. Before building any new formalism, let us settle it. The apparent loophole is instructive because it distinguishes the momentum a particle had before a measurement from the momentum it has after the measurement.

2.1 Which Velocity Have We Measured?

Suppose a first position measurement locates a particle only within a region of width δ. Wait a time T, make a second measurement with the same resolution, and let the centers of the two regions be separated by L. The inferred average velocity is

$$v_{\text{avg}} \simeq \frac{L}{T}, \qquad \Delta v_{\text{avg}} \sim \frac{\delta}{T}. \tag{2.1}$$

Numerical factors in the second estimate are unimportant. By making T large, we can make δ/T as small as we please. Have we measured position to accuracy δ and velocity almost exactly?

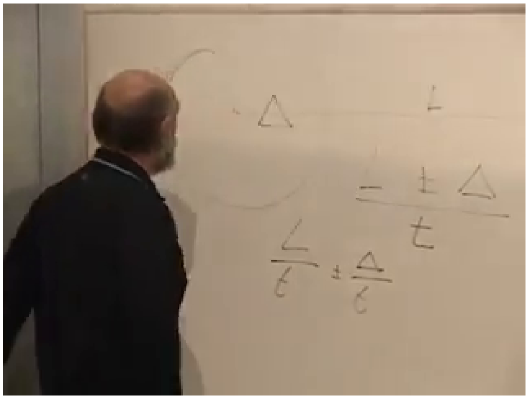

Figure 2.1: Two coarse position readings separated by a long time. The board displays the displacement L, the resolution scale δ, and the velocity uncertainty of order δ/T.

Lecture frame: 00:03:30.

Question from the audience. Can two imprecise position measurements, separated by a very long time, evade the uncertainty principle? [a]

Response. No. Equation (2.1) gives the velocity after the first position measurement. To localize the particle on the scale δ, the probe must contain wavelengths no larger than that scale. A photon capable of doing so carries momentum of order $\hbar/\delta$, and its scattering gives the particle a kick of the same order. The later trajectory therefore reveals the momentum of the disturbed state, not the momentum immediately before localization.

[a]Lecture timestamp: 00:00:17–00:04:17.

Trying to verify the result does not restore the missing information. Repeating the procedure changes the momentum again by an amount of order $\hbar/\delta$. Classical physics permits an ideal limit in which position is observed ever more gently while the velocity is left alone. Quantum mechanics does not. This is the point of the uncertainty principle, and with it settled we can begin the mathematics needed for states.

2.2 The One-Dimensional Model: Complex Numbers

The next subject is dry, but it is not difficult. We need a quick review of complex numbers only to establish notation. Write

$$z = x + iy, \qquad z^* = x - iy, \qquad i^2 = -1. \tag{2.2}$$

In the complex plane, conjugation reflects a point across the real axis. Complex numbers can be added and multiplied by arbitrary complex scalars, so $\mathbb{C}$ is itself a one-dimensional complex vector space. It is the simplest model of the structures we shall use.

In this one-dimensional model, regard z as a ket and z^* as its corresponding bra. Complex conjugation exchanges the two. If

$$z_1 = x_1 + iy_1, \qquad z_2 = x_2 + iy_2,$$

then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2), \tag{2.3}$$

$$(z_1 z_2)^* = z_1^* z_2^*. \tag{2.4}$$

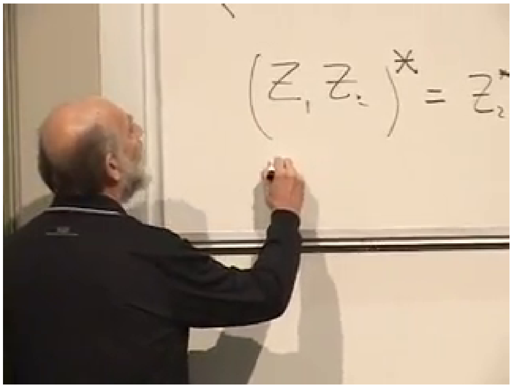

Figure 2.2: Complex conjugation of a product on the board. The corrected relation is given cleanly in Eqs. (2.4) and (2.5).

Lecture frame: 00:14:30.

For scalar numbers the factors commute. It is nevertheless useful to write the related identity

$$(z_1^* z_2)^* = z_2^* z_1, \tag{2.5}$$

because the reversal of bra and ket will survive when the intervening objects no longer commute.

Question from the audience. Can the complex number be written in exponential form? [a]

Response. Certainly. One may write $z = re^{i\theta}$, but we do not need that form yet. The Cartesian form makes conjugation and the coming vector-space rules visible with less machinery.

[a] Lecture timestamp: 00:15:12–00:15:18.

2.3 What Makes a Complex Vector Space?

A complex vector space is a collection of objects that may be added and multiplied by complex numbers without leaving the collection. If $|A\rangle$ and $|B\rangle$ are vectors and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha|A\rangle + \beta|B\rangle \tag{2.6}$$

must be another vector in the same space. The vectors need not be arrows. What matters is the algebra of addition and scalar multiplication.

Several near-examples make the definition precise. Ordinary arrows form a real vector space, but not a complex vector space unless an additional rule for multiplication by i is supplied. The real numbers form a real vector space, not a complex one. The positive real numbers do not even form a real vector space: multiplying a positive number by -1 leaves the set.

2.3.1 Functions and Columns

Complex-valued functions of a real variable form a complex vector space. If $\psi(x)$ and $\phi(x)$ are such functions, then

$$(\alpha\psi + \beta\phi)(x) = \alpha\psi(x) + \beta\phi(x) \tag{2.7}$$

is another complex-valued function.

A column vector is the same idea with a discrete

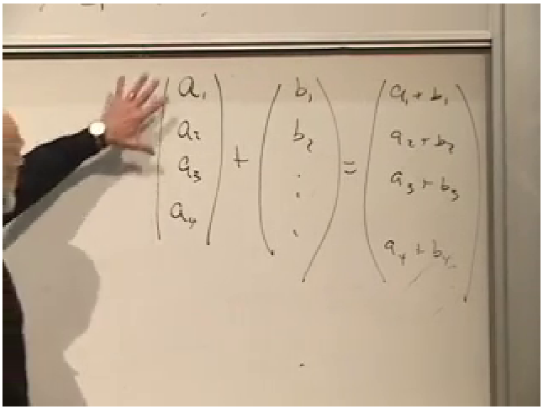

Figure 2.3: Componentwise addition of two column vectors. A column may be read as a function of a discrete index.

Lecture frame: 00:24:30.

argument. For four components,

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}, \tag{2.8}$$

and

$$\alpha|a\rangle + \beta|b\rangle = \begin{pmatrix} \alpha a_1 + \beta b_1 \\ \alpha a_2 + \beta b_2 \\ \alpha a_3 + \beta b_3 \\ \alpha a_4 + \beta b_4 \end{pmatrix}. \tag{2.9}$$

The index $1, 2, 3, 4$ plays the role of a discrete coordinate.

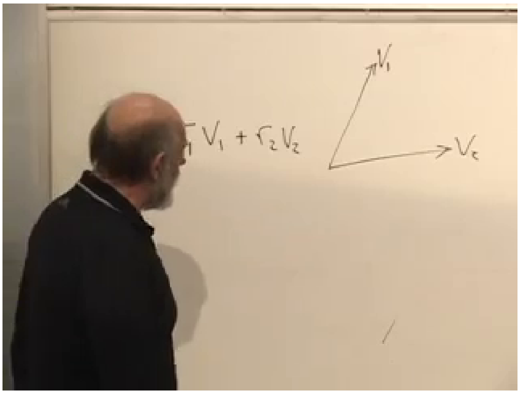

Figure 2.4: Two noncollinear vectors spanning a plane. The expression at left is partly obscured, while the geometric pair is clear.

Lecture frame: 00:28:59.

2.3.2 Dimension and Spanning

The dimension of a vector space is the minimum number of independent vectors needed to span it. One nonzero vector spans a line. Two noncollinear vectors span a plane: every vector $|v\rangle$ in that plane can be written

$$|v\rangle = r_1|v_1\rangle + r_2|v_2\rangle. \tag{2.10}$$

A third vector in the same plane adds no new direction, because it is already a linear combination of the first two. In three dimensions we need three noncoplanar vectors.

Columns with four components form a vector space

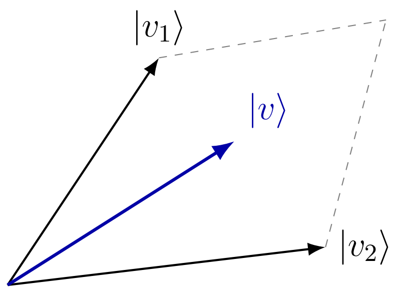

Figure 2.5: A clean reconstruction of the spanning argument: $|v\rangle = r_1|v_1\rangle + r_2|v_2\rangle$.

of dimension four. One convenient basis is

$$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad \begin{pmatrix}0\\0\\0\\1\end{pmatrix}. \tag{2.11}$$

Remove the last column and no linear combination can produce a nonzero fourth component. By contrast, one-component columns are just complex numbers and form a one-dimensional complex space.

The space of smooth complex functions has no finite spanning set; it is infinite-dimensional. The details of bases in infinite-dimensional spaces will not concern us yet. For now, the essential point is that functions and columns are both concrete representations of abstract vectors:

$$\psi(x) \longmapsto |\psi\rangle, \qquad \begin{pmatrix}\psi_1\\\psi_2\\\vdots\end{pmatrix} \longmapsto |\psi\rangle. \tag{2.12}$$

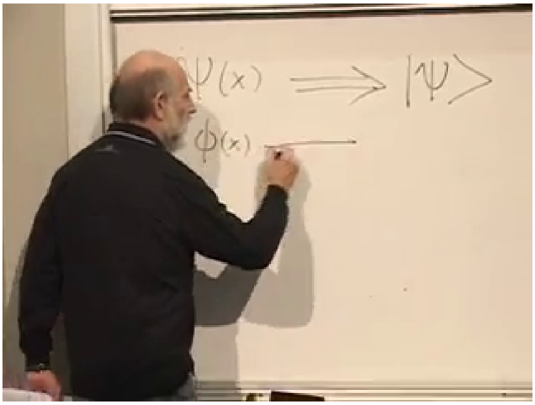

Figure 2.6: The board changes viewpoint from the function $\psi(x)$ to the abstract ket $|\psi\rangle$. The second line is still being written.

Lecture frame: 00:34:50.

2.4 Bras, Kets, and the Dual Space

To every ket $|A\rangle$ we associate a bra $\langle A|$. A bra is a linear functional: when it acts on a ket, it produces a complex number. Once an inner product has been chosen, the Riesz correspondence identifies each ket with a bra antilinearly. In finite-dimensional coordinates this is the familiar conjugate transpose:

$$|A\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{pmatrix} \quad \longleftrightarrow \quad \langle A| = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_d^* \end{pmatrix}. \tag{2.13}$$

For a function, the corresponding bra is represented by the conjugate function $\psi^*(x)$.

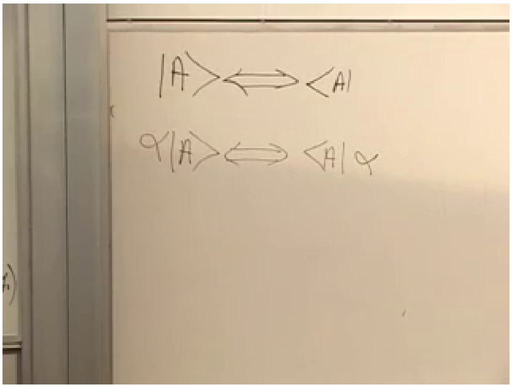

Figure 2.7: Ket-to-bra correspondence and the conjugation of a scalar coefficient.

Lecture frame: 00:41:00.

The correspondence is antilinear:

$$\alpha|A\rangle \longleftrightarrow \langle A|\alpha^*, \tag{2.14}$$

$$|A\rangle + |B\rangle \longleftrightarrow \langle A| + \langle B|. \tag{2.15}$$

Question from the audience. Is there any significance to writing α^* on the right of the bra?[a]

Response. Not while α is an ordinary complex number, because scalar multiplication commutes. It is partly a convention for keeping formulas orderly. Position will matter when noncommuting operators enter the calculation.

[a]Lecture timestamp: 00:43:24–00:43:57.

2.5 Inner Products

For ordinary real vectors, the dot product is

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta = \sum_i a_i b_i. \tag{2.16}$$

The inner product $\langle A|B\rangle$ is its complex counterpart. We use the convention standard in physics: it is linear in the ket and conjugate-linear in the bra. Thus

$$\langle A|(\beta B + \gamma C)\rangle = \beta\langle A|B\rangle + \gamma\langle A|C\rangle, \tag{2.17}$$

$$\langle \alpha A|B\rangle = \alpha^*\langle A|B\rangle. \tag{2.18}$$

It also obeys conjugate symmetry,

$$\langle A|B\rangle = \langle B|A\rangle^*. \tag{2.19}$$

Setting $B = A$ shows that $\langle A|A\rangle$ is real. Positivity is an additional axiom:

$$\langle A|A\rangle \geq 0, \qquad \langle A|A\rangle = 0 \iff |A\rangle = 0. \tag{2.20}$$

It is not a consequence of conjugate symmetry alone. Relativity supplies a useful warning: the Minkowski metric admits positive, negative, and null norms, so it is not an inner product in this positive-definite sense.

For complex-valued functions on a real interval I,

$$\langle \phi|\psi\rangle = \int_I dx\, \phi^*(x)\psi(x). \tag{2.21}$$

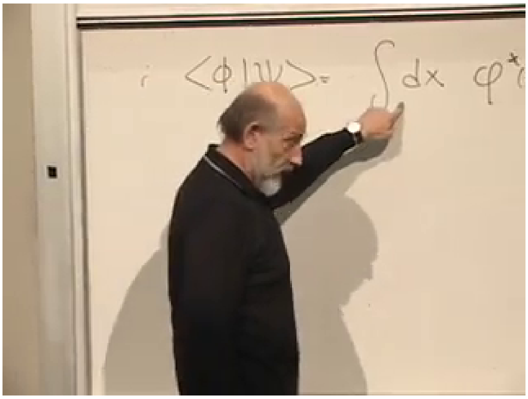

Figure 2.8: The function-space inner product being assembled on the board. The complete formula is Eq. (2.21).

Lecture frame: 00:56:30.

For columns,

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{pmatrix}, \qquad \langle b| = \begin{pmatrix} b_1^* & b_2^* & \cdots & b_d^* \end{pmatrix}, \tag{2.22}$$

and therefore

$$\langle b|a\rangle = \sum_{i=1}^{d} b_i^* a_i. \tag{2.23}$$

Question from the audience. Are the coordinates of the vector b the starred quantities?[a]

Response. The ket $|b\rangle$ has coordinates b_i. The corresponding bra $\langle b|$ has entries b_i^*. The bra symbol already indicates that conjugation has been made; one does not rename the ket coordinates themselves.

[a]Lecture timestamp: 00:57:55–00:58:37.

Question from the audience. Must the variable x in the function inner product be real?[a]

Response. In the present construction, yes. We may use several real variables, say x and y, and integrate over each of them. Functions of complex arguments can also be studied, but they are not needed for this course at this point.

[a]Lecture timestamp: 00:58:59–00:59:42.

2.6 Orthonormal Bases and Components

Choose d vectors $\{|b_i\rangle\}_{i=1}^d$ in a d-dimensional space. They are normalized when $\langle b_i|b_i\rangle = 1$, with no sum on i, and mutually orthogonal when $\langle b_i|b_j\rangle = 0$ for $i \neq j$. Both statements are contained in

$$\langle b_i|b_j\rangle = \delta_{ij}, \qquad \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{2.24}$$

An orthonormal set of d vectors in a d-dimensional space is a basis. The columns in Eq. (2.11) provide the standard example. Their corresponding bras are rows. A blackboard slip between the row and column forms was caught in the room; the corrected notation is the one used here.

Any vector can be expanded in the basis:

$$|v\rangle = \sum_{i=1}^{d} v_i |b_i\rangle. \tag{2.25}$$

Multiply from the left by $\langle b_j|$:

$$\langle b_j|v\rangle = \sum_{i=1}^{d} v_i \langle b_j|b_i\rangle \tag{2.26}$$

$$= \sum_{i=1}^{d} v_i \delta_{ji} = v_j. \tag{2.27}$$

The component v_j is the inner product with the j-th basis vector. Substitution gives the form we shall use repeatedly,

$$|v\rangle = \sum_{i=1}^{d} |b_i\rangle\langle b_i|v\rangle. \tag{2.28}$$

2.7 Classical Alternatives and Quantum Superpositions

We can now attach physical meaning to the vector space. A classical phase space is a set of possible states. Subsets support the ordinary logic of union and intersection, OR and AND. Quantum states carry additional linear structure. Adding two state vectors and multiplying them by complex

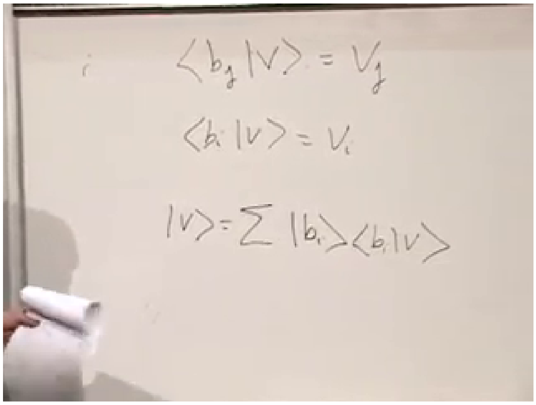

Figure 2.9: Extraction of a basis component and reconstruction of $|v\rangle$. The board preserves the same sequence as Eqs. (2.27) and (2.28).

Lecture frame: 01:13:45.

numbers can produce another physical state; mere set membership cannot express this.

Begin with a coin. Its classically distinguishable configurations are heads and tails. We may call them H and T, or label them $+1$ and -1, but those labels are names, not vector lengths. The quantum model assigns orthonormal states

$$\langle H|H\rangle = \langle T|T\rangle = 1, \qquad \langle H|T\rangle = 0. \tag{2.29}$$

The real physical example behind the coin is an electron with two distinguishable spin outcomes.

The orthogonality has operational content. Configurations that one ideal measurement can distinguish with certainty are represented by orthogonal vectors. This is one of the postulates we shall use. For a sixsided die, the same rule gives six states $|1\rangle, \ldots, |6\rangle$ satisfying

$$\langle i|j\rangle = \delta_{ij}. \tag{2.30}$$

Now comes the step that has no classical analogue:

$$|C\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle. \tag{2.31}$$

This 'confused coin' is a genuine quantum state, neither heads nor tails. It is a coherent superposition, not a classical statistical mixture whose member secretly has one definite face. Likewise,

$$|D\rangle = \sum_{i=1}^{6} \alpha_i|i\rangle \tag{2.32}$$

is a superposition of the six die outcomes.

2.8 Amplitudes, Probabilities, and Normalization

Question from the audience. How did the coefficients α_H and α_T become probabilities?[a]

Response. They do not themselves become probabilities. They are probability amplitudes, and the Born rule is a postulate:

$$P(H) = |\alpha_H|^2, \qquad P(T) = |\alpha_T|^2.$$

This is not the whole meaning of the amplitudes; their phases contain more information than these two probabilities reveal. Operationally, prepare a large ensemble of systems in the same state, make the same measurement on each, and identify the probabilities with the limiting frequencies of heads and tails.

[a]Lecture timestamp: 01:33:59–01:37:49.

If all outcomes are accounted for, their probabilities add to one:

$$|\alpha_H|^2 + |\alpha_T|^2 = 1. \tag{2.33}$$

Using Eq. (2.29), this is precisely

$$\langle C|C\rangle = 1. \tag{2.34}$$

For the die,

$$\sum_{i=1}^{6} |\alpha_i|^2 = 1, \qquad \langle D|D\rangle = 1. \tag{2.35}$$

Physical state vectors are therefore normalized.

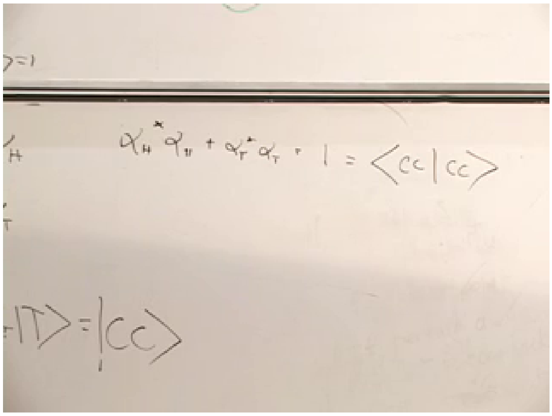

Figure 2.10: Normalization of the two-state superposition written as both a sum of squared amplitudes and the norm $\langle C|C\rangle$.

Lecture frame: 01:32:10.

Question from the audience. For equal heads and tails probabilities, are both amplitudes $1/\sqrt{2}$?[a]

Response. Their magnitudes are $1/\sqrt{2}$, but the amplitudes may be complex:

$$\alpha_H = \frac{e^{i\theta_H}}{\sqrt{2}}, \qquad \alpha_T = \frac{e^{i\theta_T}}{\sqrt{2}}.$$

Every complex number on the unit circle has unit magnitude. Multiplying such a number by $1/\sqrt{2}$ gives probability $1/2$. For six equally likely die outcomes, each amplitude has magnitude $1/\sqrt{6}$.

[a] Lecture timestamp: 01:38:02–01:39:24.

We have not yet said how to prepare an electron in the analogue of this half-heads, half-tails state. That experimental question will return when spin replaces the toy coin. For the moment it is more efficient to state the rules, learn how they work, and then ask which experiments forced physicists to adopt them. The first rules are now visible:

1. states are represented by vectors in a complex vector space;
2. perfectly distinguishable alternatives are represented by orthogonal vectors;
3. physical state vectors have unit norm; and
4. squared amplitude magnitudes give probabilities in an orthonormal measurement basis.

2.9 The First Look at Operators

There is one more mathematical object to introduce. An operation assigns a new vector to every vector in the space. Fixed-angle rotation, uniform rescaling, and reflection about an axis are familiar examples. Squaring a vector's length is also an operation, but it is not linear: a vector of length 2 is sent to one of length 4, while a vector of length 3 is sent to one of length 9.

We denote a linear operator by a hatted letter, such as $\hat{L}$. It obeys homogeneity and additivity:

$$\hat{L}\Big(\alpha|A\rangle\Big) = \alpha\,\hat{L}|A\rangle, \tag{2.36}$$

$$\hat{L}\Big(|A\rangle + |B\rangle\Big) = \hat{L}|A\rangle + \hat{L}|B\rangle. \tag{2.37}$$

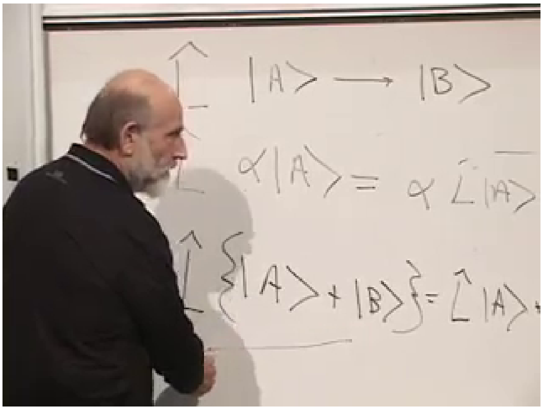

Figure 2.11: The two defining rules for a linear operator: scalar multiplication passes through the operator, and the image of a sum is the sum of the images.

Lecture frame: 01:49:20.

Equivalently, for arbitrary complex α, β,

$$\hat{L}\Big(\alpha|A\rangle + \beta|B\rangle\Big) = \alpha\hat{L}|A\rangle + \beta\hat{L}|B\rangle. \qquad (2.38)$$

Rotation by a fixed angle passes both tests. Rotating twice a vector gives twice the rotated vector. If $\mathbf{a}+\mathbf{b}$ is the diagonal of a parallelogram, rotating the whole parallelogram shows

$$R(\mathbf{a}+\mathbf{b}) = R\mathbf{a} + R\mathbf{b}. \qquad (2.39)$$

Uniform doubling and reflection are linear for the same reason. The length-squaring operation fails: scaling the input by α changes the length squared by $|\alpha|^2$, not by α.

State vectors describe the possible states of a quantum system. Linear operators will describe

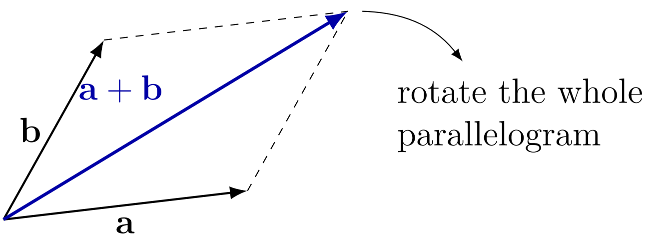

Figure 2.12: The parallelogram proof of additivity for a fixed rotation.

measurable quantities, or observables: the quantum version of asking whether the coin is heads or tails. The precise relation between an observable, its operator, and the outcomes of a measurement is the next step. We are now close enough to the formalism to begin making predictions.

CHAPTER 3

OPERATORS, MEASUREMENT, AND WAVE FUNCTIONS

Quantum mechanics is linear algebra. A state is a vector, while a measurable quantity is represented by a special kind of linear operator. The purpose of this lecture is to connect that abstract statement to calculations. We shall first turn operators into matrices, then identify Hermitian operators as observables, state the measurement postulates, and finally apply the entire structure to a particle moving on a line.

3.1 Operators, Matrix Elements, and Bases

A linear operator maps one ket into another:

$$\hat{K}\,|A\rangle = |C\rangle\,. \tag{3.1}$$

Once a bra is supplied, the result becomes a complex number,

$$\langle B|\,\hat{K}\,|A\rangle\,. \tag{3.2}$$

The order is important. The operator first acts on $|A\rangle$; the resulting ket is then paired with $\langle B|$.

Choose an orthonormal basis $\{|n\rangle\}_{n=1}^{d}$. An ordinary three-dimensional sketch helps us remember what a basis does, although a quantum state space is generally complex and may have far more than three dimensions.

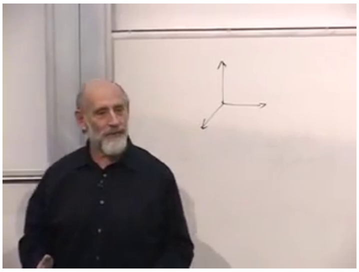

Figure 3.1: An ordinary vector sketch standing in for an orthonormal basis.

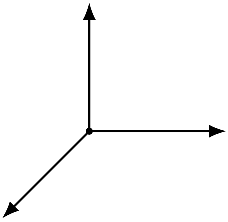

Figure 3.2: A clean reconstruction of the three-direction basis sketch.

Lecture frame: 00:04:29.

There are many possible bases. A vector drawn in one basis can always be expanded in another complete basis; only its components change.

Question from the audience. Can a vector associated with one drawn basis be represented in the other, and do the labels m and n refer to different bases?[a]

Response. Yes, the same vector can be expanded in either complete basis. In the matrix element below, however, both m and n are labels drawn from one chosen basis. They identify the row and column of the matrix representing $\hat{K}$ in that basis.

[a] Lecture timestamp: 00:05:24–00:07:40.

The basis matrix elements are

$$K_{mn} := \langle m| \hat{K} |n\rangle \,. \tag{3.3}$$

There are d^2 of them in a d-dimensional space, and together they determine the operator once the basis has been fixed.

Lecture frame: 00:06:22.

3.2 From Kets to Matrix Multiplication

Every ket can be expanded in the chosen basis:

$$|A\rangle = \sum_n A_n |n\rangle \,, \qquad A_n = \langle n \,|\, A\rangle \,. \tag{3.4}$$

Substitution gives

$$|A\rangle = \sum_n |n\rangle \langle n| \, |A\rangle \,. \tag{3.5}$$

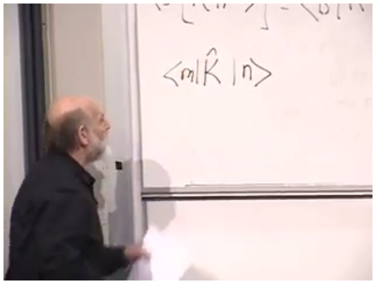

Figure 3.3: The matrix element $\langle m|\,\hat{K}\,|n\rangle$ on the board.

Since this holds for every $|A\rangle$, the operator between the summation sign and the final ket must be the identity:

$$\sum_n |n\rangle \langle n| = \mathbb{I}. \tag{3.6}$$

This is the completeness relation. The adjacent ket and bra form a projector onto one basis direction; summing all such projectors does nothing to the vector.

We can now calculate the components of $\hat{K}\,|A\rangle$:

$$\langle n|\,\hat{K}\,|A\rangle = \sum_m \langle n|\,\hat{K}\,|m\rangle \langle m\,|\,A\rangle \tag{3.7}$$

$$= \sum_m K_{nm} A_m. \tag{3.8}$$

Thus the abstract operation $\hat{K}\,|A\rangle$ is ordinary matrix multiplication when written in components.

Lecture frame: 00:13:37.

The same correspondence holds for products of operators. In $\hat{K}\hat{L}$, $\hat{L}$ acts first and $\hat{K}$ acts second.

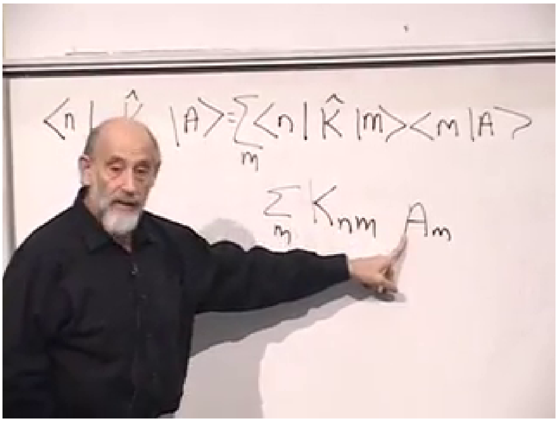

Figure 3.4: The passage from operator action to the component sum $\sum_m K_{nm}A_m$.

Inserting Eq. (3.6) between them gives

$$(KL)_{nm} = \langle n|\hat{K}\hat{L}|m\rangle \tag{3.9}$$
$$= \sum_r \langle n|\hat{K}|r\rangle\langle r|\hat{L}|m\rangle \tag{3.10}$$
$$= \sum_r K_{nr}L_{rm}. \tag{3.11}$$

This is precisely the rule for multiplying matrices. It also makes clear why operator order matters: in general $\hat{K}\hat{L} \neq \hat{L}\hat{K}$.

Lecture frame: 00:18:30.

3.3 Hermitian Operators and Their Eigenvectors

The operators associated with observables must have real measurement values. The relevant class is the Hermitian operators. Here the letter H merely stands for Hermitian; it is not yet the Hamiltonian.

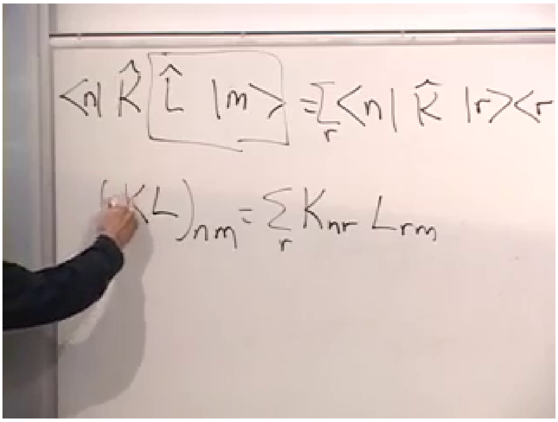

Figure 3.5: Insertion of a complete basis between two operators.

Definition 3.1. An operator $\hat{H}$ is Hermitian when

$$\langle B| \hat{H} |A\rangle = \Big(\langle A| \hat{H} |B\rangle\Big)^* \qquad (3.12)$$

for vectors in its domain. In finite dimensions this is equivalent to saying that $\langle A| \hat{H} |A\rangle$ is real for every $|A\rangle$.

Most vectors change direction under a general operator. An eigenvector is a special direction that does not:

$$\hat{H} |\lambda\rangle = \lambda |\lambda\rangle . \qquad (3.13)$$

Only its length and phase may change, through the scalar eigenvalue λ.

Question from the audience. Does every operation have eigenvectors? [a]

Response. Over a finite-dimensional complex vector space, every linear operator has at least one eigenvector, but a general operator need not have enough eigenvectors to form a basis, much less an orthonormal basis. A real rotation can also lack real eigenvectors. The special result needed here is stronger: a Hermitian operator has a complete orthonormal eigenbasis. In infinite-dimensional spaces, continuous spectra require generalized eigenvectors, as the position example will show.

[a] Lecture timestamp: 00:29:53–00:30:18.

The finite-dimensional spectral theorem supplies three facts:

1. every eigenvalue of a Hermitian operator is real;
2. eigenvectors with distinct eigenvalues are orthogonal;
3. the space has a complete orthonormal basis of eigenvectors.

The first two statements follow directly from Hermiticity. If $\hat{H}\,|\lambda\rangle = \lambda\,|\lambda\rangle$, then

$$\langle\lambda|\,\hat{H}\,|\lambda\rangle = \lambda\,\langle\lambda\,|\,\lambda\rangle\,. \tag{3.14}$$

The left side is real and the norm is positive, so λ is

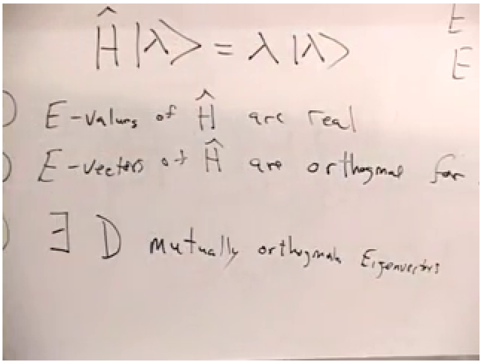

Figure 3.6: The eigenvalue equation and the three spectral facts for a Hermitian operator.

real. For two eigenvectors,

$$\langle\lambda_1|\,\hat{H}\,|\lambda_2\rangle = \lambda_2\,\langle\lambda_1\,|\,\lambda_2\rangle\,, \tag{3.15}$$

$$\langle\lambda_1|\,\hat{H}\,|\lambda_2\rangle = \lambda_1\,\langle\lambda_1\,|\,\lambda_2\rangle\,. \tag{3.16}$$

Consequently,

$$(\lambda_2 - \lambda_1)\,\langle\lambda_1\,|\,\lambda_2\rangle = 0. \tag{3.17}$$

Distinct eigenvalues therefore give orthogonal eigenvectors. If an eigenvalue is degenerate, an orthonormal basis can be chosen within its eigenspace. Finally, every eigenvector can be rescaled to unit norm.

Lecture frame: 00:33:40.

Question from the audience. Is a rotation Hermitian? [a]

Response. A finite rotation is generally unitary, or orthogonal in a real vector space, rather than Hermitian. Its infinitesimal generator is anti-Hermitian; equivalently, the generator can be written as $-i$ times a Hermitian operator. In a real basis, symmetric matrices are the analogues of Hermitian matrices, while infinitesimal rotation generators are antisymmetric.

[a] Lecture timestamp: 00:35:35–00:36:55.

Question from the audience. In which vector space do the eigenvectors form a basis? [a]

Response. They form a basis of the same complex vector space on which $\hat{H}$ acts. If we choose that eigenbasis for our coordinates, the matrix of $\hat{H}$ is diagonal, with its eigenvalues along the diagonal.

[a] Lecture timestamp: 00:37:35–00:40:05.

3.4 Five Measurement Postulates

The basic rules can now be stated in the same language.

1. A physical state is represented by a ket in a complex vector space. A coin has a two-dimensional state space, a die has a six-dimensional one, and a particle on a line requires an infinite-dimensional space.

2. Every observable is represented by a Hermitian

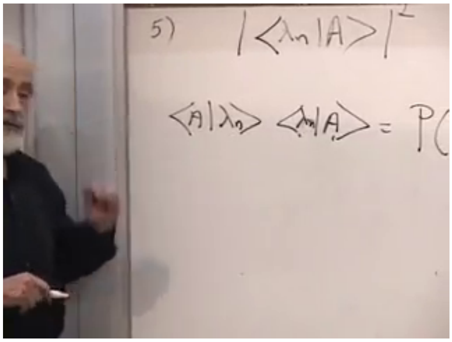

Figure 3.7: The fifth postulate written as a squared projection onto an eigenvector.

operator.

3. The possible outcomes of measuring an observable are the eigenvalues of its operator.

4. An eigenvector $|\lambda_n\rangle$ is a state in which that observable has the definite value λ_n.

5. If the normalized state is $|A\rangle$, the probability of obtaining λ_n is

$$P(\lambda_n \mid A) = |\langle\lambda_n \mid A\rangle|^2 . \tag{3.18}$$

Lecture frame: 00:54:50.

If the eigenvectors form an orthonormal basis, normalization guarantees that the probabilities sum to one:

$$\sum_n P(\lambda_n \mid A) = \sum_n |\langle\lambda_n \mid A\rangle|^2 = \langle A \mid A\rangle = 1. \tag{3.19}$$

Question from the audience. Is a state itself an observable? [a]

Response. No. Position, momentum, spin components, and fields are examples of observables. A state is not a number read from an apparatus; it encodes the probabilities for the possible readings. A single measurement therefore does not reveal the complete state.

[a] Lecture timestamp: 00:42:05–00:42:50.

Question from the audience. If eigenvalues are real, what about experimental error bars?[a]

Response. Real eigenvalues do not eliminate experimental uncertainty. An apparatus may report a real interval or a real value with an error bar. The point is that a measurement outcome is not an imaginary number.

[a] Lecture timestamp: 00:45:33–00:46:28.

Question from the audience. Do we observe eigenvectors, and what does it mean for an observable to be definite?[a]

Response. The apparatus records an eigenvalue, not an eigenvector. An observable is definite when repeated ideal measurements on identically prepared systems give the same value with probability one. For example, a spin state can be definite along one axis and probabilistic along another, because it is an eigenstate of one spin operator but not of the other.

[a] Lecture timestamp: 00:47:41–00:50:29.

3.5 A Particle on a Line

We now leave finite-dimensional examples and consider a particle moving on the real line. Its states are represented by complex functions $\psi(x)$. Functions form a vector space, and the natural inner product is

$$\langle \phi \,|\, \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x)\, dx. \tag{3.20}$$

We shall initially work with square-integrable, sufficiently well-behaved functions and impose whatever boundary conditions are needed by the operators.

Question from the audience. Is the inner product of two wave functions always real? [a]

Response. No. The inner product $\langle \phi \,|\, \psi \rangle$ can be complex. What must be real and nonnegative is a vector's inner product with itself:

$$\langle \psi \,|\, \psi \rangle = \int_{-\infty}^{\infty} |\psi(x)|^2\, dx.$$

[a] Lecture timestamp: 00:58:13–00:59:07.

3.5.1 The position operator

The position operator multiplies a wave function by its argument:

$$(\hat{x}\psi)(x) = x\psi(x). \tag{3.21}$$

It is Hermitian on its proper domain because

$$\langle \psi |\, \hat{x} \,| \psi \rangle = \int_{-\infty}^{\infty} x|\psi(x)|^2\, dx \tag{3.22}$$

is real.

The position eigenvalue equation is

$$\hat{x}\,|\lambda\rangle = \lambda\,|\lambda\rangle\,, \tag{3.23}$$

or, in the x-representation,

$$(x - \lambda)\psi_\lambda(x) = 0. \tag{3.24}$$

For every $x \neq \lambda$, the factor $x - \lambda$ is nonzero, so the eigenfunction must vanish. Its support is concentrated at the single point $x = \lambda$.

Question from the audience. Have we assumed that the norm integral converges? [a]

Response. Yes. Physical state vectors are normalized, so their wavefunctions are square-integrable. For the elementary manipulations here we also assume sufficiently tame behavior at large distance. Square integrability by itself does not force pointwise decay in every mathematical sense, but the chosen operator domains will supply the boundary behavior needed below.

[a] Lecture timestamp: 01:05:02–01:05:42.

No ordinary square-integrable function can be nonzero at only one point and have unit norm. The position eigenfunction is therefore a generalized function:

$$\langle x\,|\,\lambda\rangle = \delta(x - \lambda). \tag{3.25}$$

The lecture pictures $\delta(x - \lambda)$ as a rectangle of width ϵ and height $1/\epsilon$, then lets ϵ shrink. This choice

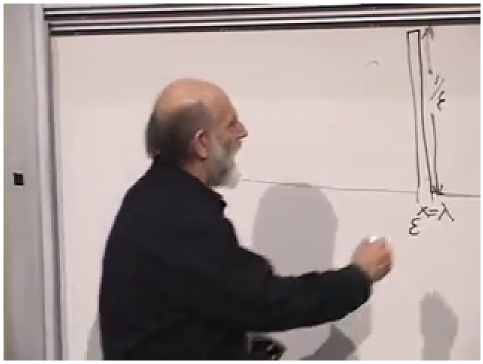

Figure 3.8: A narrow unit-area approximation to $\delta(x-\lambda)$.

gives unit area,

$$\int_{-\infty}^{\infty} \delta(x-\lambda)\, dx = 1, \tag{3.26}$$

but it does not give unit Hilbert-space norm: the integral of the square of the rectangle grows like $1/\epsilon$. The delta function is not an ordinary normalized state.

Lecture frame: 01:08:33.

Question from the audience. Are these position eigenfunctions orthonormal? [a]

Response. They are orthogonal in the generalized continuous sense, but not normalizable one by one as ordinary vectors. The precise replacement for Kronecker-delta normalization is

$$\langle x \,|\, x' \rangle = \delta(x - x').$$

Position eigenkets are delta-normalized generalized eigenvectors.

[a] Lecture timestamp: 01:09:13–01:11:22.

There is one such generalized eigenvector for every real position, so the spectrum of $\hat{x}$ is continuous. Completeness now takes the form

$$\int_{-\infty}^{\infty} |x\rangle \langle x| \; dx = \mathbb{I}. \tag{3.27}$$

3.5.2 What the wave function means

The delta function samples an ordinary function:

$$\int_{-\infty}^{\infty} \delta(x - \lambda) f(x) \, dx = f(\lambda). \tag{3.28}$$

Applying this to a state gives

$$\langle \lambda \,|\, \psi \rangle = \int_{-\infty}^{\infty} \delta(x - \lambda)\psi(x) \, dx = \psi(\lambda). \tag{3.29}$$

Relabeling the eigenvalue by x,

$$\psi(x) = \langle x \,|\, \psi \rangle \,. \tag{3.30}$$

The wave function is therefore the component of the abstract state ket in the position basis.

The fifth postulate then says that

$$\rho(x) = |\psi(x)|^2 \tag{3.31}$$

is the position probability density. A probability belongs to an interval, not to one isolated point:

$$P(a \leq x \leq b) = \int_a^b |\psi(x)|^2 \, dx. \tag{3.32}$$

Question from the audience. Has ψ changed from an arbitrary function into one special function?[a]

Response. No. Any normalized state wavefunction in the allowed domain supplies a probability distribution through $|\psi(x)|^2$. The wavefunction itself may be complex and may carry signs and phases. Its absolute square is real and nonnegative. Those phases are essential because position is not the only observable we may measure; replacing ψ by the probability density alone would discard information needed for other measurements.

[a] Lecture timestamp: 01:18:43–01:19:28.

3.6 Differentiation and the Momentum Operator

Differentiation is another linear operation on functions:

$$\frac{d}{dx}(\alpha\psi + \beta\phi) = \alpha\frac{d\psi}{dx} + \beta\frac{d\phi}{dx}. \tag{3.33}$$

The derivative itself is not Hermitian. With boundary terms chosen to vanish, it is anti-Hermitian:

$$\left(\frac{d}{dx}\right)^{\dagger} = -\frac{d}{dx}. \tag{3.34}$$

Multiplication by $-i$ converts it into a Hermitian operator,

$$\hat{K} = -i\frac{d}{dx}. \tag{3.35}$$

The key calculation is integration by parts. For functions in a common domain with $\phi^*(x)\psi(x) \to 0$ at both ends,

$$\langle\phi|\,\hat{K}\,|\psi\rangle = -i\int_{-\infty}^{\infty} \phi^*(x)\frac{d\psi}{dx}\,dx \tag{3.36}$$

$$= -i\,[\phi^*(x)\psi(x)]_{-\infty}^{\infty} + i\int_{-\infty}^{\infty} \frac{d\phi^*}{dx}\psi(x)\,dx \tag{3.37}$$

$$= \left\langle \hat{K}\phi \,\middle|\, \psi \right\rangle. \tag{3.38}$$

Thus $\hat{K}$ is Hermitian on that domain. The boundary conditions are part of the operator; without them, the formal differential expression alone is not enough to establish Hermiticity.

Lecture frame: 01:31:37.

3.6.1 Plane-wave eigenfunctions

The eigenvalue equation

$$-i\frac{d\psi_k}{dx} = k\psi_k \tag{3.39}$$

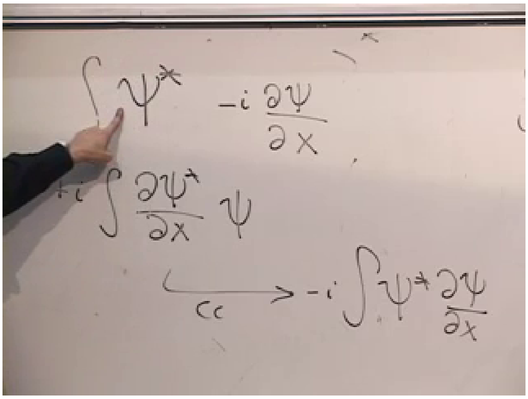

Figure 3.9: The integration-by-parts step that moves the derivative and changes its sign.

has the solution

$$\psi_k(x) \propto e^{ikx} = \cos(kx) + i\sin(kx). \tag{3.40}$$

Indeed,

$$-i\frac{d}{dx}e^{ikx} = -i(ik)e^{ikx} = ke^{ikx}. \tag{3.41}$$

Lecture frame: 01:34:50.

A plane wave is the opposite of a position eigenstate. Its magnitude is constant:

$$|e^{ikx}|^2 = 1. \tag{3.42}$$

Formally, it gives no preferred position at all. It is also not square-integrable on the whole line, so, like the delta-localized position eigenket, it is a generalized eigenstate rather than a normalizable physical state by itself.

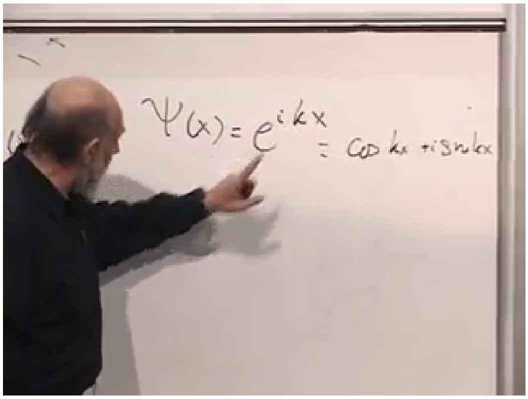

Figure 3.10: The plane-wave eigenfunction of $\hat{K} = -i\, d/dx$.

Question from the audience. Did the Hermiticity argument not assume decay at infinity, while the plane waves fail to decay?[a]

Response. Yes. The tension is real and should not be hidden. Plane waves belong to the generalized continuous spectrum. One may approximate them by long, normalizable wave packets, work in a large periodic box and then take its size to infinity, or use delta normalization. The formal plane waves remain useful because they expose the spectral structure exactly.

[a]Lecture timestamp: 01:38:41–01:39:47.

3.7 Why Wave Number Measures Momentum

One full oscillation of e^{ikx} occurs when its phase changes by 2π. If L denotes the wavelength, then

$$kL = 2\pi, \qquad L = \frac{2\pi}{|k|}. \tag{3.43}$$

The sign of k distinguishes the two propagation directions, while its magnitude fixes the wavelength.

Einstein's photon relation and de Broglie's extension to material particles suggest

$$p = \frac{h}{L}. \tag{3.44}$$

Using $L = 2\pi/|k|$ for the magnitude and retaining the sign for direction,

$$p = \hbar k, \qquad \hat{p} = \hbar\hat{K} = -i\hbar\frac{d}{dx}. \tag{3.45}$$

Lecture frame: 01:43:50.

At this stage, Eq. (3.45) is a well-motivated identification rather than a completed derivation from classical mechanics. To justify the name physically, we must introduce time evolution and show that localized wave packets move in the classical limit with velocity related to their k-content.

Question from the audience. Does large k mean a large or a small wavelength? [a]

Response. A large $|k|$ gives rapid variation and a short wavelength; a small $|k|$ gives slow variation and a long wavelength, because $L = 2\pi/|k|$.

[a] Lecture timestamp: 01:54:03–01:54:30.

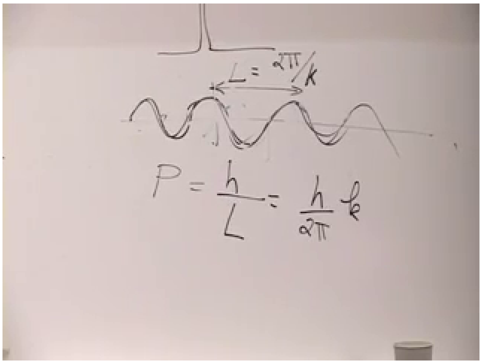

Figure 3.11: Wavelength and the de Broglie identification $p = \hbar k$.

3.8 Position, Momentum, and Incompatible Bases

The position and momentum eigenfunctions could hardly look more different. A position eigenfunction is concentrated at one point and contains no definite k. A momentum eigenfunction extends over the entire line and contains no definite position. The two families form different generalized bases in the same state space.

This is the first glimpse of the uncertainty principle in the language of state vectors. A sharply localized wave packet requires a superposition of many values of k, while a single value of k produces complete delocalization. The precise inequality will come later; the clash of the two eigenbases is already visible.

There is also a useful geometric way to picture the complex plane wave. At each point x, e^{ikx} is a

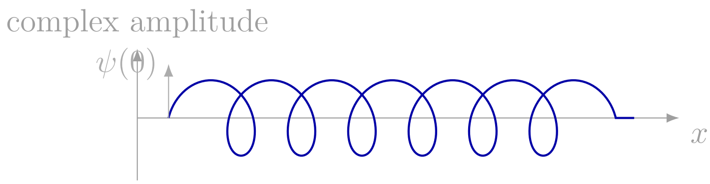

Figure 3.12: Transcript-based schematic of the plane-wave phase winding as a helix along the spatial axis.

point on the unit circle in the complex plane. As x increases, that point winds around the circle. If the spatial axis is drawn perpendicular to the complex plane, the resulting curve is a helix. Large $|k|$ winds rapidly; small $|k|$ winds slowly.

Question from the audience. Can e^{ikx} be visualized as a function twisting around the x-axis?[a]

Response. Yes. The real and imaginary parts are the two perpendicular coordinates of a unit vector in the complex plane. Advancing along the spatial axis rotates that vector through phase kx, producing the helix described above.

[a]Lecture timestamp: 01:50:04–01:53:05.

Question from the audience. Why was wave-function collapse absent from the five postulates, and what is the probability distribution for momentum? [a]

Response. The lecture has so far stated only which outcomes are possible and how their probabilities are assigned. The physical measurement process and its effect on the state have not yet been introduced. Likewise, the momentum-space probability rule requires expanding the state in the $|k\rangle$ basis; that construction is deferred to the next step of the course.

[a]Lecture timestamp: 01:53:15–01:54:02.

Question from the audience. Classical momentum involves motion in time. Why has a spatial derivative appeared before any time derivative? [a]

Response. The identification must ultimately be checked through time evolution. A normalizable wave packet forms a localized lump. In the appropriate heavy-particle or classical limit, that lump should move according to the classical equations, with its velocity related to the wave-number content of the packet. Time has not yet entered the theory, so that dynamical check remains ahead.

[a]Lecture timestamp: 01:54:31–01:56:03.

CHAPTER 4

CONTINUUM STATES, QUANTIZED MOMENTUM, AND INTERFERENCE

4.1 When a Discrete Label Becomes Continuous

Begin with an observable whose possible values are labeled by integers $n = 1, 2, \ldots$. Its probabilities obey

$$\sum_n P(n) = 1, \tag{4.1}$$

and the corresponding basis vectors may be chosen orthonormally:

$$\langle m \,|\, n \rangle = \delta_{mn}. \tag{4.2}$$

The Kronecker delta is one when its two labels agree and zero otherwise. It can equally well be regarded as a function of the difference $m - n$.

For a state $|\psi\rangle$, the probability of obtaining n is

$$P(n) = |\langle n \,|\, \psi \rangle|^2 \,. \tag{4.3}$$

These three statements are the pattern we must preserve when the label is no longer discrete.

Let the measured quantity now be the position x of a particle on a line. For an ordinary continuous distribution, a single point has zero probability. The useful object is therefore a probability density $\rho(x)$.

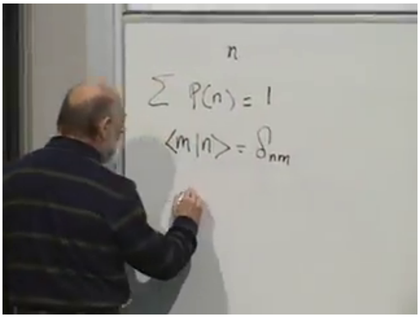

Figure 4.1: Discrete probability normalization and basis orthonormality.

Lecture frame: 00:03:11.

The probability of finding the particle in a short interval is

$$\text{Prob}(x_1 \le x \le x_2) = \int_{x_1}^{x_2} \rho(x)\, dx, \tag{4.4}$$

$$\int_{-\infty}^{\infty} \rho(x)\, dx = 1. \tag{4.5}$$

Thus sums become integrals, and probability masses become densities.

4.2 The Dirac Delta

The continuum analogue of the Kronecker delta is the Dirac delta distribution. Its informal picture is an infinitely narrow spike at $x = y$, zero away from that point, with unit area:

$$\delta(x-y) = 0 \quad (x \ne y), \qquad \int_{-\infty}^{\infty} \delta(x-y)\, dx = 1. \tag{4.6}$$

No ordinary function has these properties. The precise definition is its action on a sufficiently regular

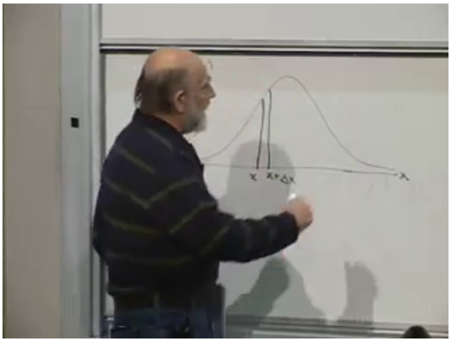

Figure 4.2: For a continuous coordinate, probability belongs to an interval and is represented by area under a density.

Lecture frame: 00:05:59.

test function:

$$\int_{-\infty}^{\infty} \delta(x-y) f(x)\, dx = f(y). \tag{4.7}$$

The delta is therefore best understood as a distribution under an integral, not as a function with a literal infinite value.

A useful approximation makes the spike picture explicit. Let $\delta_\epsilon(x-y)$ be a rectangle of width ϵ and height $1/\epsilon$, centered at y. Its area is one, and for continuous f,

$$\int \delta_\epsilon(x-y) f(x)\, dx \longrightarrow f(y) \qquad (\epsilon \to 0). \tag{4.8}$$

The unit area must not be confused with a unit Hilbert-space norm: $\int |\delta_\epsilon|^2\, dx = 1/\epsilon$ diverges in the limit.

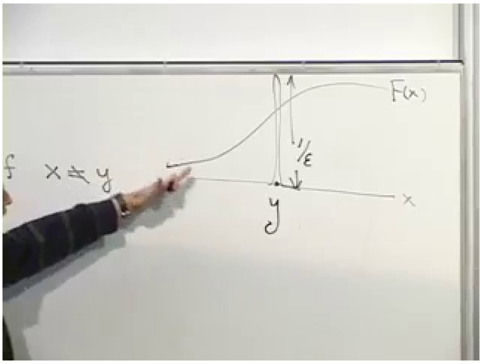

Figure 4.3: A narrow unit-area spike samples the value of a smooth function at $x = y$.

Lecture frame: 00:12:00.

Position basis states therefore obey delta normalization:

$$\langle x \,|\, y\rangle = \delta(x-y), \qquad \int_{-\infty}^{\infty} dx\ |x\rangle\,\langle x| = \mathbf{1}. \quad (4.9)$$

Distinct positions are orthogonal. The formal quantity $\langle x \,|\, x\rangle = \delta(0)$ is divergent because $|x\rangle$ is a generalized eigenstate, not a normalizable vector in the Hilbert space.

4.3 The Position Amplitude and Born Density

The abstract probability rule has not changed. In the discrete case, $|\langle n \,|\, \psi\rangle|^2$ is the probability for the outcome n. In the continuous case,

$$\rho(y) = |\langle y \,|\, \psi\rangle|^2 \quad (4.10)$$

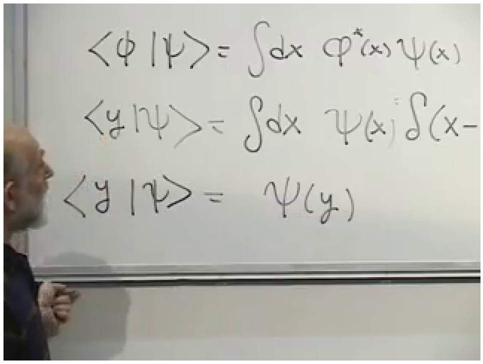

Figure 4.4: Delta sampling identifies the position amplitude $\langle y \,|\, \psi \rangle$ with the wavefunction value $\psi(y)$. *Lecture frame: 00:22:00.*

is the probability density at y. To see what this means in the position representation, write

$$\psi(x) := \langle x \,|\, \psi \rangle \,. \tag{4.11}$$

The wavefunction of the generalized position state $|y\rangle$ is $\langle x \,|\, y \rangle = \delta(x - y)$. Hence

$$\langle y \,|\, \psi \rangle = \int dx \; \langle y \,|\, x \rangle \, \langle x \,|\, \psi \rangle \tag{4.12}$$

$$= \int dx \, \delta(x - y)\psi(x) = \psi(y). \tag{4.13}$$

Born's rule is therefore

$$\rho(y) = |\psi(y)|^2 \,, \tag{4.14}$$

$$\mathrm{Prob}(y \in [a, b]) = \int_a^b |\psi(y)|^2 \; dy. \tag{4.15}$$

Question from the audience. Could the first probability postulate be stated once more for the continuous case?[a]

Response. For a discrete observable, the probability of outcome n is $|\langle n \,|\, \psi\rangle|^2$. For position, the same expression $|\langle y \,|\, \psi\rangle|^2$ is a density. Since $\langle y \,|\, \psi\rangle = \psi(y)$, the probability of an interval is the integral of $|\psi(y)|^2$ over that interval. Dirac's notation was designed so that the discrete and continuous statements have the same form.

[a]Lecture timestamp: 00:22:39–00:26:37.

4.4 A Lattice Explains Continuum Normalization

The divergent self-overlap of $|x\rangle$ becomes transparent if the line is temporarily replaced by a lattice with spacing a. Let $x_n = na$, and let $|n\rangle$ be ordinary normalized site states:

$$\langle n \,|\, m\rangle = \delta_{nm}, \qquad \sum_n |n\rangle \langle n| = \mathbf{1}. \tag{4.16}$$

Define continuum-normalized states at the lattice points by

$$|x_n\rangle := \frac{1}{\sqrt{a}} |n\rangle . \tag{4.17}$$

Their overlap and completeness relations are

$$\langle x_n \,|\, x_m\rangle = \frac{\delta_{nm}}{a}, \qquad \sum_n a \, |x_n\rangle \langle x_n| = \mathbf{1}. \tag{4.18}$$

As $a \to 0$, the sum $\sum_n a$ becomes an integral, while δ_{nm}/a becomes $\delta(x-y)$. The height $1/a$ and width a again give unit area.

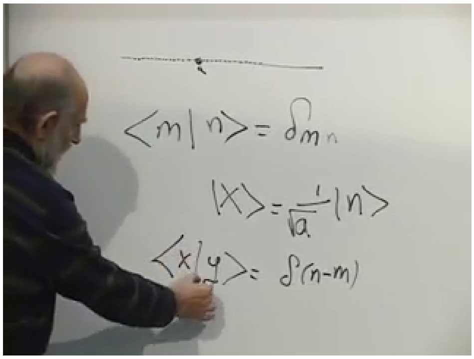

Figure 4.5: Rescaling lattice kets by $a^{-1/2}$ produces Dirac-delta normalization in the continuum limit.

Lecture frame: 00:30:00.

Question from the audience. How can the overlap of a state with itself be infinite without ruining the probability interpretation? [a]

Response. The position eigenstates are not physical normalized states; they are generalized basis vectors. On a lattice their normalized ancestors are perfectly ordinary. The factor $a^{-1/2}$ converts a discrete probability into a density, and it converts δ_{nm} into δ_{nm}/a. Finite wave packets remain normalized, and every delta appears under an integral where its unit-area property gives a finite answer.

[a] Lecture timestamp: 00:26:38–00:34:24.

4.5 A Particle on a Circle

Now constrain the particle to a circle of radius R. Cutting the circle at one point turns it into the

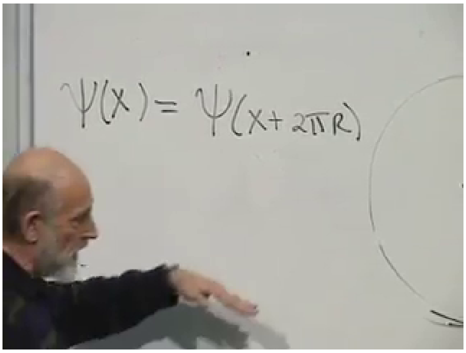

Figure 4.6: Cutting the circle gives an interval whose wavefunction values must match after one circumference.

Lecture frame: 00:38:00.

interval

$$0 \leq x < 2\pi R, \tag{4.19}$$

but the two ends represent the same physical point. The wavefunction must therefore satisfy

$$\psi(x + 2\pi R) = \psi(x). \tag{4.20}$$

Periodic functions form a complex vector space: scalar multiples and sums of periodic functions are still periodic.

On this space the momentum operator is

$$\hat{p} = -i\hbar \frac{d}{dx}, \tag{4.21}$$

with periodic boundary conditions included in its domain. Its eigenvalue equation,

$$-i\hbar \frac{d\psi_p}{dx} = p\psi_p, \tag{4.22}$$

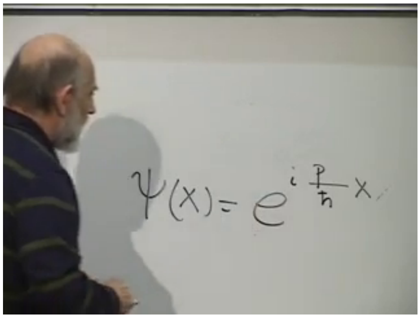

Figure 4.7: The momentum eigenfunction is a complex plane wave.

Lecture frame: 00:41:40.

has the plane-wave solution

$$\psi_p(x) = Ce^{ipx/\hbar}. \tag{4.23}$$

The name momentum will acquire its full physical justification only after time evolution and the motion of wave packets are developed. For now it is the eigenvalue of the translation generator.

Normalization on the finite circle fixes C:

$$1 = \int_0^{2\pi R} \left|\psi_p(x)\right|^2 \, dx = 2\pi R \left|C\right|^2, \tag{4.24}$$

$$\psi_p(x) = \frac{1}{\sqrt{2\pi R}} e^{ipx/\hbar}. \tag{4.25}$$

The constant magnitude means that a momentum eigenstate has uniform position density around the circle.

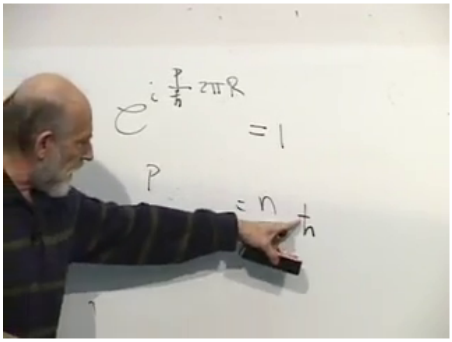

Figure 4.8: Single-valuedness around the circle restricts momentum to $p_n = n\hbar/R$.

Lecture frame: 00:50:00.

4.6 Periodicity Quantizes Momentum

The normalized plane wave is not automatically an allowed state. It must also obey the periodic boundary condition:

$$e^{ip(x+2\pi R)/\hbar} = e^{ipx/\hbar}, \tag{4.26}$$

$$e^{i2\pi Rp/\hbar} = 1. \tag{4.27}$$

The phase gained in one circuit must be an integer multiple of 2π:

$$\frac{2\pi Rp}{\hbar} = 2\pi n, \qquad n \in \mathbb{Z}. \tag{4.28}$$

Thus

$$p_n = \frac{n\hbar}{R}, \qquad \psi_n(x) = \frac{1}{\sqrt{2\pi R}} e^{inx/R}. \tag{4.29}$$

The allowed functions are mutually orthogonal:

$$\int_0^{2\pi R} \psi_m^*(x)\psi_n(x)\, dx = \delta_{mn}. \tag{4.30}$$

Here position is continuous while momentum is discrete. In this problem the discreteness is a consequence of compactness and periodicity; discrete spectra in other systems can arise from other boundary conditions or from a confining potential.

The spacing between neighboring momenta is

$$\Delta p = \frac{\hbar}{R}. \tag{4.31}$$

As $R \to \infty$, this spacing tends to zero. A continuum-normalized momentum basis may be obtained by the same rescaling used for position:

$$\begin{aligned} |p_n\rangle_{\mathrm{c}} &:= \frac{1}{\sqrt{\Delta p}}\,|n\rangle\,, && (4.32)\\ {}_{\mathrm{c}}\langle p_m \,|\, p_n\rangle_{\mathrm{c}} &= \frac{\delta_{mn}}{\Delta p} && (4.33)\\ &\longrightarrow \delta(p-p'). && (4.34) \end{aligned}$$

For orbital motion on the circle, the angular momentum is

$$L = Rp, \tag{4.35}$$

and therefore

$$L_n = Rp_n = n\hbar. \tag{4.36}$$

The radius cancels because angular momentum measures phase winding around an angle. This conclusion applies to a scalar orbital wavefunction with $\psi(\phi + 2\pi) = \psi(\phi)$. Half-integer spin belongs to a different representation and is a later subject.

Question from the audience. Does $R \to \infty$ recover classical mechanics? [a]

Response. No. It only makes the momentum spectrum dense. Classical behavior requires more than closely spaced eigenvalues; in particular, it requires suitable wave packets and a dynamical limit in which position and momentum can both be narrow compared with the relevant macroscopic scales.

[a]Lecture timestamp: 00:56:58–00:57:20.

Question from the audience. Why does R remain in the momentum spacing but disappear from the angular-momentum spectrum? [a]

Response. A fixed phase winding n around a larger circle has a smaller tangential wave number and hence smaller $p_n = n\hbar/R$. Multiplying by the lever arm R restores $L_n = n\hbar$. The quantized object is the action associated with a full angular circuit.

[a]Lecture timestamp: 00:57:37–00:58:48.

4.7 Compatibility and Common Eigenstates

A classical particle may be assigned a position and a momentum at the same instant. Quantum mechanically, a position eigenstate is localized like a delta distribution, whereas a momentum eigenstate is a plane wave with uniform position density. These states cannot be the same.

The algebraic question is whether two Hermitian operators $\hat{A}$ and $\hat{B}$ possess a complete common eigenbasis. If

$$\hat{A}\left|n\right\rangle = a_n \left|n\right\rangle , \qquad \hat{B}\left|n\right\rangle = b_n \left|n\right\rangle , \tag{4.37}$$

then

$$\hat{A}\hat{B}\left|n\right\rangle = a_n b_n \left|n\right\rangle = \hat{B}\hat{A}\left|n\right\rangle . \tag{4.38}$$

Define the commutator

$$[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{4.39}$$

Compatible observables commute. Conversely, commuting Hermitian matrices can be simultaneously diagonalized, including within degenerate eigenspaces. For unbounded operators, the same statement requires compatible domains and the corresponding spectral conditions; those analytic details are suppressed here.

4.8 The Canonical Commutator

For position and momentum,

$$(\hat{x}\psi)(x) = x\psi(x), \qquad (\hat{p}\psi)(x) = -i\hbar\psi'(x). \tag{4.40}$$

Apply their commutator to an arbitrary differentiable wavefunction:

$$\begin{aligned} [\hat{x}, \hat{p}]\psi &= x\left(-i\hbar\frac{d\psi}{dx}\right) + i\hbar\frac{d}{dx}\Big(x\psi\Big) && (4.41)\\ &= -i\hbar x\psi' + i\hbar\Big(\psi + x\psi'\Big) && (4.42)\\ &= i\hbar\psi. && (4.43) \end{aligned}$$

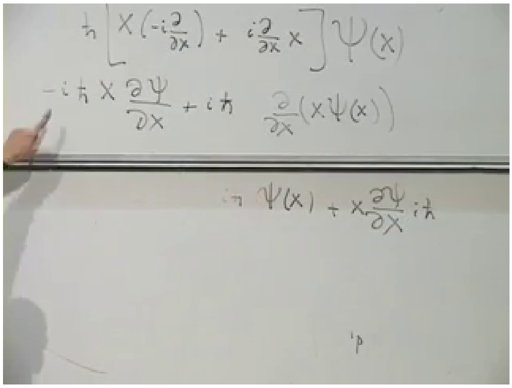

Figure 4.9: The product rule supplies the uncancelled term in $[\hat{x}, \hat{p}]\psi = i\hbar\psi$.

Lecture frame: 01:13:00.

Since this holds for every state in the common domain,

$$[\hat{x}, \hat{p}] = i\hbar\,\mathbf{1}. \tag{4.44}$$

The commutator is nonzero on every nonzero state. If a vector were a common eigenvector of $\hat{x}$ and $\hat{p}$, their commutator would annihilate it, contradicting (4.44). Position and momentum therefore have no common eigenvector. The uncertainty relation will later quantify this incompatibility.

Question from the audience. Is every state an eigenvector of the commutator, and can $i\hbar$ be regarded as an observable value? [a]

Response. Every vector is indeed an eigenvector of $[\hat{x}, \hat{p}] = i\hbar\mathbf{1}$, with the same eigenvalue. But the commutator of two Hermitian operators is anti-Hermitian, so it is not itself an observable. The Hermitian combination $-i[\hat{x}, \hat{p}] = \hbar\mathbf{1}$ contains no state-dependent measurement information. Its importance is that it fixes the algebra relating position and momentum.

[a] Lecture timestamp: 01:15:33–01:16:45.

Question from the audience. Why is $x\psi(x)$ not an eigenvalue times $\psi(x)$? [a]

Response. An eigenvalue is one scalar constant for the whole vector. The coordinate x varies across the function. A generic Gaussian, for example, is therefore not an eigenfunction of the position operator. Only a generalized state supported at one point satisfies $x\delta(x - x_0) = x_0\delta(x - x_0)$.

[a] Lecture timestamp: 01:16:51–01:19:07.

4.9 One Aperture and Two Alternatives

The two-slit experiment turns the linear structure of states into a visible probability pattern. An incoming particle meets a barrier with two small openings, and a screen records the position y at which it arrives. With only one opening available, the outgoing wave

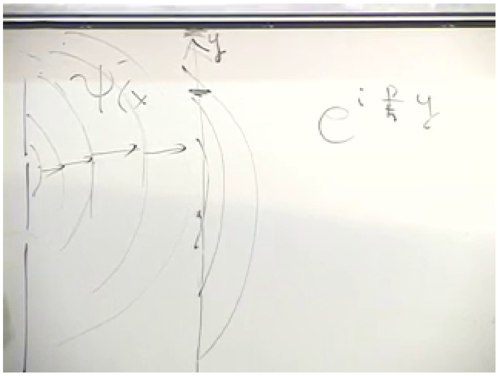

Figure 4.10: A small opening produces an outgoing wave whose local phase gradient changes across the screen.

Lecture frame: 01:23:00.

spreads. In three dimensions a schematic far-field contribution has the form

$$\psi(r) \sim \frac{e^{ikr}}{r}, \qquad r = \sqrt{\ell^2 + y^2}, \tag{4.45}$$

where ℓ is the distance from aperture to screen. Over a small region of the screen it may be approximated by a local plane wave.

Now open both holes. Let $\psi_1(y)$ and $\psi_2(y)$ be the amplitudes associated with the two alternatives. The state-space rule is

$$\psi(y) = \psi_1(y) + \psi_2(y), \tag{4.46}$$

not the addition of the separate probabilities. Locally, suppressing slowly varying envelopes and temporarily setting $\hbar = 1$, write

$$\psi_1 = e^{ipy}, \qquad \psi_2 = e^{iqy}. \tag{4.47}$$

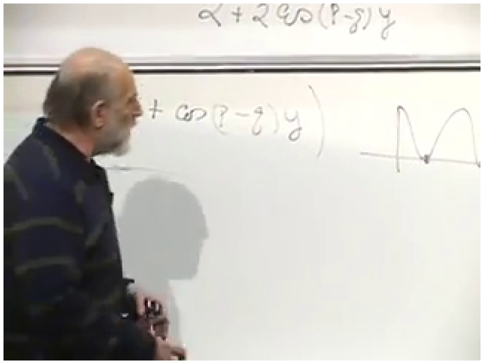

Figure 4.11: The cross terms in the squared sum generate a nonnegative oscillating detection density.
Lecture frame: 01:30:00.

Then

$$|\psi_1 + \psi_2|^2 = (e^{ipy} + e^{iqy})(e^{-ipy} + e^{-iqy}) \tag{4.48}$$
$$= 2 + e^{i(p-q)y} + e^{-i(p-q)y} \tag{4.49}$$
$$= 2 + 2\cos\big((p-q)y\big). \tag{4.50}$$

The cross terms retain the relative phase. They create alternating bright and dark fringes even though either local plane wave separately has constant magnitude.

In the far field, if the slit separation is d, the path difference at angle θ is approximately $d\sin\theta$. For wavelength $\lambda = 2\pi/k$, adjacent bright fringes satisfy

$$d\sin\theta_m = m\lambda. \tag{4.51}$$

At small angles, $y \approx \ell\theta$, so

$$\Delta y \approx \frac{\lambda\ell}{d}. \tag{4.52}$$

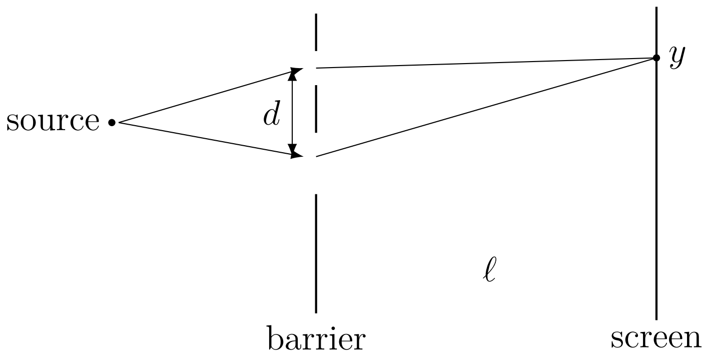

Figure 4.12: The relative path length from two separated openings changes with the screen coordinate.

Moving the holes farther apart makes the fringes closer; moving them together spreads the fringes apart.

Question from the audience. What happens in the local model if $p = q$? [a]

Response. Then the relative phase is constant: $\psi_1 + \psi_2 = 2e^{ipy}$. The intensity is the constant 4, in the chosen normalization. There are no alternating fringes, but the two contributions are everywhere fully constructive.

[a]Lecture timestamp: 01:32:25–01:33:08.

Question from the audience. Is this the Fraunhofer diffraction pattern? [a]

Response. It is the far-field two-aperture part of that pattern. The local beat model isolates the rapidly varying two-path fringes. A full Fraunhofer calculation also retains the finite width of each opening, which multiplies those fringes by the broader one-aperture diffraction envelope.

[a] Lecture timestamp: 01:38:39–01:39:57.

4.10 Which-Path Information Must Wait

Suppose a detector at the barrier records which opening the electron used. The detector must then become correlated with the alternative. Schematically,

$$|\Psi\rangle = |\text{path } 1\rangle\,|D_1\rangle + |\text{path } 2\rangle\,|D_2\rangle\,. \tag{4.53}$$

When the detector records are distinguishable, the electron alone no longer has the same interference term. A full explanation requires composite systems and entanglement, so it belongs after the tensor-product structure has been developed.

Question from the audience. Why does a detector placed at the openings destroy the interference pattern?[a]

Response. Detecting the path means leaving different imprints in another physical system. The electron and detector must then be described as one composite state, with each path correlated with a different detector record. The interference disappears when those records are distinguishable. The mechanism is entanglement, whose derivation must wait until composite systems are available.

[a]Lecture timestamp: 01:40:04–01:41:53.

4.11 Phase Gradients and Elastic Deflection

For a wave of the form $e^{i\phi(y)}$, the local vertical momentum is the phase gradient

$$p_y(y) = \hbar \frac{d\phi}{dy}. \tag{4.54}$$

Near the central axis the distance $r = \sqrt{\ell^2 + y^2}$ changes slowly with y, so the vertical phase gradient is small. Farther from the axis, the wave has a larger vertical momentum component. This does not require an increase in kinetic energy. In an elastic scattering picture, the magnitude of the momentum remains fixed:

$$p_x^2 + p_y^2 = p^2. \tag{4.55}$$

An increase in $|p_y|$ is accompanied by a decrease in $|p_x|$; the momentum vector changes direction rather than length.

Question from the audience. Where does the energy come from when the particle acquires vertical momentum?[a]

Response. It need not acquire additional energy. The barrier can redirect the momentum elastically, much as a stationary wall changes the direction of a bouncing ball. The outgoing wavelength, and therefore the magnitude of the momentum, can remain the same while its horizontal and vertical components change.

[a]Lecture timestamp: 01:41:59–01:44:12.

4.12 Particles Arrive One at a Time

The wavefunction describes both alternatives, but each detection event is a localized spot. Send one electron at a time and the screen records one spot at a time. Repeat the preparation many times and the accumulated spots approach the density $|\psi_1 + \psi_2|^2$. The density cannot be negative because it is an absolute square, but it can vanish where the two amplitudes cancel.

Question from the audience. Is it then meaningless to speak of a corpuscular particle in this experiment?[a]

Response. No. The particle aspect is visible in each localized detection. The wave aspect is visible in the probability law governing many such detections. A single event is a spot; the interference pattern is the statistical arrangement of many spots.

[a]Lecture timestamp: 01:44:19–01:45:35.

Question from the audience. How does this compare with the interference of light?[a]

Response. The same distinction appears with photons. An intense beam draws the fringe pattern quickly. If the beam is attenuated enough, the screen registers separated photon events, yet those events accumulate into the same interference distribution. Reducing the flux changes the number of particles present at once, not the underlying probability law.

[a]Lecture timestamp: 01:45:39–01:47:17.

4.13 The One-Aperture Envelope

A single opening does not give a featureless amplitude. A convenient local form is

$$\psi_1(y) = \rho(y)e^{i\phi(y)}, \tag{4.56}$$

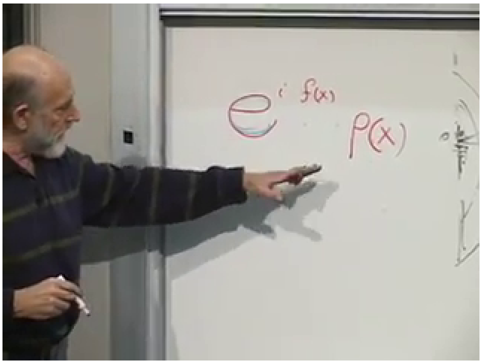

Figure 4.13: A rapidly varying phase multiplied by a smooth envelope gives a smooth one-aperture probability density.

Lecture frame: 01:49:00.

where $\rho(y)$ changes slowly while the phase may change rapidly. Its detection density is

$$|\psi_1(y)|^2 = \rho(y)^2. \tag{4.57}$$

The rapid phase cancels against its complex conjugate, leaving a smooth envelope. With a second displaced opening,

$$\psi_2(y) = \rho_2(y)e^{i\phi_2(y)}, \tag{4.58}$$

and the probability becomes

$$|\psi_1 + \psi_2|^2 = \rho_1^2 + \rho_2^2 + 2\rho_1\rho_2 \cos(\phi_1 - \phi_2). \tag{4.59}$$

The smooth one-aperture envelopes are modulated by the relative-phase fringes.

Question from the audience. If the amplitude oscillates behind one opening, why does the screen show only a broad blob?[a]

Response. The oscillation is in complex phase. Multiplication by the complex conjugate removes that phase, leaving the slowly varying $\rho(y)^2$. With two openings, a relative phase remains in the cross term, and that is what produces fringes.

[a]Lecture timestamp: 01:47:28–01:51:36.

A finite aperture can also diffract by itself. Different points across a sharp opening contribute different phases, so a single-slit envelope may contain maxima and minima of its own. This is distinct from the finer two-slit modulation, although a real experiment displays both at once.

Question from the audience. Can the two edges of one sharp opening create their own interference pattern?[a]

Response. Yes. A finite opening is a continuum of secondary contributions, and its edges make the phase structure especially visible. The resulting single-aperture diffraction determines the broad envelope; the separation of two openings determines the additional two-path fringes.

[a]Lecture timestamp: 01:51:37–01:52:41.

4.14 Visibility, Wavelength, and Idealization

Exact dark fringes require coherent contributions of equal magnitude. If the two apertures have unequal amplitudes A_1 and A_2, then

$$I(y) = |A_1|^2 + |A_2|^2 + 2\,|A_1 A_2| \cos \Delta\phi(y). \quad (4.60)$$

The minimum is $(|A_1| - |A_2|)^2$, so unequal illumination fills in the dark fringes. Noise, finite source coherence, detector resolution, and imperfect geometry do the same. Under ideal conditions the minima can nevertheless be extremely dark.

Question from the audience. Are there screen positions that will never be hit, even when particles are sent one at a time? [a]

Response. In the ideal equal-amplitude model, yes: destructive interference makes the probability exactly zero at the dark fringes. Real apparatus introduces background, unequal amplitudes, and finite resolution, so observed minima are generally small rather than mathematically perfect.

[a] Lecture timestamp: 01:52:43–01:53:43.

Question from the audience. Do unequal hole sizes matter, and how small must the openings be?[a]

Response. Small, similarly illuminated openings give the clearest two-path pattern. Unequal sizes alter the two amplitudes and reduce fringe visibility. The relevant scales are the opening widths, their separation, the propagation distance, and the de Broglie wavelength $\lambda = h/p$. Long-wavelength particles diffract through larger structures; short-wavelength electrons require correspondingly finer geometry.

[a]Lecture timestamp: 01:53:44–01:55:54.

Passing through a sufficiently small opening prepares an approximate outgoing state centered at that opening. In this limited sense the later pattern can become insensitive to many details of the source. Coherence still matters: the source and apparatus must not carry distinguishable which-path information.

Question from the audience. Does the particle forget its source after passing through the small opening?[a]

Response. The opening can prepare a new approximate outgoing state, so many earlier details cease to affect the downstream shape. That is an idealization, not complete independence from the source: momentum spread, coherence, and correlations with other systems can still matter.

[a]Lecture timestamp: 01:55:55–01:56:41.

4.15 Reuse Is Not Reversal

The final question separates two ideas that are easy to confuse. One may collect an electron, prepare it again, and repeat the experiment with that same particle; repeated trials do not require newly created electrons. Reversing an entire detected event is a different matter. The screen and all other correlated degrees of freedom would have to be included in the reversal.

Question from the audience. Could the same electron be sent through repeatedly, and is that the same as reversing the experiment? [a]

Response. The same electron may be recovered, re-prepared, and used for another trial, producing the same statistics as any identically prepared electron. Reversing a completed detection is not the same operation. To decide whether the full evolution can be reversed, one must first introduce Hamiltonians, time evolution, and the detector as part of the quantum system.

[a] Lecture timestamp: 01:56:42–01:59:16.

That is where the calculation properly stops. The kinematics has supplied continuous bases, delta normalization, quantized momentum on a circle, noncommuting position and momentum, and interference from added amplitudes. The next step is dynamics: a Hamiltonian must say how these states evolve in time.

CHAPTER 5

FOURIER DUALITY, WAVE PACKETS, AND PHOTON POLARIZATION

5.1 A Periodic Line as a Regulator

Begin with a line segment of length L, but identify its two ends. A particle that leaves at one end reappears at the other. This is equivalent mathematically to motion on a circle, although we need not imagine the particle literally travelling around a physical ring. The construction is a regulator: it makes momentum discrete and normalization ordinary, and the infinite line is recovered by taking L to infinity.

The endpoint identification requires

$$\psi(x+L) = \psi(x). \tag{5.1}$$

It is convenient to write $L = 2\pi r$. The symbol r then resembles a circle radius, but only the circumference $2\pi r$ matters.

Momentum eigenfunctions are plane waves. Restoring units before simplifying the notation,

$$e^{ipx/\hbar} = e^{ikx}, \qquad k := \frac{p}{\hbar}. \tag{5.2}$$

Thus k is the wave number, while the physical momentum is $p = \hbar k$. On one period, a normalized

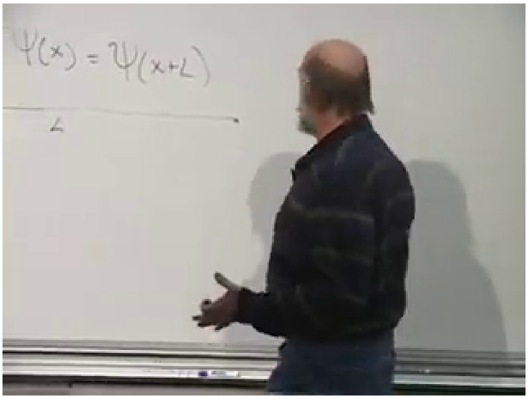

Figure 5.1: The periodicity condition and the finite interval of length L.

Lecture frame: 00:02:17.

plane wave is

$$\psi_k(x) = \langle x \mid k \rangle_{\text{box}} = \frac{e^{ikx}}{\sqrt{2\pi r}}, \qquad 0 \le x < 2\pi r. \tag{5.3}$$

Question from the audience. Why does the normalization contain a square root? [a]

Response. Normalization is imposed on the absolute square. Since $|e^{ikx}|^2 = 1$,

$$\int_0^{2\pi r} |e^{ikx}|^2 \, dx = 2\pi r.$$

Dividing the amplitude by $\sqrt{2\pi r}$ divides its absolute square by $2\pi r$, leaving unit norm.

[a] Lecture timestamp: 00:04:29–00:05:26.

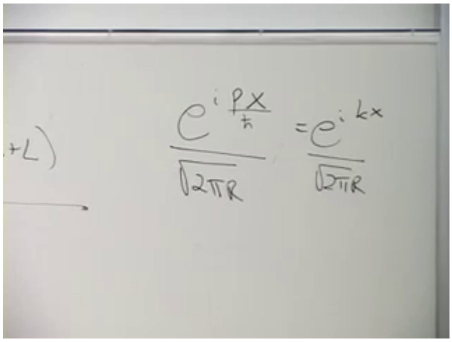

Figure 5.2: The normalized plane wave written in $p/\hbar$ and k notation.

Lecture frame: 00:04:20.

Periodicity now quantizes the wave number:

$$e^{ik(x+2\pi r)} = e^{ikx}, \tag{5.4}$$

$$e^{i2\pi rk} = 1, \tag{5.5}$$

$$k = \frac{n}{r}, \qquad n \in \mathbb{Z}. \tag{5.6}$$

The allowed values are separated by

$$\Delta k = \frac{1}{r}. \tag{5.7}$$

As r grows, the spectrum becomes dense. A dense quantum spectrum is not by itself a classical limit; it is only the continuum limit of this kinematical basis.

5.2 Orthogonality Before the Continuum

The momentum operator is Hermitian on periodic functions with the appropriate domain, so eigenfunctions with different eigenvalues are orthogonal. Here the theorem can be checked directly. For $k_n = n/r$ and $k_m = m/r$,

$$\langle k_m \mid k_n \rangle_{\text{box}} = \frac{1}{2\pi r} \int_0^{2\pi r} e^{i(k_n - k_m)x}\, dx \tag{5.8}$$
$$= \delta_{mn}. \tag{5.9}$$

When $m = n$, the integrand is one. When $m \neq n$, it completes an integer number of oscillations, and its positive and negative contributions cancel.

Question from the audience. Is orthogonality here a definition or a result? [a]

Response. It is a result. Distinct eigenvalues of a Hermitian operator have orthogonal eigenvectors. The integral above verifies that general theorem for the periodic plane waves; it does not define orthogonality.

[a] Lecture timestamp: 00:11:24–00:12:54.

The states $|k_n\rangle_{\text{box}}$ therefore form an orthonormal basis. At finite r, the correct language is discrete: sums over n and Kronecker deltas δ_{mn}. We now stretch this basis into the continuum without losing its normalization bookkeeping.

5.3 From Kronecker to Dirac Normalization

Because $\Delta k = 1/r$, define continuum-normalized states at the sampled wave numbers by

$$|k_n\rangle := \frac{1}{\sqrt{\Delta k}}\, |k_n\rangle_{\text{box}} = \sqrt{r}\, |k_n\rangle_{\text{box}}. \tag{5.10}$$

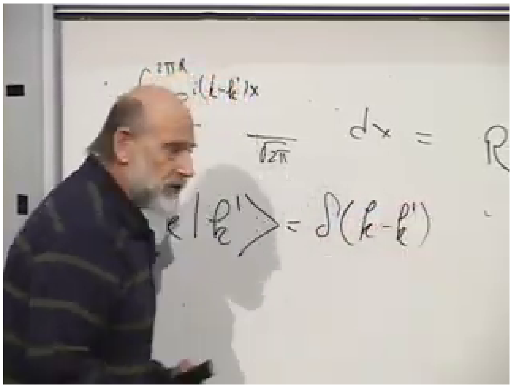

Figure 5.3: The board transition from normalized plane waves to $\delta(k-k')$ normalization.

Lecture frame: 00:16:50.

Their position-space kernels are

$$\langle x \mid k_n \rangle = \sqrt{r}\,\frac{e^{ik_nx}}{\sqrt{2\pi r}} = \frac{e^{ik_nx}}{\sqrt{2\pi}}. \qquad (5.11)$$

The factor of $\sqrt{r}$ has not vanished. It has converted a unit-normalized discrete state into a delta-normalized continuum state. Indeed,

$$\langle k_m \mid k_n \rangle = \frac{\delta_{mn}}{\Delta k}, \qquad (5.12)$$

which tends distributionally to

$$\langle k' \mid k \rangle = \delta(k-k'). \qquad (5.13)$$

The peak grows like $1/\Delta k = r$ while its width shrinks like Δk, leaving unit area.

Question from the audience. Where did the factor $\sqrt{r}$ go in the overlap? [a]

Response. There is one factor from the bra and one from the ket. Together they multiply the old overlap by r. Thus unequal sampled wave numbers remain orthogonal, while the equal-wave-number overlap grows from 1 to $r = 1/\Delta k$, exactly the height needed for a Dirac delta in the continuum limit.

[a] Lecture timestamp: 00:17:37–00:18:39.

Completeness transforms consistently:

$$\mathbf{1} = \sum_n |k_n\rangle_{\text{box box}}\langle k_n| \tag{5.14}$$

$$= \sum_n \Delta k \; |k_n\rangle \langle k_n| \longrightarrow \int_{-\infty}^{\infty} dk \; |k\rangle \langle k| \,. \tag{5.15}$$

The rescaling, the delta function, and the replacement of a sum by an integral are three parts of one operation.

5.4 Dyads and the Resolution of Identity

An inner product such as $\langle a \,|\, b\rangle$ is a number. Reversing the order creates a different kind of object:

$$|b\rangle \langle a| \,. \tag{5.16}$$

This outer product, or dyad, is an operator. Acting on a third vector,

$$\Big(|b\rangle \langle a|\Big) \, |c\rangle = \langle a \,|\, c\rangle \; |b\rangle \,. \tag{5.17}$$

The bra extracts the component of $|c\rangle$ along $|a\rangle$, and the ket places that coefficient along $|b\rangle$.

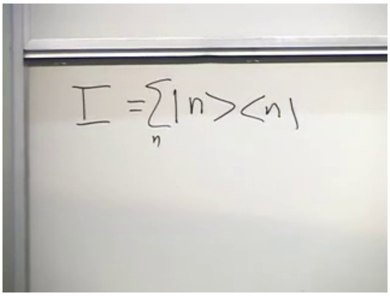

Figure 5.4: A complete orthonormal basis resolves the identity into projectors.

Lecture frame: 00:25:02.

For any orthonormal basis $\{|n\rangle\}$,

$$|v\rangle = \sum_n v_n \,|n\rangle\,, \qquad v_n = \langle n\,|\,v\rangle\,. \tag{5.18}$$

Substitution gives

$$|v\rangle = \sum_n |n\rangle\,\langle n|\,|v\rangle\,. \tag{5.19}$$

Since the relation holds for every $|v\rangle$,

$$\mathbf{1} = \sum_n |n\rangle\,\langle n|\,. \tag{5.20}$$

Question from the audience. Does the repeated symbol n refer to one special state?[a]

Response. No. It is a dummy summation label that runs over every vector in the chosen basis. Another complete orthonormal basis gives another resolution of the same identity. For a continuous label, the sum becomes an integral and Kronecker normalization becomes Dirac normalization.

[a]Lecture timestamp: 00:25:58–00:27:40.

In particular,

$$\mathbf{1} = \int dx\ |x\rangle\langle x| = \int dk\ |k\rangle\langle k|\,. \tag{5.21}$$

These two forms of the identity are the mechanism behind the Fourier transform.

5.5 One State in Two Representations

For an abstract state $|\psi\rangle$, define

$$\psi(x) := \langle x\,|\,\psi\rangle\,, \qquad \widetilde{\psi}(k) := \langle k\,|\,\psi\rangle\,. \tag{5.22}$$

They are not two states. They are the components of one state in two different bases. The change-of-basis kernel is

$$\langle x\,|\,k\rangle = \frac{e^{ikx}}{\sqrt{2\pi}}, \qquad \langle k\,|\,x\rangle = \frac{e^{-ikx}}{\sqrt{2\pi}}. \tag{5.23}$$

Insert the position resolution of identity into $\langle k\,|\,\psi\rangle$:

$$\widetilde{\psi}(k) = \int dx\ \langle k\,|\,x\rangle\langle x\,|\,\psi\rangle \tag{5.24}$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\, e^{-ikx}\psi(x). \tag{5.25}$$

Insert the momentum resolution into $\langle x\,|\,\psi\rangle$:

$$\psi(x) = \int dk\ \langle x\,|\,k\rangle\langle k\,|\,\psi\rangle \tag{5.26}$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dk\, e^{ikx}\widetilde{\psi}(k). \tag{5.27}$$

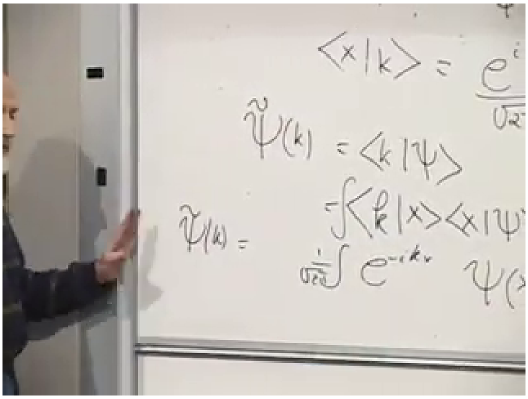

Figure 5.5: Inserting the position basis turns the momentum amplitude into a Fourier integral.
Lecture frame: 00:34:30.

These are the forward and inverse Fourier transforms. Fourier analysis is older than quantum mechanics; quantum mechanics interprets the transform as a change between position and momentum descriptions.

Question from the audience. Can k be negative, and what does the symmetry between x and k mean physically?[a]

Response. Negative k means momentum in the opposite direction. The deeper symmetry is that position and momentum are conjugate variables. Their reciprocity is natural in Fourier analysis and later appears in classical mechanics through canonical coordinates and Poisson brackets.

[a]Lecture timestamp: 00:38:43–00:39:43.

The operator rules display the same reciprocity. In

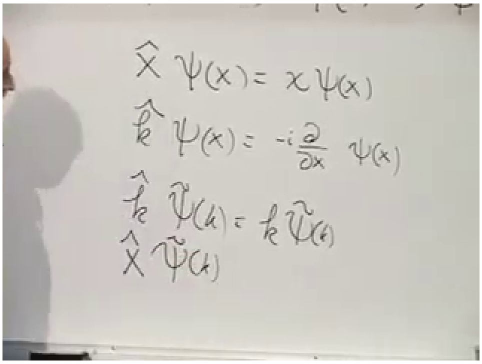

Figure 5.6: Multiplication and differentiation exchange roles in the x- and k-representations.

Lecture frame: 00:42:15.

the position representation,

$$\hat{X}\psi(x) = x\psi(x), \tag{5.28}$$

$$\hat{K}\psi(x) = -i\frac{\partial}{\partial x}\psi(x). \tag{5.29}$$

In the wave-number representation,

$$\hat{K}\widetilde{\psi}(k) = k\widetilde{\psi}(k), \tag{5.30}$$

$$\hat{X}\widetilde{\psi}(k) = i\frac{\partial}{\partial k}\widetilde{\psi}(k). \tag{5.31}$$

The physical momentum operator is $\hat{P} = \hbar\hat{K}$.

Question from the audience. Why is the state represented by $\psi(x)$ or $\widetilde{\psi}(k)$, rather than by one probability amplitude depending on both x and k?[a]

Response. These are two complete representations of the same state, and they are related invertibly by a Fourier transform. Position and momentum are incompatible observables: no normalizable state has both exactly sharp. A definite-k state is spread over all x, while a definite-position state is spread over all k. Quantum mechanics does not assign simultaneous sharp values merely because either representation can be used.

[a] Lecture timestamp: 00:43:13–00:44:36.

Question from the audience. Is the classical conservation of phase-space volume related to quantum time evolution?[a]

Response. Yes, at the structural level emphasized in the lecture. Hamiltonian evolution is a canonical transformation and preserves phase-space volume. It cannot collapse a finite cloud of distinct initial conditions to one point. Quantum evolution is unitary and preserves inner products, so distinctions between state vectors are likewise maintained. The detailed dynamical statement is still ahead of us.

[a] Lecture timestamp: 00:44:38–00:46:39.

Question from the audience. Where does the differential form $\hat{K} = -i\partial_x$ come from?[a]

Response. It is the position-representation form introduced for the momentum generator. Acting on a plane wave,

$$-i\frac{\partial}{\partial x}e^{ikx} = ke^{ikx},$$

so e^{ikx} is an eigenfunction with eigenvalue k. Fourier transformation then gives the reciprocal rule $\hat{X} = i\partial_k$ in momentum space.

[a]Lecture timestamp: 00:46:43–00:48:08.

5.6 Wave Packets and the Classical Bridge

At this stage, the operators have been named position and momentum, but the connection with classical motion has not yet been proved. The bridge is a wave packet. Its magnitude is concentrated near some position x_0, and its Fourier transform is concentrated near some wave number k_0. A typical form is

$$\psi(x) = f(x - x_0)e^{ik_0x}, \tag{5.32}$$

where f is a smooth localized envelope. The oscillations inside the envelope carry the approximate momentum, while the envelope locates the particle.

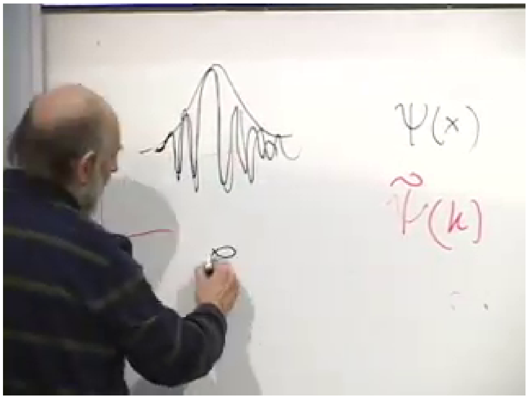

Figure 5.7: A localized envelope modulates a nearly monochromatic plane wave, producing a packet localized in both representations.

Lecture frame: 00:51:25.

Question from the audience. Must the packet be Gaussian? [a]

Response. No. A Gaussian is the cleanest example because its Fourier transform is also Gaussian and it can minimize $\Delta x\,\Delta k \geq 1/2$. The lecture needs only a smooth packet localized in both representations; its exact shape is not essential.

[a] Lecture timestamp: 00:50:23–00:50:34.

Under a sufficiently smooth potential, a narrow packet can retain its integrity for a useful interval. Its center moves through position space, and the center of its momentum distribution moves through momentum space. The theorem promised in the lecture is the Ehrenfest relation: under the conditions in which the packet remains narrow, these expectation values approximately obey the classical equations of motion. The packet may also spread or distort, so the classical description is an approximation, not a new postulate.

Question from the audience. What establishes that this quantum quantity really is classical momentum?[a]

Response. Not the name alone. The connection is established when a localized wave packet evolves according to the Schrodinger equation and its centers in x and $p = \hbar k$ follow the corresponding classical trajectory. That result is announced here and derived only after quantum dynamics has been introduced.

[a]Lecture timestamp: 00:48:13–00:54:02.

5.7 Incompatibility in a Two-State System

For complex vectors, $|A\rangle \langle B|$ is generally not Hermitian, and

$$\Big(|A\rangle \langle B|\Big)^\dagger = |B\rangle \langle A| \, . \tag{5.33}$$

Hermitian conjugation is the operator analogue of complex conjugation.

Question from the audience. What is the outer product of two complex vectors? [a]

Response. It is an operator, usually a non-Hermitian one. Reversing the vectors and converting each ket to its adjoint bra gives its Hermitian conjugate. The special dyad $|A\rangle\langle A|$ is Hermitian and, for normalized $|A\rangle$, is a projector.

[a] Lecture timestamp: 00:54:15–00:55:32.

The larger point is incompatibility. Classically one specifies position *and* momentum. Quantum mechanically one may use the position representation *or* the momentum representation. The state contains the probability information needed for either experiment, but it does not make incompatible quantities simultaneously definite.

To see the same logic without infinite-dimensional function spaces, turn to photon polarization. Ignore the photon's position and retain only two polarization degrees of freedom.

5.8 From Electromagnetic Waves to a Polarizer

A classical light wave travelling along the z-axis has electric and magnetic fields transverse to its motion and perpendicular to one another. For a plane wave they oscillate in phase. The polarization direction is defined to be the direction in which the electric field oscillates; the opposite direction along the same axis is not a different linear polarization.

A wire-grid polarizer makes this physical. If its

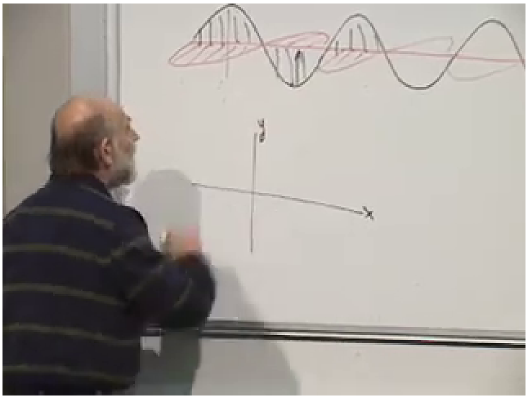

Figure 5.8: Perpendicular electric and magnetic oscillations in a plane-polarized electromagnetic wave.

Lecture frame: 01:00:45.

conducting wires are vertical, a vertical electric field drives currents in the wires. Those currents reflect or absorb that component. A horizontal electric field cannot drive a horizontal current because no conducting path runs that way, so it is transmitted. Thus the transmission axis is perpendicular to the wires. More generally, an ideal polarizer transmits the electric-field component along one axis and removes the orthogonal component.

The wave description gives continuously variable field amplitudes. The single-photon description gives discrete events. A photon incident on a polarizer is either transmitted or it is absorbed or reflected. Its energy does not split into two fractional photons.

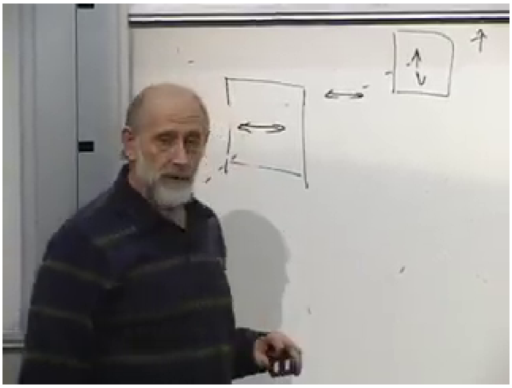

Figure 5.9: Two polarizers illustrate preparation followed by a polarization measurement.

Lecture frame: 01:10:00.

5.9 Preparation and Measurement

A photon that emerges from an ideal vertical polarizer is vertically polarized. If it then meets a second vertical polarizer, it passes with probability one. If it meets a horizontal polarizer, it is blocked with probability one. The first apparatus prepares a state; the second tests it. The same physical device can therefore serve as a state preparer or as a measurement apparatus, depending on its role in the experiment.

Question from the audience. Is a reflected photon itself polarized? [a]

Response. It may be. The lecture discards that output channel and counts only transmission. One may instead build a fuller experiment that detects both channels, but the two-outcome polarization logic for the transmitted beam is unchanged.

[a]Lecture timestamp: 01:06:14–01:06:38.

Choose horizontal and vertical axes called x and y. In this section they label polarization, not the particle's position. Represent the two orthogonal states by

$$|x\rangle_{\text{pol}} \leftrightarrow \begin{pmatrix}1\\0\end{pmatrix}, \qquad |y\rangle_{\text{pol}} \leftrightarrow \begin{pmatrix}0\\1\end{pmatrix}. \tag{5.34}$$

They satisfy

$$\langle x \,|\, x\rangle = \langle y \,|\, y\rangle = 1, \qquad \langle x \,|\, y\rangle = 0. \tag{5.35}$$

They are orthogonal because a single ideal polarization experiment can distinguish them perfectly.

5.10 States, Outcomes, and the Polarization Operator

The state vectors are not themselves observables. To define an observable, associate numerical records with the two possible outcomes. Following the lecture, record $+1$ for x-polarization and -1 for y-polarization. The corresponding operator obeys

$$\hat{P}_{xy}\,|x\rangle = +\,|x\rangle\,, \qquad \hat{P}_{xy}\,|y\rangle = -\,|y\rangle\,. \tag{5.36}$$

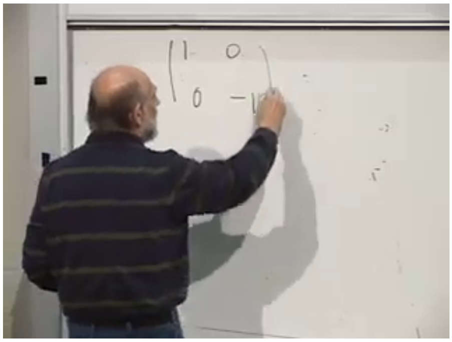

Figure 5.10: The horizontal-vertical polarization observable is diagonal in the $(|x\rangle, |y\rangle)$ basis.
Lecture frame: 01:19:45.

In the $(|x\rangle, |y\rangle)$ basis,

$$\hat{P}_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |x\rangle \langle x| - |y\rangle \langle y| . \tag{5.37}$$

Question from the audience. If polarization is observable, why is the state not called an observable?[a]

Response. The state describes the preparation and supplies probability amplitudes. The observable is the Hermitian operator tied to a measurement setup; its eigenvectors are states with definite outcomes, and its eigenvalues are the numbers recorded for those outcomes.

[a]Lecture timestamp: 01:14:56–01:16:03.

Knowing the action of $\hat{P}_{xy}$ on the two basis vectors determines its action on every superposition. This

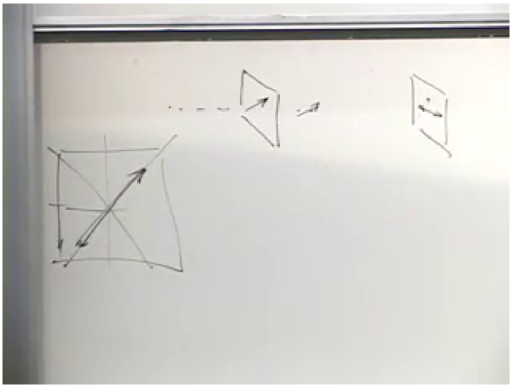

Figure 5.11: A diagonal preparation followed by a horizontal or vertical polarization test.

Lecture frame: 01:23:45.

is why specifying eigenvectors and eigenvalues is enough to specify an observable.

5.11 Rotate the Apparatus

Nothing prevents us from rotating a polarizer by 45°. A photon that emerges from it is neither in $|x\rangle$ nor in $|y\rangle$, but an x- or y-oriented test gives either result with equal probability. The normalized state is

$$|+\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}}. \tag{5.38}$$

The orthogonal linear polarization is

$$|-\rangle = \frac{|x\rangle - |y\rangle}{\sqrt{2}}. \tag{5.39}$$

The relative minus sign makes the two diagonal

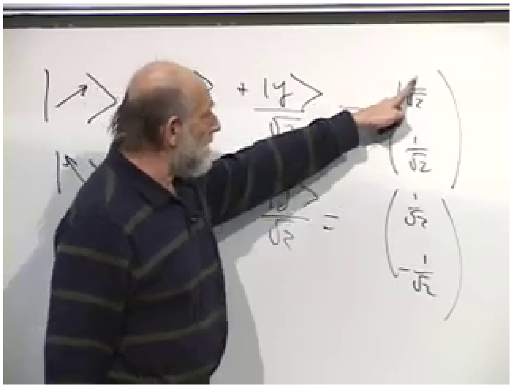

Figure 5.12: The two diagonal polarization states form a second orthonormal basis.

Lecture frame: 01:29:00.

states orthogonal:

$$\langle - \,|\, + \rangle = \frac{1}{2}\Big(\langle x \,|\, x\rangle + \langle x \,|\, y\rangle - \langle y \,|\, x\rangle - \langle y \,|\, y\rangle\Big) = 0. \tag{5.40}$$

In column form,

$$|+\rangle \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |-\rangle \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}. \tag{5.41}$$

The Born rule confirms the equal probabilities:

$$\text{Prob}(y|+) = |\langle y \,|\, +\rangle|^2 = \left|\frac{1}{\sqrt{2}}\Big(\langle y \,|\, x\rangle + \langle y \,|\, y\rangle\Big)\right|^2 \tag{5.42}$$

$$= \frac{1}{2}, \tag{5.43}$$

$$\text{Prob}(x|+) = \frac{1}{2}. \tag{5.44}$$

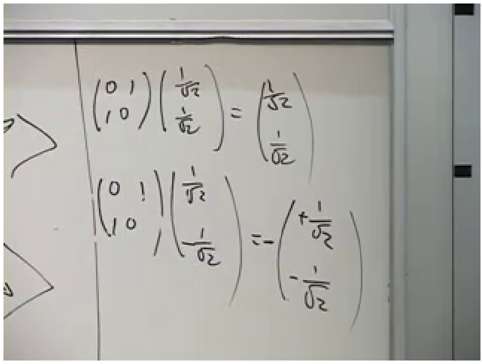

Figure 5.13: The off-diagonal matrix returns $|+\rangle$ and reverses $|-\rangle$, giving the required eigenvalues.
Lecture frame: 01:38:30.

For $|-\rangle$, one amplitude changes sign, but its absolute square is again $1/2$. Amplitudes carry signs and phases; probabilities do not retain them directly.

Define the rotated observable by

$$\hat{P}_{45}\,|+\rangle = +\,|+\rangle\,, \qquad \hat{P}_{45}\,|-\rangle = -\,|-\rangle\,. \tag{5.45}$$

In the original x, y basis,

$$\hat{P}_{45} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |+\rangle\langle+| - |-\rangle\langle-|\,. \tag{5.46}$$

The eigenvectors of $\hat{P}_{xy}$ are $|x\rangle\,, |y\rangle$; those of $\hat{P}_{45}$ are $|+\rangle\,, |-\rangle$. No nonzero vector is an eigenvector of both operators. Equivalently,

$$[\hat{P}_{xy}, \hat{P}_{45}] \neq 0. \tag{5.47}$$

The two polarization observables are incompatible, just as position and momentum are incompatible, but now the entire statement fits inside a two-dimensional vector space.

For an arbitrary linear-polarization angle θ, the announced generalization is

$$|\theta\rangle = \cos\theta \; |x\rangle + \sin\theta \; |y\rangle \,, \tag{5.48}$$

$$|\theta_\perp\rangle = -\sin\theta \; |x\rangle + \cos\theta \; |y\rangle \,. \tag{5.49}$$

The corresponding observable has these two states as eigenvectors. Rotating the axis by 90° merely exchanges the outcomes and reverses the sign of the same observable. Distinct nonorthogonal axes give genuinely incompatible tests. Complex relative coefficients introduce circular and elliptical polarization; that extension is reserved for the next lecture.

5.12 What One Experiment Records

The closing discussion distinguishes an outcome from a probability. A single photon gives one definite record: transmitted or blocked, represented here by $+1$ or -1. The probability of either record is inferred from many repetitions with identically prepared photons.

Question from the audience. Is the probability that a photon passes a polarizer itself an observable?[a]

Response. Not in the single-shot technical sense used here. One run produces an outcome, not a probability. Probability is estimated from the relative frequency of outcomes over an ensemble of repeated preparations.

[a]Lecture timestamp: 01:42:27–01:45:03.

The probability for a particular eigenstate $|a\rangle$ is obtained directly from the overlap,

$$\mathrm{Prob}(a|\psi) = |\langle a \mid \psi\rangle|^2 . \tag{5.50}$$

One does not first apply the observable to $|\psi\rangle$ and then square an eigenvalue. The observable identifies the possible values and the eigenvectors associated with them; the state's overlaps with those eigenvectors determine the probabilities.

Question from the audience. Should the observable act on the state before the probability is calculated?[a]

Response. No. To find the probability of a particular result, take the inner product of the prepared state with the eigenvector for that result, then square its magnitude. Applying the operator is useful for expectation values and eigenvalue equations, but it is not an extra step in the basic Born-rule calculation.

[a]Lecture timestamp: 01:45:03–01:48:02.

Question from the audience. Is an observable synonymous with a Hermitian operator? [a]

Response. Within the finite-dimensional framework of this lecture, yes: observables are represented by Hermitian operators, and every Hermitian operator may be associated with an observable. In infinite-dimensional problems, the precise mathematical condition is self-adjointness with a specified domain.

[a] Lecture timestamp: 01:48:04–01:48:14.

The numerical labels +1 and −1 are a convention. We could record 1 and 0, or any two distinct real numbers, provided the convention is used consistently. Choosing 1 and 0 produces the projector

$$\Pi_x = |x\rangle\langle x| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{5.51}$$

For a normalized state $|\psi\rangle$,

$$\begin{aligned} \langle\psi|\,\Pi_x\,|\psi\rangle &= \langle\psi|\,x\rangle\,\langle x\,|\,\psi\rangle \qquad (5.52)\\ &= |\langle x\,|\,\psi\rangle|^2 . \qquad (5.53) \end{aligned}$$

The expectation value equals the probability, but the result of one projector measurement is still 1 or 0.

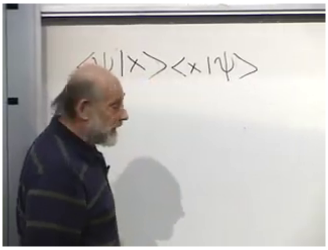

Figure 5.14: A projector expectation gives the x-polarization probability.

Lecture frame: 01:50:30.

Question from the audience. If the eigenvalue labels are arbitrary, what does the apparatus actually define?[a]

Response. The apparatus defines the mutually exclusive outcomes and the states that give them with certainty. We then choose real numbers to record those outcomes. With the $+1, -1$ convention the observable is a difference of projectors; with the $1, 0$ convention it is one projector, whose expectation value is the ensemble probability for that outcome.

[a]Lecture timestamp: 01:48:16–01:55:23.

An observable is therefore inseparable from a specified experimental question. The polarizer orientation fixes the eigenvectors, the notebook convention fixes the eigenvalues, and one interaction yields one recorded event. Repetition reveals the probabilities. In this two-state example, the abstract language of vectors, operators, incompatible bases, and Born amplitudes has become a concrete laboratory procedure.

CHAPTER 6

LINEAR AND CIRCULAR POLARIZATION, OPERATORS, AND EXPECTATION VALUES

6.1 The Smallest Useful Quantum System

Take a photon moving in the z-direction with definite momentum and set aside its translational motion. What remains is its transverse polarization. Classically we picture an electric field oscillating along an axis in the xy-plane. For a single photon that transverse degree of freedom becomes a particularly clean quantum system: the state space is only two-dimensional, but it already contains superposition, incompatible measurements, complex amplitudes, and probabilistic outcomes.

Begin with an ideal linear polarizer. The conducting direction of a wire-grid polarizer supports a current and therefore suppresses the electric field component parallel to the wires. The perpendicular component is transmitted. Consequently the *transmission axis*, rather than the wire direction, labels the polarizer.

The same apparatus has two roles. If a photon emerges from a polarizer, the apparatus has prepared a state. If a prepared photon is sent into a second polarizer, that apparatus tests the state. Equal preparation and test axes give certain transmission; orthogonal axes give certain rejection.

Question from the audience. Could ordinary classical particles imitate this first experiment?[a]

Response. Yes. One could imagine bullets carrying little flags and a mechanical slit that passes one flag orientation while blocking the other. Nothing nonclassical has appeared while the only preparations and tests are the same two orthogonal directions.

[a]Lecture timestamp: 00:09:01–00:09:25.

The difficulty begins with rotational symmetry. A polarizer can be turned to any angle, so a photon can be prepared along a continuum of linear polarization axes. A fixed analyzer nevertheless has only two outcomes: the photon passes or it does not. No single photon divides into a transmitted fraction and a rejected fraction. Continuous preparations and discrete outcomes must coexist.

This is the central puzzle. A classical wave divides continuously, and the flagged-bullet model remains deterministic. A photon at an intermediate angle instead gives one indivisible event whose probability depends on the relative orientation of the apparatus.

Mathematically this polarization problem is isomorphic to a spin-one-half system: both use a two-dimensional complex state space and noncommuting two-outcome observables. Polarization axes are somewhat easier to picture, so they provide the cleaner geometry for the present discussion.

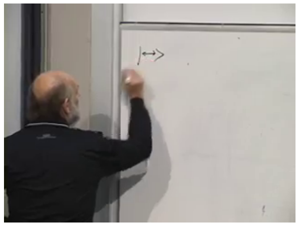

Figure 6.1: The double-headed mark records a polarization axis without assigning it a preferred direction.

Lecture frame: 00:14:32.

6.2 The Axial Basis and Its Observable

Choose horizontal and vertical transmission axes and call them x and y. Linear polarization is axial: horizontal-to-the-right is not different from horizontal-to-the-left. Reversing the drawn arrow leaves the same polarization axis.

Represent the two orthogonal states by

$$|x\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |y\rangle = \begin{pmatrix}0\\1\end{pmatrix}. \tag{6.1}$$

They obey

$$\langle x\,|\,x\rangle = \langle y\,|\,y\rangle = 1, \qquad \langle x\,|\,y\rangle = 0. \tag{6.2}$$

These are two basis vectors, not the only two state vectors. Every normalized superposition

$$|\psi\rangle = a\,|x\rangle + b\,|y\rangle\,, \qquad |a|^2 + |b|^2 = 1, \tag{6.3}$$

is a possible pure polarization state. Multiplication of the entire vector by one common phase does not change the physical state.

Question from the audience. Are the entries $1, 0$ and $0, 1$ completely arbitrary?[a]

Response. The coordinate convention is chosen with the axes, but the vectors must be normalized and mutually orthogonal. Those conditions encode total probability one and perfect distinguishability. Their common overall phase is irrelevant, so several column vectors can represent the same physical ray.

[a]Lecture timestamp: 00:16:50–00:18:38.

To turn a test into an observable, attach numbers to its outcomes. Record $+1$ when an x-analyzer transmits the photon and -1 when it rejects it. The labels are conventional; 7 and 12 would distinguish the outcomes too. The symmetric choice ± 1 gives

$$\hat{P}_0 \left|x\right\rangle = + \left|x\right\rangle , \qquad \hat{P}_0 \left|y\right\rangle = - \left|y\right\rangle , \tag{6.4}$$

and therefore

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \left|x\right\rangle \left\langle x\right| - \left|y\right\rangle \left\langle y\right| . \tag{6.5}$$

The operator can be constructed from known eigenstates and recorded values, as we have just done. In other problems the order is reversed: one begins with an operator and solves for its eigenvectors and eigenvalues. For example, the wave-number operator $-i\, d/dx$ is given first, and its plane-wave eigenstates are then found. Either route expresses the same postulate: an observable has definite value λ in a state satisfying $\hat{A} \left|\lambda\right\rangle = \lambda \left|\lambda\right\rangle$.

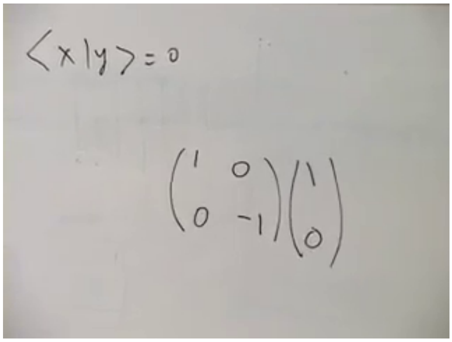

Figure 6.2: Orthogonality and the first matrix action in the $(|x\rangle, |y\rangle)$ basis.

Lecture frame: 00:22:00.

6.3 A Second Basis at Forty-Five Degrees

Turn the preparation polarizer through 45°. The transmitted state has equal x- and y-components:

$$|d_+\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{6.6}$$

Its orthogonal companion, the backslash polarization, can be chosen as

$$|d_-\rangle = \frac{|x\rangle - |y\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{6.7}$$

Putting the minus sign on the other component multiplies the whole state by -1 and changes no physics. Directly,

$$\langle d_- \mid d_+\rangle = \frac{1}{2}(1-1) = 0. \tag{6.8}$$

The amplitude that a slash-polarized photon passes an x-analyzer is

$$\langle x \mid d_+ \rangle = \frac{1}{\sqrt{2}}. \tag{6.9}$$

The probability is the absolute square,

$$\mathrm{Prob}(x|d_+) = |\, \langle x \mid d_+ \rangle \,|^2 = \frac{1}{2}, \qquad \mathrm{Prob}(y|d_+) = \frac{1}{2}. \tag{6.10}$$

The amplitude and the probability are not interchangeable. Amplitudes may be negative or complex; probabilities are nonnegative real numbers.

Now attach $+1$ to the slash outcome and -1 to the backslash outcome. The corresponding observable is

$$\hat{P}_{\pi/4} = |d_+\rangle \langle d_+| - |d_-\rangle \langle d_-| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{6.11}$$

Exchanging which diagonal outcome is called $+1$ simply replaces this operator by $-\hat{P}_{\pi/4}$. Indeed,

$$\hat{P}_{\pi/4} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \hat{P}_{\pi/4} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = - \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{6.12}$$

Question from the audience. Does knowing the state determine what is observable?[a]

Response. No. The state determines which observables, if any, have definite values. In $|d_+\rangle$, the diagonal observable is definite while $\hat{P}_0$ gives two outcomes with equal probability. Similarly, a position delta state makes position definite, whereas a plane wave makes momentum definite; a generic wavefunction usually makes neither definite.

[a]Lecture timestamp: 00:35:14–00:37:32.

6.4 Arbitrary Linear Polarization

Let the transmission axis make angle θ with the x-axis. Its state and orthogonal partner are

$$|\theta\rangle = \cos\theta\ |x\rangle + \sin\theta\ |y\rangle \tag{6.13}$$

$$= \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \tag{6.14}$$

$$|\theta_\perp\rangle = |\theta + \pi/2\rangle = -\sin\theta\ |x\rangle + \cos\theta\ |y\rangle \tag{6.15}$$

$$= \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}. \tag{6.16}$$

Both have unit norm, and

$$\langle\theta\,|\,\theta_\perp\rangle = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0. \tag{6.17}$$

Also $|\theta + \pi\rangle = -|\theta\rangle$, confirming that a linear polarization axis has period π, not 2π.

Suppose the first polarizer prepares $|\theta\rangle$ and the

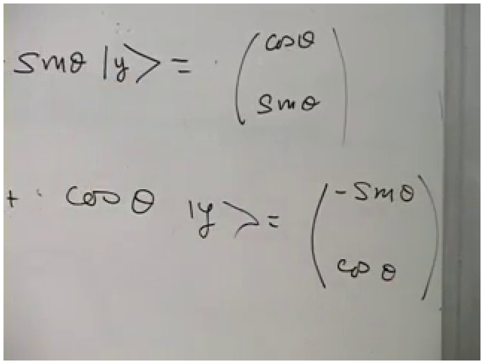

Figure 6.3: The arbitrary linear-polarization state and its orthogonal companion written in the axial basis.

Lecture frame: 00:43:30.

second tests the x-direction. Then

$$\mathrm{Prob}(x|\theta) = |\,\langle x \,|\, \theta\rangle\,|^2 = \cos^2\theta, \tag{6.18}$$

$$\mathrm{Prob}(y|\theta) = \sin^2\theta. \tag{6.19}$$

Their sum is one. Reversing the inner product does not change the probability because $\langle \theta \,|\, x\rangle = \langle x \,|\, \theta\rangle^*$.

For two arbitrary axes α and β, the needed trigonometric identity is

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta. \tag{6.20}$$

The related sum, sine, and double-angle identities are useful enough to keep beside it.

The transition amplitude is

$$\langle \beta \,|\, \alpha\rangle = \begin{pmatrix}\cos\beta & \sin\beta\end{pmatrix}\begin{pmatrix}\cos\alpha \\ \sin\alpha\end{pmatrix} \tag{6.21}$$

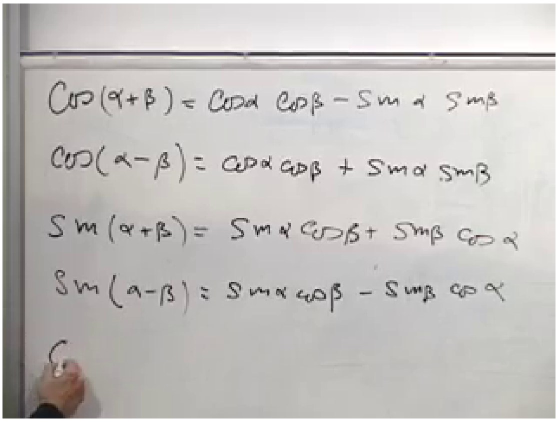

Figure 6.4: The trigonometric identities used to compare two rotated polarization bases.

Lecture frame: 00:54:30.

$$= \cos\alpha\cos\beta + \sin\alpha\sin\beta \tag{6.22}$$
$$= \cos(\alpha - \beta). \tag{6.23}$$

Therefore the single-photon form of Malus's law is

$$\boxed{\mathrm{Prob}(\beta|\alpha) = |\langle\beta\,|\,\alpha\rangle|^2 = \cos^2(\alpha-\beta).} \tag{6.24}$$

Only the relative angle appears, as rotational symmetry requires. At equal angles the probability is one; at orthogonal angles it is zero. For every intermediate angle each photon still gives one of two discrete records.

6.5 How an Extra Polarizer Opens a Closed Path

Prepare $|\theta\rangle$ and test it directly with the orthogonal analyzer $\theta + \pi/2$. The transmission probability

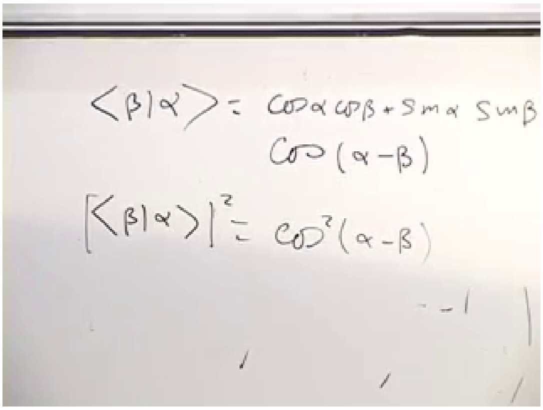

Figure 6.5: The overlap is an amplitude; its absolute square is the transmission probability.

Lecture frame: 00:57:30.

vanishes:

$$\text{Prob}(\theta + \pi/2|\theta) = \cos^2(\pi/2) = 0. \quad (6.25)$$

Now insert an x-polarizer between them. Passing the intermediate polarizer has probability $\cos^2\theta$, and passage prepares the new state $|x\rangle$. Conditional on that event, transmission through the last analyzer has probability

$$\text{Prob}(\theta + \pi/2|x) = \cos^2(\theta + \pi/2) = \sin^2\theta. \quad (6.26)$$

The probability of completing both stages is therefore

$$\text{Prob}(\theta \longrightarrow x \longrightarrow \theta + \pi/2) = \cos^2\theta\,\sin^2\theta. \quad (6.27)$$

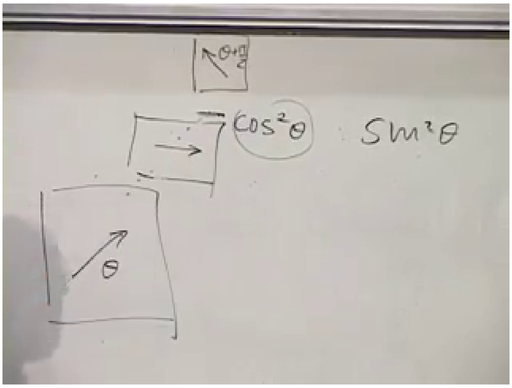

Figure 6.6: An intermediate horizontal polarizer changes the state and produces the nonzero product $\cos^2\theta\,\sin^2\theta$.

Lecture frame: 01:02:30.

Question from the audience. How can inserting another obstacle increase the chance of transmission?[a]

Response. Without the middle apparatus the initial and final states are orthogonal. The intermediate polarizer is not a passive filter on an unchanged state: successful passage prepares $|x\rangle$, which has a nonzero overlap with the final state. The two conditional probabilities multiply. Adding a measurement can therefore open a route that was absent in the direct experiment.

[a]Lecture timestamp: 00:59:29–01:03:40.

Question from the audience. Why can crossed polarizers show colors when stressed transparent material is placed between them? [a]

Response. The exchange in the lecture leaves the mechanism open and notes that real optical elements can have wavelength-dependent efficiency. More specifically, stressed transparent materials are often birefringent: they give two polarization components a wavelength-dependent relative phase. The final analyzer converts that phase delay into wavelength-dependent intensity. The photon frequency is not changed by an ideal static polarizer; different frequencies are transmitted with different probabilities.

[a] Lecture timestamp: 01:03:42–01:05:00.

6.6 The Observable at Angle θ

The general analyzer must have $|\theta\rangle$ and $|\theta_\perp\rangle$ as eigenvectors:

$$\hat{P}_\theta |\theta\rangle = + |\theta\rangle , \qquad \hat{P}_\theta |\theta_\perp\rangle = - |\theta_\perp\rangle . \tag{6.28}$$

Its spectral form is already the answer:

$$\hat{P}_\theta = |\theta\rangle \langle\theta| - |\theta_\perp\rangle \langle\theta_\perp| . \tag{6.29}$$

In the x, y basis,

$$\hat{P}_\theta = \begin{pmatrix} \cos^2\theta - \sin^2\theta & 2\sin\theta\cos\theta \\ 2\sin\theta\cos\theta & \sin^2\theta - \cos^2\theta \end{pmatrix} \tag{6.30}$$

$$= \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} . \tag{6.31}$$

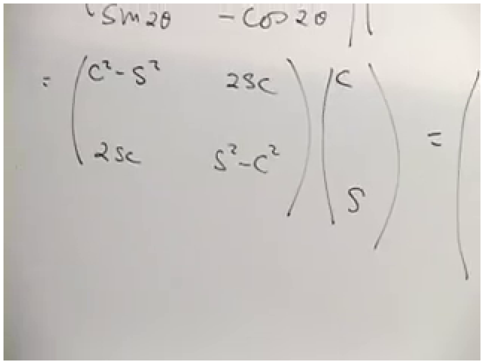

Figure 6.7: The board check of the general polarization matrix on $(\cos\theta, \sin\theta)^T$.

Lecture frame: 01:10:30.

Let $c = \cos\theta$ and $s = \sin\theta$. Acting on the proposed $+1$ eigenvector gives

$$\hat{P}_\theta \begin{pmatrix} c \\ s \end{pmatrix} = \begin{pmatrix} c^2 - s^2 & 2sc \\ 2sc & s^2 - c^2 \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \tag{6.32}$$

$$= \begin{pmatrix} c(c^2 + s^2) \\ s(c^2 + s^2) \end{pmatrix} = \begin{pmatrix} c \\ s \end{pmatrix}. \tag{6.33}$$

Rotating the analyzer by $90°$ exchanges the two outcomes:

$$\hat{P}_{\theta+\pi/2} = -\hat{P}_\theta. \tag{6.34}$$

The trigonometric reason is that both $\cos(2\theta + \pi)$ and $\sin(2\theta + \pi)$ acquire a minus sign. The physical reason is simpler: the state formerly labelled $+1$ is now labelled -1, and conversely.

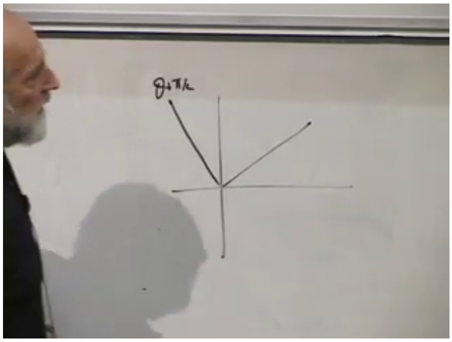

Figure 6.8: The two orthogonal analyzer axes whose outcome labels are exchanged by a 90° rotation.

Lecture frame: 01:15:15.

6.7 Complex Amplitudes and Circular Polarization

Real coefficients describe every linear polarization, but not every vector in a complex two-dimensional Hilbert space. Consider

$$|c_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |c_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{6.35}$$

Complex conjugation is essential:

$$\langle c_+ \,|\, c_+\rangle = \frac{1}{2}\Big(1 + (-i)i\Big) = 1, \tag{6.36}$$

and

$$\langle c_- \,|\, c_+\rangle = \frac{1}{2}\Big(1 + i^2\Big) = 0. \tag{6.37}$$

The tempting product without conjugating the bra gives the wrong answer.

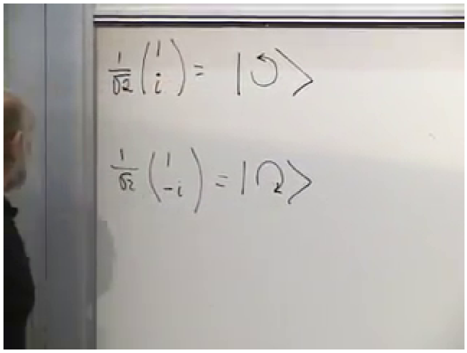

Figure 6.9: The two normalized complex vectors that form the circular polarization basis.

Lecture frame: 01:21:00.

To see their wave meaning, restore a dimensionally explicit phase $\phi = kz - \omega t$, with $\omega = ck$. Equal in-phase components,

$$E_x = E_0 \cos\phi, \qquad E_y = E_0 \cos\phi, \tag{6.38}$$

make a linearly polarized wave at 45°. A relative quarter-cycle shift gives

$$E_x = E_0 \sin\phi, \qquad E_y = E_0 \cos\phi. \tag{6.39}$$

At a fixed point in space, the tip of $\boldsymbol{E} = E_x\hat{\boldsymbol{x}} + E_y\hat{\boldsymbol{y}}$ moves around a circle. Reversing the sign of one quadrature reverses the sense of rotation. The magnetic field is transverse as well; in vacuum, $\boldsymbol{B} = \hat{\boldsymbol{z}} \times \boldsymbol{E}/c$.

Which state is called right- or left-circular depends on the propagation, time, and viewing conventions. The invariant content here is the orthogonal pair $(1, \pm i)^T/\sqrt{2}$.

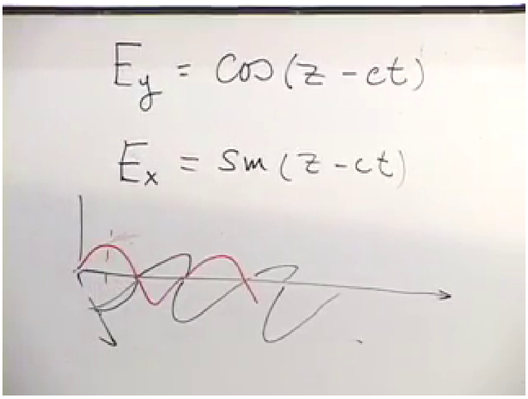

Figure 6.10: The x- and y-components are displaced by one quarter cycle.

Lecture frame: 01:24:30.

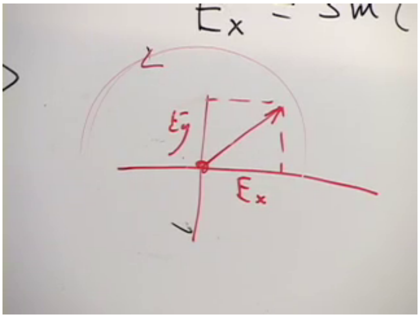

Figure 6.11: At fixed z, the two quadratures make the electric-field vector rotate.

Lecture frame: 01:25:30.

Question from the audience. What optical device creates the relative phase required for circular polarization?[a]

Response. A quarter-wave plate. Its two principal axes accumulate different phases; a linear state with equal components along those axes emerges with a relative phase of $\pi/2$, producing circular polarization. Unequal component magnitudes generally produce an ellipse.

[a]Lecture timestamp: 01:26:37–01:27:05.

For any linear analyzer,

$$\langle \theta \mid c_+ \rangle = \frac{1}{\sqrt{2}}\Big(\cos\theta + i\sin\theta\Big). \tag{6.40}$$

Hence

$$|\langle \theta \mid c_+ \rangle|^2 = \frac{1}{2}(\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta) \tag{6.41}$$

$$= \frac{1}{2}(\cos^2\theta + \sin^2\theta) = \frac{1}{2}. \tag{6.42}$$

A circularly polarized photon therefore passes every linear analyzer with probability one half. Its statistics reveal no preferred linear axis.

The general normalized pair $a, b \in \mathbb{C}$, modulo a common phase, describes elliptical polarization. A real relative phase gives linear polarization. Equal magnitudes with relative phase $\pm\pi/2$ give the two circular polarizations. Intermediate magnitudes and phases give ellipses.

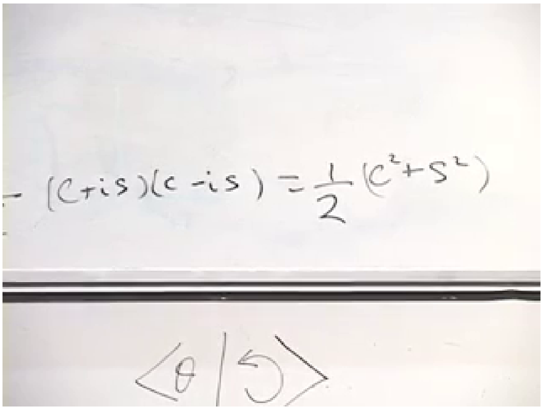

Figure 6.12: The complex-conjugate product removes the phase and leaves transmission probability 1/2.

Lecture frame: 01:29:50.

Question from the audience. What observable distinguishes the two circular states? [a]

Response. The lecture leaves this construction as an exercise. With $|c_+\rangle$ assigned eigenvalue $+1$ and $|c_-\rangle$ assigned -1, the spectral rule gives

$$\hat{P}_c = |c_+\rangle\langle c_+| - |c_-\rangle\langle c_-| = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Changing the handedness convention or exchanging the labels multiplies the operator by -1.

[a]Lecture timestamp: 01:32:19–01:32:36.

6.8 Compatible and Incompatible Polarization Tests

The axial, diagonal, arbitrary linear, and circular observables do not usually share eigenvectors. In the present two-state space,

$$\hat{P}_\theta = \cos 2\theta\, \sigma_z + \sin 2\theta\, \sigma_x, \tag{6.43}$$

so two linear analyzers satisfy

$$[\hat{P}_\alpha, \hat{P}_\beta] = 2i \sin\big(2(\beta - \alpha)\big) \sigma_y. \tag{6.44}$$

They commute only when their axes are the same or orthogonal, modulo the π-periodicity of a linear polarization axis. In the orthogonal case $\hat{P}_{\theta+\pi/2} = -\hat{P}_\theta$, so measuring one immediately fixes the other with the opposite label.

Question from the audience. Can x- and y-polarization be measured simultaneously, and what changes at 45°? [a]

Response. The x and y outcome labels belong to one common basis and are negatives of one another in the ± 1 convention. Knowing one fixes the other. The x- and 45°-analyzers are genuinely different: their operators do not commute and have no common eigenstate. The same incompatibility holds between a linear analyzer and the circular observable.

[a] Lecture timestamp: 01:33:31–01:35:20.

Incompatibility is therefore not vagueness about an apparatus. Each apparatus is perfectly definite. It is

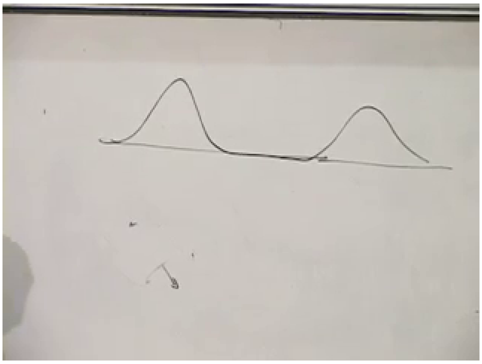

Figure 6.13: A symmetric two-lobe profile can have average position zero even though its central probability is very small.

Lecture frame: 01:36:09.

the impossibility of one state being a simultaneous eigenstate of two noncommuting tests.

6.9 Expectation Value Means Average Value

The standard term *expectation value* can be misleading. Imagine a position probability density concentrated in two symmetric, separated lobes. Its average position is zero, even if the probability of finding the particle near zero is tiny or vanishes. The average need not be a likely single outcome.

Let $\hat{K}$ be a Hermitian observable with a complete orthonormal eigenbasis,

$$\hat{K}\,|n\rangle = \lambda_n\,|n\rangle\,, \qquad \sum_n |n\rangle\,\langle n| = \mathbf{1}. \tag{6.45}$$

For a normalized state $|\psi\rangle$, define

$$\left\langle \hat{K} \right\rangle_{\psi} := \langle\psi|\, \hat{K}\, |\psi\rangle\,. \tag{6.46}$$

Question from the audience. Should there be a complex-conjugation star on $\hat{K}$?[a]

Response. No additional star is needed. Turning $|\psi\rangle$ into $\langle\psi|$ already takes the complex conjugate transpose. The observable itself is Hermitian, $\hat{K}^\dagger = \hat{K}$, which ensures that the expectation value is real.

[a]Lecture timestamp: 01:38:17–01:38:41.

Insert the eigenbasis resolution of identity between $\hat{K}$ and $|\psi\rangle$:

$$\begin{aligned} \left\langle \hat{K} \right\rangle_{\psi} &= \langle\psi|\, \hat{K} \left(\sum_n |n\rangle \langle n| \right) |\psi\rangle && (6.47)\\ &= \sum_n \langle\psi|\, \hat{K}\, |n\rangle \langle n \,|\, \psi\rangle && (6.48)\\ &= \sum_n \lambda_n \langle\psi \,|\, n\rangle \langle n \,|\, \psi\rangle && (6.49)\\ &= \sum_n \lambda_n |\langle n \,|\, \psi\rangle|^2. && (6.50) \end{aligned}$$

The factor

$$p_n = |\langle n \,|\, \psi\rangle|^2 \tag{6.51}$$

is the probability of obtaining λ_n. Thus $\left\langle \hat{K} \right\rangle_{\psi} = \sum_n p_n \lambda_n$ is exactly an ordinary weighted average of the possible outcomes.

Return finally to the axial polarization observable

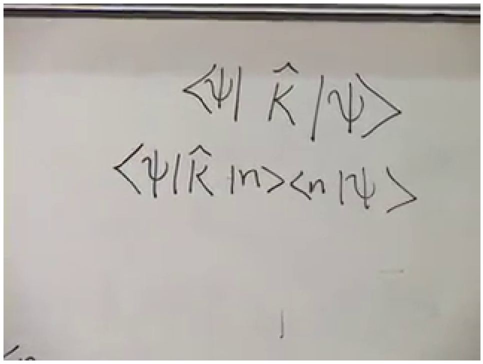

Figure 6.14: The expectation-value sandwich after inserting the eigenbasis resolution of identity.

Lecture frame: 01:39:45.

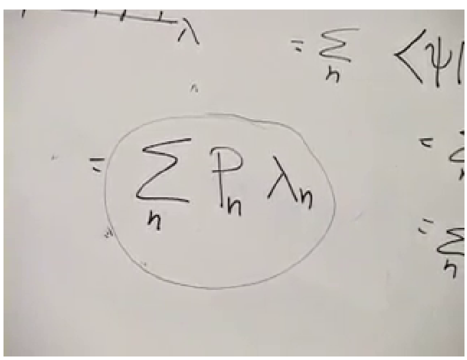

Figure 6.15: Possible eigenvalues weighted by their probabilities give the average $\sum_n p_n \lambda_n$.

Lecture frame: 01:42:50.

in the state $|\theta\rangle$:

$$\left\langle \hat{P}_0 \right\rangle_\theta = \langle\theta| \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} |\theta\rangle \tag{6.52}$$

$$= \begin{pmatrix} \cos\theta & \sin\theta \end{pmatrix} \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix} \tag{6.53}$$

$$= \cos^2\theta - \sin^2\theta = \cos 2\theta. \tag{6.54}$$

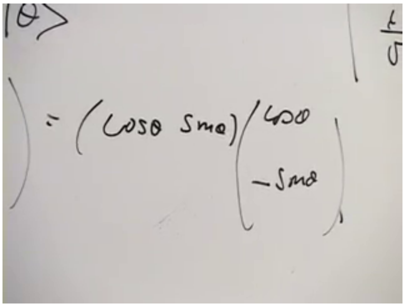

Figure 6.16: The axial polarization expectation value in the state $|\theta\rangle$.

Lecture frame: 01:45:50.

The same answer follows directly from the two outcomes:

$$(+1)\cos^2\theta + (-1)\sin^2\theta = \cos 2\theta. \tag{6.55}$$

No single detector click reports $\cos 2\theta$. Each run still records $+1$ or -1; $\cos 2\theta$ emerges only as the average over repeated identically prepared photons. That distinction between a state, a single-shot outcome, a probability, and an ensemble average completes the two-state polarization example.

CHAPTER 7

GLOBAL PHASE, POLARIZATION ELLIPSES, AND UNITARY EVOLUTION

7.1 Completing the Polarization Observable

We begin with the unfinished exercise from the preceding lecture. The horizontal and vertical polarization states are

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{7.1}$$

and the observable that records $+1$ for horizontal transmission and -1 for vertical transmission is

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{7.2}$$

At 45°, the two eigenstates

$$|d_\pm\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} \tag{7.3}$$

belong to

$$\hat{P}_{\pi/4} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{7.4}$$

The homework was to construct the observable for the circular basis

$$|c_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad |c_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{7.5}$$

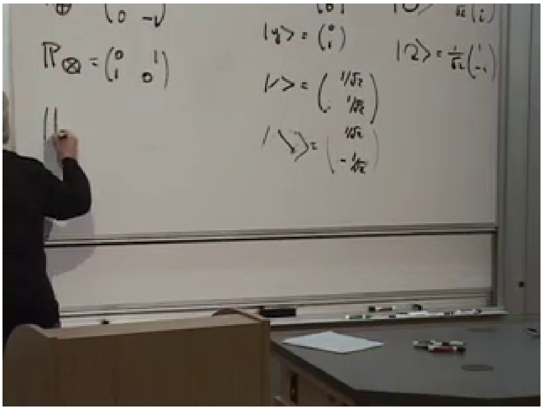

Figure 7.1: The three polarization bases and the circular observable at the end of the homework review.

Lecture frame: 00:05:20.

The spectral rule gives the answer immediately:

$$\hat{P}_c = |c_+\rangle \langle c_+| - |c_-\rangle \langle c_-| \tag{7.6}$$

$$= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{7.7}$$

Indeed,

$$\hat{P}_c |c_\pm\rangle = \pm |c_\pm\rangle . \tag{7.8}$$

The only delicate point in the check is $i^2 = -1$, together with complex conjugation when a ket becomes a bra.

Orthogonality has an operational meaning throughout this discussion. Two orthogonal polarization states can be distinguished with certainty in one ideal analyzer: an apparatus that transmits one rejects the other. This is why the inner product is not decorative notation. It records a measurable relation between preparations.

7.2 What States and Observables Are For

It is useful to pause and compare our present position with the opening classical discussion. For a coin or die, the state is one member of a finite set. A classical observable is simply a numerical function on that set. We might assign $+1$ to heads and -1 to tails, while the die arrives with the labels $1, \ldots, 6$. Relabelling the outcomes changes the numerical convention, not the underlying states.

The same idea survives in continuous classical mechanics. A state is a point (q, p) in phase space, and an observable is a function

$$F = F(q, p). \tag{7.9}$$

Position q, momentum p, and $p + q$ are elementary examples. As the phase-space point moves, the numerical value of each function changes.

Quantum mechanics replaces the set of states by a complex vector space and the numerical functions by Hermitian operators. Before discussing motion in that space, however, we should isolate the two principal questions that a state vector already answers.

First, for a normalized state $|\psi\rangle$ and observable $\hat{K}$, the ensemble average is

$$\left\langle \hat{K} \right\rangle_\psi = \langle\psi| \hat{K} |\psi\rangle . \tag{7.10}$$

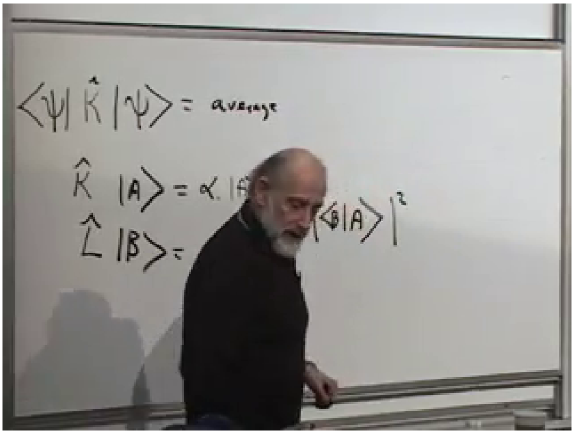

Figure 7.2: Expectation value, eigenstate preparations, and the squared overlap as the two physical uses of a state vector.

Lecture frame: 00:14:20.

Prepare the same state repeatedly, measure K on each member of the ensemble, and the empirical mean approaches this number.

Second, suppose

$$\hat{K}\,|a\rangle = \alpha\,|a\rangle\,, \qquad \hat{L}\,|b\rangle = \beta\,|b\rangle\,. \tag{7.11}$$

If the preparation makes $K = \alpha$ definite and the subsequent measurement asks whether $L = \beta$, then

$$\text{Prob}(b|a) = |\,\langle b\,|\,a\rangle\,|^2 = \langle a\,|\,b\rangle\,\langle b\,|\,a\rangle\,. \tag{7.12}$$

Reversing the preparation and test gives the same numerical probability, because $|\,\langle a\,|\,b\rangle\,|^2 = |\,\langle b\,|\,a\rangle\,|^2$.

Question from the audience. Does the order of the two overlaps in the probability matter?[a]

Response. The two amplitudes are complex conjugates: $\langle a \mid b \rangle = \langle b \mid a \rangle^*$. Their product is therefore the same nonnegative real number in either order. The correction on the board changes the amplitude, but not the transition probability.

[a]Lecture timestamp: 00:14:54–00:15:18.

Average values and transition probabilities both refer to repeated experiments. A single trial returns one allowed outcome; the probability and the average emerge from an ensemble. This common structure leads directly to the question of phase.

7.3 A Physical State Is a Ray

7.3.1 Pure Phases

A physical state vector is normalized:

$$\langle \psi \mid \psi \rangle = 1. \tag{7.13}$$

If $|\psi\rangle = \sum_n c_n |n\rangle$ in an orthonormal basis, this is $\sum_n |c_n|^2 = 1$, the statement that the total probability is one.

Normalization does not remove every redundancy. To see what remains, write a complex number in polar form,

$$z = x + iy = re^{i\chi} = r(\cos\chi + i\sin\chi). \tag{7.14}$$

Multiplication multiplies moduli and adds angles:

$$r_1 e^{i\chi_1} r_2 e^{i\chi_2} = r_1 r_2 e^{i(\chi_1+\chi_2)}. \tag{7.15}$$

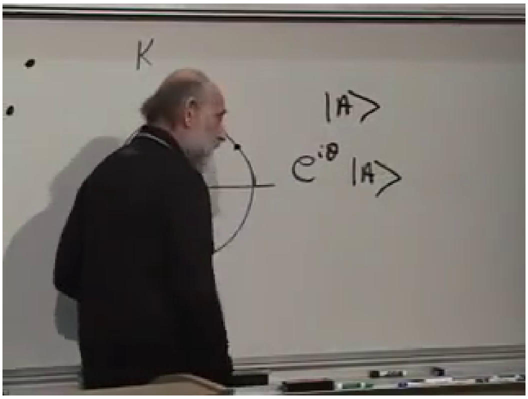

Figure 7.3: The unit circle beside the comparison of $|A\rangle$ and $e^{i\chi}\,|A\rangle$.

Lecture frame: 00:21:40.

Complex conjugation reflects the number across the real axis,

$$z^* = re^{-i\chi}. \tag{7.16}$$

A complex number of unit modulus is a *pure phase*:

$$e^{i\chi}, \qquad e^{i\chi}e^{-i\chi} = 1. \tag{7.17}$$

The word phase may refer either to the angle χ or to the unit complex number $e^{i\chi}$; the context tells us which is intended.

7.3.2 Why the Overall Phase Cancels

Mathematically, $|\psi\rangle$ and $e^{i\chi}\,|\psi\rangle$ are different vectors. Physically, they represent the same pure state. Let

$$|\psi'\rangle = e^{i\chi}\,|\psi\rangle\,. \tag{7.18}$$

The corresponding bra is

$$\langle\psi'| = e^{-i\chi}\,\langle\psi|\,. \tag{7.19}$$

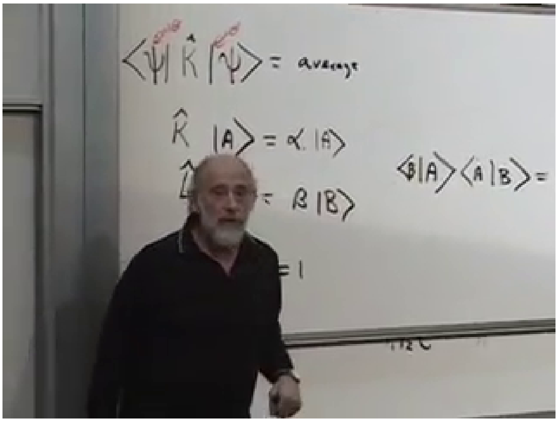

Figure 7.4: The two phase factors cancel in both the operator sandwich and the squared overlap.

Lecture frame: 00:23:35.

Every expectation value is unchanged:

$$\langle\psi'|\,\hat{K}\,|\psi'\rangle = e^{-i\chi}\,\langle\psi|\,\hat{K}e^{i\chi}\,|\psi\rangle \tag{7.20}$$

$$= \langle\psi|\,\hat{K}\,|\psi\rangle\,. \tag{7.21}$$

The phase is an ordinary scalar and commutes through $\hat{K}$.

The transition probability is equally insensitive. If $|b'\rangle = e^{i\eta}\,|b\rangle$, then

$$|\,\langle b'\,|\,a\rangle\,|^2 = e^{-i\eta}\,\langle b\,|\,a\rangle\,e^{i\eta}\,\langle a\,|\,b\rangle \tag{7.22}$$

$$= |\,\langle b\,|\,a\rangle\,|^2. \tag{7.23}$$

Independent phase choices for either state disappear from the absolute square.

The physical object is therefore not one normalized vector but the equivalence class

$$|\psi\rangle \sim e^{i\chi}\,|\psi\rangle\,. \tag{7.24}$$

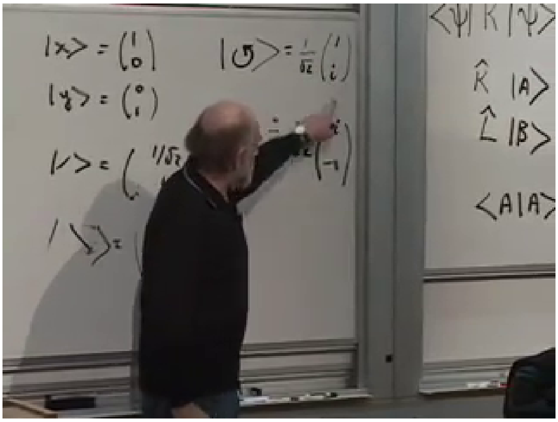

Figure 7.5: Multiplying both circular-polarization components by i changes the vector but not the physical ray.

Lecture frame: 00:28:40.

Such an equivalence class is called a *ray* in Hilbert space. In particular,

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \sim \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \tag{7.25}$$

because $-1 = e^{i\pi}$. Likewise,

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \sim \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ -1 \end{pmatrix}, \tag{7.26}$$

because the second vector is i times the first.

Question from the audience. What notation should express that two vectors have the same physical significance without saying that they are equal? [a]

Response. Ordinary equality is too strong, because the vectors really differ. The standard statement is that they represent the same ray, written here as $|\psi\rangle \sim e^{i\chi}\,|\psi\rangle$. What is preserved is every expectation value and transition probability.

[a]Lecture timestamp: 00:25:49–00:27:50.

7.4 Two Parameters Describe Polarization

Begin with the general two-component vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad \alpha, \beta \in \mathbb{C}. \tag{7.27}$$

The two complex entries contain four real parameters. Normalization removes one:

$$|\alpha|^2 + |\beta|^2 = 1. \tag{7.28}$$

The irrelevant common phase removes another. Except at the coordinate pole $\alpha = 0$, choose that phase so that $\alpha = a$ is real and nonnegative. Then

$$\beta = \sqrt{1-a^2}\, e^{i\phi}, \qquad 0 \le a \le 1, \tag{7.29}$$

and

$$|\psi\rangle = \begin{pmatrix} a \\ \sqrt{1-a^2}\, e^{i\phi} \end{pmatrix}. \tag{7.30}$$

Exactly two real parameters remain: a and the relative phase ϕ.

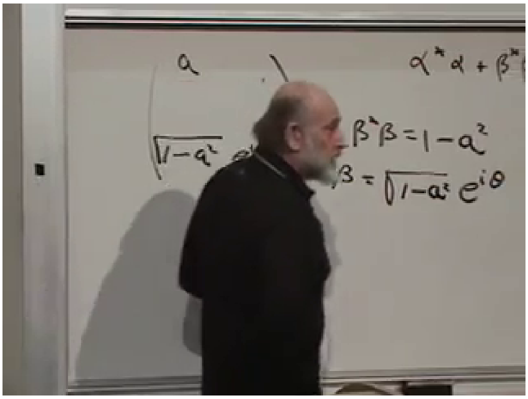

Figure 7.6: Normalization and phase fixing reduce two complex amplitudes to two physical real parameters.

Lecture frame: 00:32:45.

An equivalent parameterization makes the geometry especially transparent:

$$|\psi\rangle = \begin{pmatrix} \cos(\vartheta/2) \\ e^{i\phi}\sin(\vartheta/2) \end{pmatrix}, \qquad 0 \le \vartheta \le \pi. \tag{7.31}$$

The rays of a two-state system form a sphere, often called the Bloch sphere or, for polarization, the Poincare sphere. The lecture does not require that name, but its two-parameter count is precisely this geometry.

7.5 From the State Vector to a Polarization Ellipse

For a monochromatic wave moving along z, write the transverse electric field as

$$\boldsymbol{E}(z,t) = \mathrm{Re}\left[E_0\,(\alpha\,\hat{\boldsymbol{x}} + \beta\,\hat{\boldsymbol{y}})\,e^{i(kz-\omega t)}\right]. \tag{7.32}$$

The normalized pair $(\alpha, \beta)^T$ is the Jones vector. If the two components have the same phase, the tip of $\boldsymbol{E}$ moves back and forth along a line. A quarter-cycle relative phase with equal magnitudes produces a circle. General amplitudes and relative phases produce an ellipse.

The examples on the board may be written dimensionally as

$$E_x = E_0 \cos(kz - \omega t), \tag{7.33}$$
$$E_y = E_0 \sin(kz - \omega t), \tag{7.34}$$

which traces a circle at fixed z, and

$$E_x = E_0 \cos(kz - \omega t), \tag{7.35}$$
$$E_y = \frac{E_0}{2} \sin(kz - \omega t), \tag{7.36}$$

which traces an ellipse with its major axis along x.

The ellipse needs two independent real parameters. One gives the orientation of its major axis. The other gives its shape together with the sense of rotation. The lecture informally calls the latter a signed eccentricity. In standard optics it is cleaner to use the *signed ellipticity angle* ϵ:

$$-\frac{\pi}{4} \le \epsilon \le \frac{\pi}{4}, \qquad |\tan \epsilon| = \frac{\text{minor semiaxis}}{\text{major semiaxis}}. \tag{7.37}$$

The sign records handedness. Linear polarization has $\epsilon = 0$; circular polarization has $|\epsilon| = \pi/4$, where the major-axis direction becomes meaningless.

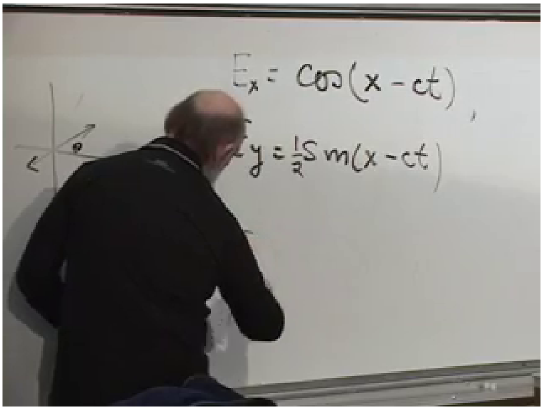

Figure 7.7: Equal quadrature amplitudes give a circle; reducing one amplitude gives an ellipse.

Lecture frame: 00:35:25.

Question from the audience. Do clockwise and counterclockwise polarization require a third parameter?[a]

Response. No. The shape parameter may carry a sign. As an ellipse is squeezed through the linear-polarization limit, changing the sign reverses the sense in which the field traverses the curve. Standard optics expresses this with signed ellipticity rather than literal signed geometric eccentricity.

[a]Lecture timestamp: 00:37:52–00:39:00.

7.5.1 Reading the Ellipse from Analyzer Probabilities

Let

$$b = \sqrt{1 - a^2}, \qquad |\psi\rangle = \begin{pmatrix} a \\ be^{i\phi} \end{pmatrix}, \tag{7.38}$$

and compare it with a linearly polarized analyzer state

$$|\theta\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}. \tag{7.39}$$

The two conjugate amplitudes are

$$\langle \theta \,|\, \psi \rangle = a\cos\theta + be^{i\phi}\sin\theta, \tag{7.40}$$

$$\langle \psi \,|\, \theta \rangle = a\cos\theta + be^{-i\phi}\sin\theta. \tag{7.41}$$

Question from the audience. Should one of the phase factors in the overlap be complex conjugated?[a]

Response. Yes. The sign depends on the bra-ket order. In $\langle \psi \,|\, \theta \rangle$, the coefficient from $|\psi\rangle$ is conjugated, giving $e^{-i\phi}$. The opposite order gives $e^{i\phi}$. Their absolute squares agree.

[a] Lecture timestamp: 00:41:48–00:42:02.

The transmission probability is

$$P_\theta = |\, \langle \theta \,|\, \psi \rangle \,|^2 \tag{7.42}$$

$$= a^2\cos^2\theta + b^2\sin^2\theta + 2ab\cos\phi\,\sin\theta\cos\theta \tag{7.43}$$

$$= \frac{1}{2}\left[1 + (a^2 - b^2)\cos 2\theta + 2ab\cos\phi\,\sin 2\theta\right]. \tag{7.44}$$

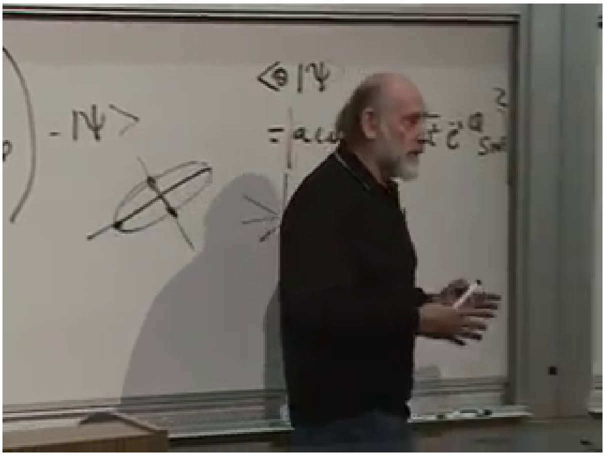

Figure 7.8: The overlap amplitude and the geometric search for the linear-analyzer direction of maximum transmission.

Lecture frame: 00:45:20.

The lecture leaves this multiplication and maximization as homework. Finishing that step makes its geometric prescription exact. The major-axis angle θ_{maj} obeys

$$2\theta_{\mathrm{maj}} = \operatorname{atan2}\left(2ab\cos\phi,\ a^2 - b^2\right) \pmod{2\pi}. \tag{7.45}$$

Thus the analyzer most likely to transmit the photon points along the major axis of the polarization ellipse.

It is useful to collect the same information in the normalized Stokes parameters

$$S_1 = a^2 - b^2, \qquad S_2 = 2ab\cos\phi, \qquad S_3 = 2ab\sin\phi, \tag{7.46}$$

for which $S_1^2 + S_2^2 + S_3^2 = 1$. With a fixed propagation

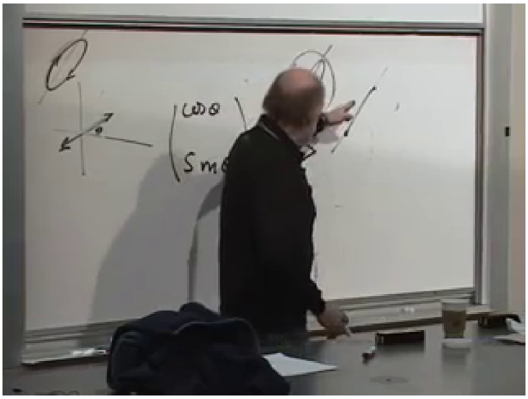

Figure 7.9: Squeezing through the linear limit reverses the traversal sense, which is encoded by the sign of ellipticity.

Lecture frame: 00:46:40.

and time convention,

$$2\theta_{\mathrm{maj}} = \mathrm{atan2}(S_2, S_1), \qquad \sin 2\epsilon = S_3. \tag{7.47}$$

Changing the convention for viewing the wave reverses the sign assigned to handedness, but not the two-parameter count.

Question from the audience. Orientation and ordinary eccentricity still seem to omit clockwise versus counterclockwise motion. Where is it stored? [a]

Response. It is stored in the sign of the second parameter. The major-axis angle gives the orientation; signed ellipticity gives both the axis ratio and the traversal sense. At the linear limit the sign can change, and at the circular limit the axis angle is irrelevant.

[a] Lecture timestamp: 00:45:31–00:47:14.

7.6 Time Evolution Must Preserve Distinctions

7.6.1 The Classical Analogy

We now turn from the description of states to their motion. A finite classical state set admits only jumps if time is viewed microscopically, but the quantum vector space contains continuous paths even when a measurement has only two possible outcomes. A linear polarization axis can rotate continuously with time although every fixed analyzer still records only $+1$ or -1.

For a closed reversible classical system, evolution is one-to-one. It may permute the states, but it cannot send two distinct initial states to the same final state. If both 5 and 2 on a die evolved to 3, the final state would not reveal which initial state had been prepared. The distinction would have been erased.

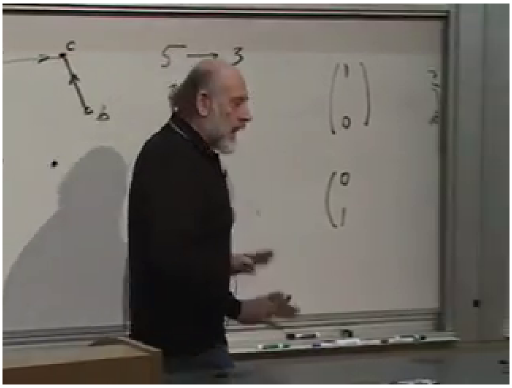

Figure 7.10: A forbidden many-to-one rule would send both 5 and 2 to 3, erasing the initial distinction.

Lecture frame: 00:51:15.

Quantum states have more structure than simply same or different. For normalized vectors, the invariant transition probability

$$|\langle B \mid A \rangle|^2 \tag{7.48}$$

measures their operational overlap. Orthogonal states have zero overlap and can be distinguished with certainty. States on the same ray have $|\langle B \mid A \rangle| = 1$. Intermediate values mean that no single ideal test distinguishes the preparations with certainty.

7.6.2 Preserving the Inner Product

Prepare two identical systems in $|A\rangle$ and $|B\rangle$, and let both evolve for the same time:

$$|A\rangle \longrightarrow |A'\rangle\,, \qquad |B\rangle \longrightarrow |B'\rangle\,. \tag{7.49}$$

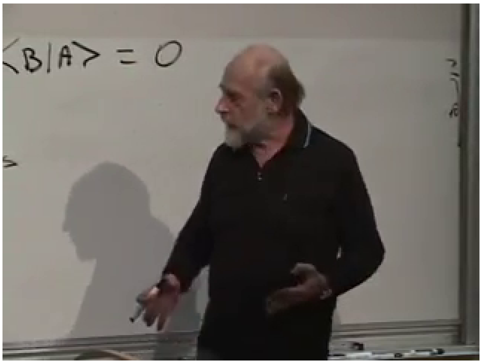

Figure 7.11: Zero overlap represents orthogonal, perfectly distinguishable polarization states.

Lecture frame: 00:54:40.

Closed-system quantum evolution preserves their inner product:

$$\boxed{\langle B' \mid A' \rangle = \langle B \mid A \rangle \,.} \tag{7.50}$$

Consequently equal rays remain equal rays, orthogonal states remain orthogonal, and every transition probability remains unchanged. This is the quantum form of preserving distinctions.

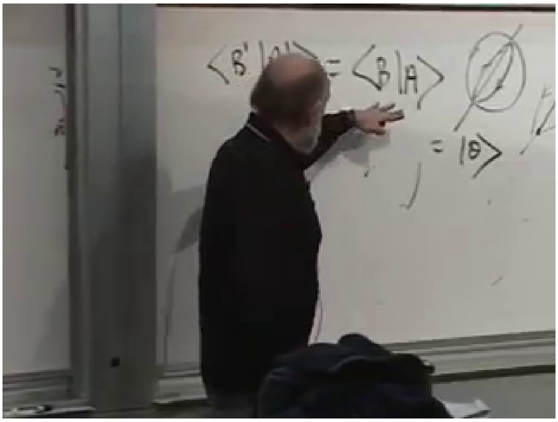

Figure 7.12: The overlap of two states is unchanged when both undergo the same closed-system evolution.

Lecture frame: 00:57:40.

Question from the audience. If one photon fails a horizontal analyzer, how do we know it was vertical rather than a 45° state that happened to fail? [a]

Response. From one event, we do not. Orthogonality is tested by certainty across identically prepared trials. Every genuinely vertical photon is rejected by an ideal horizontal analyzer; a 45° state is rejected only with probability one half. The ensemble distinguishes the preparations.

[a] Lecture timestamp: 00:58:17–00:59:00.

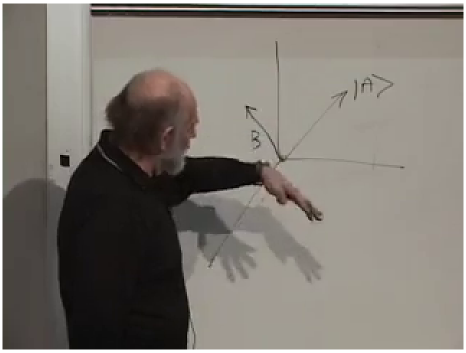

Figure 7.13: The geometric analogy: vectors move while their lengths and mutual angles remain fixed.

Lecture frame: 01:01:25.

Question from the audience. Does the uncertainty principle force a small chance that a perfectly horizontal photon passes a vertical polarizer? [a]

Response. No. One polarization observable can be prepared with arbitrary precision in the ideal theory. The uncertainty concerns incompatible observables: no state has definite outcomes for two genuinely different analyzer axes at once. It does not blur an eigenstate of one chosen axis.

[a] Lecture timestamp: 00:59:01–01:00:31.

In an ordinary real vector space, a transformation preserving all inner products looks like a rigid rotation or reflection: vectors move, but their lengths and mutual angles do not. The complex quantum version is a unitary transformation.

7.7 Operators on Bras and the Adjoint

Let $\hat{L}$ be a linear operator on kets:

$$\hat{L}\,|A\rangle = |C\rangle\,. \tag{7.51}$$

Linearity means

$$\hat{L}(\alpha\,|A\rangle + \beta\,|B\rangle) = \alpha\hat{L}\,|A\rangle + \beta\hat{L}\,|B\rangle\,. \tag{7.52}$$

For any bra $\langle B|$, the matrix element

$$\langle B|\,\hat{L}\,|A\rangle \tag{7.53}$$

may be read associatively as the inner product of $\langle B|$ with $\hat{L}\,|A\rangle$, or as the bra $\langle B|\,\hat{L}$ applied to $|A\rangle$.

Inserting a complete orthonormal basis makes the latter bra explicit:

$$\langle B|\,\hat{L} = \sum_n \langle B|\,\hat{L}\,|n\rangle\,\langle n|\,. \tag{7.54}$$

The coefficients are numbers, so the right side is a linear combination of bras.

Question from the audience. Does this construction assume that $\hat{L}$ is Hermitian?[a]

Response. No. The matrix element and the bra $\langle B|\,\hat{L}$ are defined for a general linear operator. Hermiticity is the additional condition $\hat{L}^\dagger = \hat{L}$, introduced later.

[a]Lecture timestamp: 01:05:51–01:06:00.

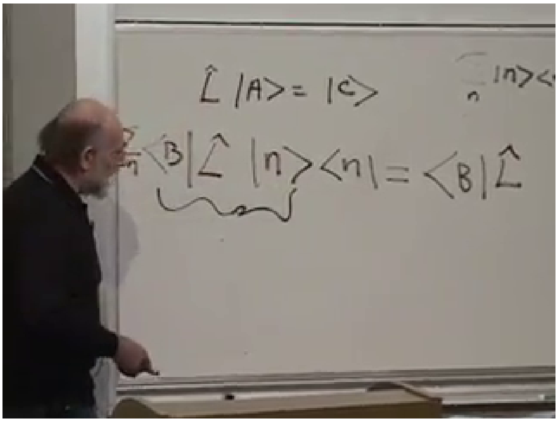

Figure 7.14: A completeness insertion defines the bra $\langle B|\,\hat{L}$ from the known action of $\hat{L}$ on kets.

Lecture frame: 01:06:50.

Question from the audience. Are we assuming associativity when the brackets are regrouped?[a]

Response. Yes. Operator composition and its action on vectors are associative wherever the products are defined: $\langle B|\,(\hat{L}\,|A\rangle) = (\langle B|\,\hat{L})\,|A\rangle$. This is what lets the notation be manipulated consistently.

[a] Lecture timestamp: 01:06:01–01:06:18.

7.7.1 Hermitian Conjugation

Taking the adjoint of Equation (7.51) gives

$$\langle A|\,\hat{L}^\dagger = \langle C|\,. \tag{7.55}$$

The defining matrix-element identity is

$$\boxed{\langle B|\,\hat{L}\,|A\rangle = \left(\langle A|\,\hat{L}^\dagger\,|B\rangle\right)^*.} \tag{7.56}$$

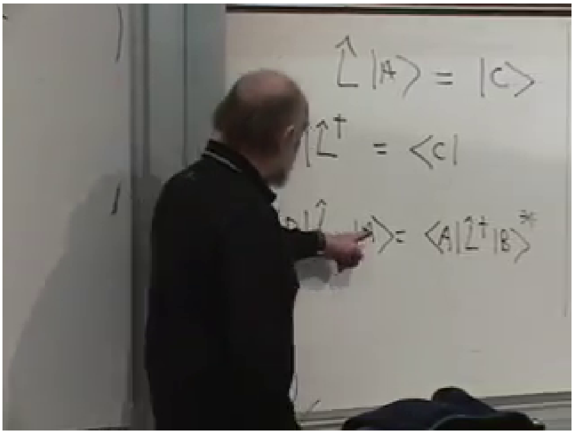

Figure 7.15: The adjoint relation obtained by reversing the bra-ket order and complex conjugating the matrix element.

Lecture frame: 01:12:50.

It combines two operations already familiar from inner products: interchange bra and ket, then complex conjugate.

In an orthonormal basis, define

$$L_{mn} = \langle m| \hat{L} |n\rangle . \tag{7.57}$$

Then

$$(\hat{L}^{\dagger})_{mn} = L^{*}_{nm}. \tag{7.58}$$

Thus the matrix of $\hat{L}^{\dagger}$ is the complex-conjugate transpose:

$$L^{\dagger} = (L^{*})^{T}. \tag{7.59}$$

Rows and columns are interchanged, and every entry is conjugated.

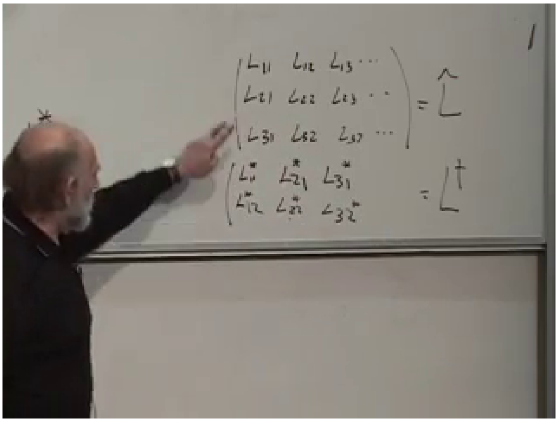

Figure 7.16: The matrix rule for the adjoint: transpose and complex conjugate.

Lecture frame: 01:17:25.

Question from the audience. Are we assuming a finite-dimensional state space here? [a]

Response. For the displayed matrices, yes. The same adjoint identity extends to Hilbert spaces, but infinite-dimensional operators require care about domains and completeness. The finite-dimensional case contains the algebra needed for the present argument without those analytic qualifications.

[a] Lecture timestamp: 01:13:44–01:14:05.

7.8 Hermitian Operators Represent Observables

An operator is Hermitian when

$$\hat{L} = \hat{L}^{\dagger}. \tag{7.60}$$

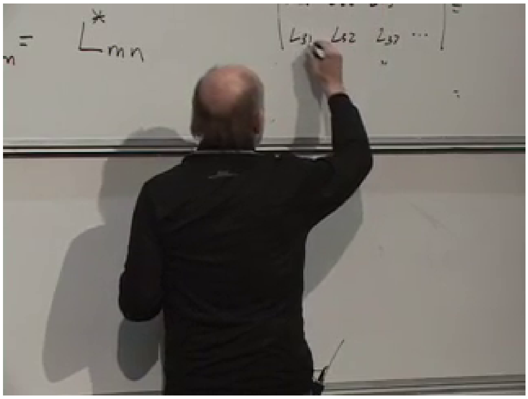

Figure 7.17: The conjugate-symmetry condition $L_{mn} = L^*_{nm}$ for a Hermitian matrix.

Lecture frame: 01:19:30.

In components,

$$L_{mn} = L^*_{nm}. \tag{7.61}$$

The diagonal entries are therefore real,

$$L_{nn} = L^*_{nn} \in \mathbb{R}, \tag{7.62}$$

while entries reflected across the diagonal are complex conjugates.

In finite dimensions, the spectral theorem gives the properties needed of an observable:

1. every eigenvalue is real;
2. eigenvectors belonging to distinct eigenvalues are orthogonal;
3. an orthonormal eigenbasis exists.

The eigenvectors need not have real components. Circular polarization is the immediate counterexample: the Hermitian matrix $\hat{P}_c$ has eigenvectors containing $\pm i$.

The expectation value of a Hermitian operator is real:

$$\left(\langle\psi|\,\hat{L}\,|\psi\rangle\right)^* = \langle\psi|\,\hat{L}^\dagger\,|\psi\rangle \tag{7.63}$$

$$= \langle\psi|\,\hat{L}\,|\psi\rangle\,. \tag{7.64}$$

Conversely, in finite dimensions, if $\langle\psi|\,\hat{L}\,|\psi\rangle$ is real for every $|\psi\rangle$, then $\hat{L}$ is Hermitian.

Question from the audience. How large is the matrix representing an observable? [a]

Response. It matches the dimension of the state space under study. Polarization alone is two-dimensional. A full photon state also carries continuous momentum or position labels and is generally infinite-dimensional. Finite truncations are often used for calculation, but finite systems such as an isolated spin degree of freedom are not merely approximations.

[a] Lecture timestamp: 01:20:28–01:22:28.

The three polarization observables make a compact check:

$$\hat{P}_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{P}_{\pi/4} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{P}_c = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{7.65}$$

Each equals its adjoint. They are, respectively, the Pauli matrices $\sigma_z, \sigma_x, \sigma_y$.

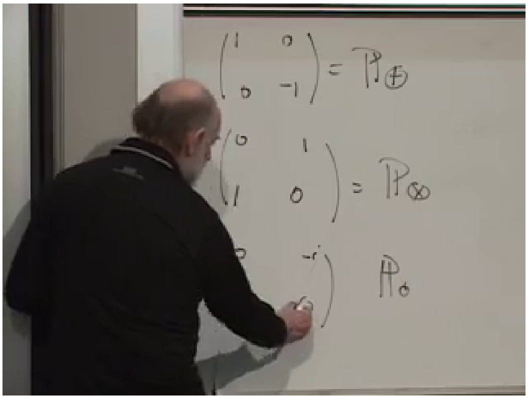

Figure 7.18: The axial, diagonal, and circular polarization observables are all Hermitian.

Lecture frame: 01:24:45.

Question from the audience. Why is Hermiticity the right condition for an observable? [a]

Response. Laboratory outcomes and their averages are real numbers. Hermiticity guarantees real eigenvalues and real expectation values. In finite dimensions the converse condition on all quadratic forms also recovers Hermiticity, so the mathematical requirement is tightly matched to the operational one.

[a] Lecture timestamp: 01:25:08–01:26:20.

Question from the audience. Must the eigenvectors of an observable span the whole state space?[a]

Response. Yes in the finite-dimensional Hermitian case: the spectral theorem supplies a complete orthonormal eigenbasis, including an orthonormal choice inside degenerate eigenspaces. A general non-Hermitian operator may have nonorthogonal eigenvectors or may fail to have a complete eigenbasis.

[a]Lecture timestamp: 01:26:22–01:27:43.

7.9 Unitary Operators Preserve Quantum Relations

Hermitian operators represent quantities that can be measured. They are not, in general, the transformations that move states while preserving their relations. Let one linear operator $\hat{U}$ act on every state:

$$\hat{U}\left|A\right\rangle = \left|A'\right\rangle, \qquad \hat{U}\left|B\right\rangle = \left|B'\right\rangle. \tag{7.66}$$

The transformed bras are

$$\left\langle A'\right| = \left\langle A\right|\hat{U}^\dagger, \qquad \left\langle B'\right| = \left\langle B\right|\hat{U}^\dagger. \tag{7.67}$$

Therefore

$$\left\langle B' \mid A'\right\rangle = \left\langle B\right|\hat{U}^\dagger\hat{U}\left|A\right\rangle. \tag{7.68}$$

Demanding equality with $\left\langle B \mid A\right\rangle$ for every pair $\left|A\right\rangle, \left|B\right\rangle$ forces

$$\boxed{\hat{U}^\dagger\hat{U} = \mathbf{1}.} \tag{7.69}$$

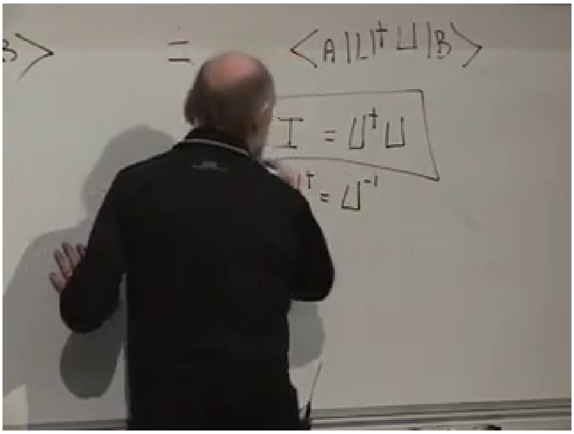

Figure 7.19: Preservation of every inner product leads to the unitary condition $\hat{U}^\dagger \hat{U} = \mathbf{1}$.

Lecture frame: 01:34:50.

In finite dimensions this condition makes $\hat{U}$ invertible and implies

$$\hat{U}^\dagger = \hat{U}^{-1}, \qquad \hat{U}\hat{U}^\dagger = \hat{U}^\dagger \hat{U} = \mathbf{1}. \tag{7.70}$$

In an infinite-dimensional Hilbert space, the definition of a unitary operator includes both equalities; $\hat{U}^\dagger \hat{U} = \mathbf{1}$ alone defines an isometry that need not be onto.

A unitary transformation is not the identity. It may rotate polarization, implement a symmetry, or carry a state from one time to another. Its adjoint undoes it, and throughout the motion

$$\langle UB \,|\, UA \rangle = \langle B |\, \hat{U}^\dagger \hat{U} \,| A \rangle = \langle B \,|\, A \rangle \,. \tag{7.71}$$

Thus every norm, orthogonality relation, and transition probability is preserved.

Question from the audience. Does preserving relationships mean anything broader than preserving inner products?[a]

Response. Mathematically, inner-product preservation is the precise condition used here. Physically it preserves the absolute squares of those inner products and therefore every transition probability between simultaneously evolved preparations.

[a]Lecture timestamp: 01:35:13–01:35:53.

The lecture closes just as the eigenvalue structure of a unitary operator is introduced:

$$\hat{U} |n\rangle = \lambda_n |n\rangle . \tag{7.72}$$

The unanswered question has a short consequence of unitarity:

$$\begin{aligned} 1 = \langle n \mid n \rangle &= \langle n | \hat{U}^\dagger \hat{U} |n\rangle & (7.73) \\ &= |\lambda_n|^2 \langle n \mid n \rangle , & (7.74) \end{aligned}$$

so

$$|\lambda_n| = 1, \qquad \lambda_n = e^{i\omega_n}. \tag{7.75}$$

Finite-dimensional unitary matrices are normal and therefore possess a complete orthonormal eigenbasis, but their eigenvalues are phases rather than necessarily real numbers.

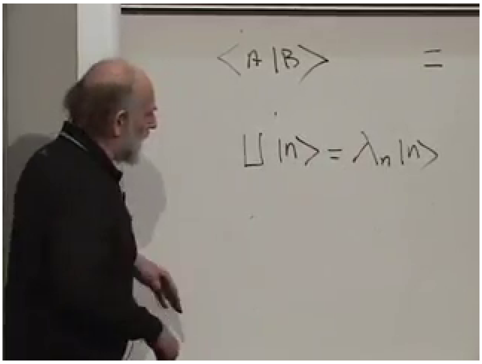

Figure 7.20: The final board question: the eigenvalues of a unitary operator lie on the unit circle.

Lecture frame: 01:37:20.

We have reached the threshold of dynamics. Closed systems evolve by unitary transformations. The next step is to discover how continuous unitary evolution is generated by a Hermitian Hamiltonian and how that statement becomes the Schrodinger equation.

CHAPTER 8

HAMILTONIAN TIME EVOLUTION, INTERFERENCE, AND THE SCHRÖDINGER EQUATION

8.1 The Classical Hamiltonian Generates a Flow

We now begin the serious study of change with time. It is useful to start with the classical picture, not because we shall need every detail of classical mechanics, but because one structural idea survives intact: the Hamiltonian is the generator of time evolution.

For a system with finitely many classical states, a law of evolution may be drawn as arrows from one state to the next. A coin could pass through a sequence such as heads, tails, heads, and so on. Since there is no continuous path between two isolated states, this picture naturally describes updates at discrete sampling times.

Continuous classical motion requires a continuous state space. A particle on a line has, at each instant, a position x and momentum p. Its state is therefore a point in the two-dimensional phase space with coordinates (x, p). As time passes, that point follows a flow line.

The flow is determined by a function of phase space

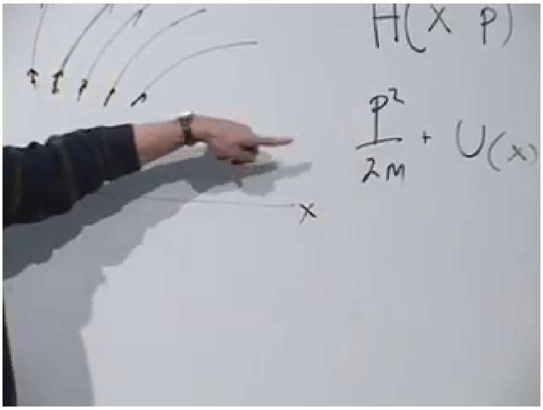

Figure 8.1: The classical Hamiltonian for a particle on a line, with the kinetic and potential terms separated.

Lecture frame: 00:05:22.

called the Hamiltonian. For a particle of mass m in a potential $V(x)$, the standard example is

$$H(x,p) = \frac{p^2}{2m} + V(x). \tag{8.1}$$

The momentum-dependent term is the kinetic energy and the position-dependent term is the potential energy. For many particles there is one momentum conjugate to each coordinate, but one pair is enough to display the structure.

Hamilton's equations turn this energy function into motion:

$$\dot{x} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial x}. \tag{8.2}$$

For Equation (8.1), $\partial H/\partial p = p/m$, which is the velocity. The second equation states that the rate of change of momentum is the force, $-\partial V/\partial x$.

Now let $F(x,p)$ be any classical observable. The chain rule gives

$$\frac{dF}{dt} = \frac{\partial F}{\partial x}\dot{x} + \frac{\partial F}{\partial p}\dot{p} \tag{8.3}$$

$$= \frac{\partial F}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial H}{\partial x}. \tag{8.4}$$

The combination on the right is the Poisson bracket:

$$\{F,H\}_{\mathrm{PB}} = \frac{\partial F}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial H}{\partial x}. \tag{8.5}$$

Thus

$$\boxed{\frac{dF}{dt} = \{F,H\}_{\mathrm{PB}}.} \tag{8.6}$$

One need not memorize the bracket at this stage. Its message is more important than its detailed form: once H is known, every phase-space quantity has a determined time derivative. The Hamiltonian generates the flow.

8.2 Quantum Evolution Preserves Relations Between States

The quantum statement begins from a different picture. A state is a vector, and the physically meaningful relation between two normalized states is their inner product. Orthogonal states can be distinguished with certainty; nonorthogonal states have a nonzero transition probability. Closed-system evolution must preserve all of these relations.

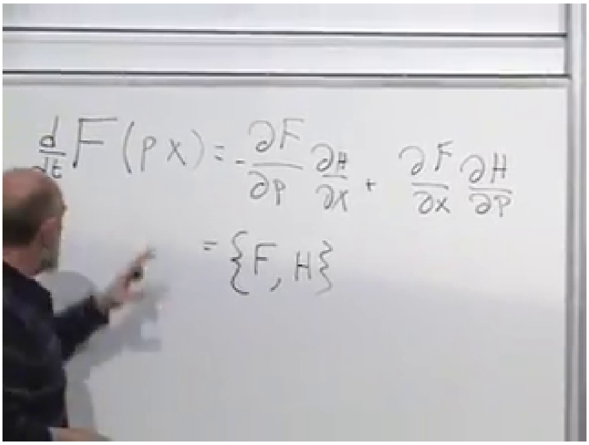

Figure 8.2: Hamilton's equations inserted into the chain rule produce the Poisson-bracket law for an arbitrary observable.

Lecture frame: 00:08:38.

Suppose that after an elapsed time T,

$$|A\rangle \longmapsto \hat{U}(T)\,|A\rangle\,, \qquad |B\rangle \longmapsto \hat{U}(T)\,|B\rangle\,. \quad (8.7)$$

The corresponding transformed bra is

$$\langle B| \longmapsto \langle B|\,\hat{U}^{\dagger}(T). \quad (8.8)$$

Preservation of the inner product requires

$$\langle B\,|\,A\rangle = \langle B|\,\hat{U}^{\dagger}(T)\hat{U}(T)\,|A\rangle\,. \quad (8.9)$$

This must hold for every pair $|A\rangle\,, |B\rangle$. Therefore

$$\boxed{\hat{U}^{\dagger}(T)\hat{U}(T) = \mathbf{1}.} \quad (8.10)$$

In finite dimensions this also implies $\hat{U}(T)\hat{U}^{\dagger}(T) = \mathbf{1}$. The evolution operator is unitary.

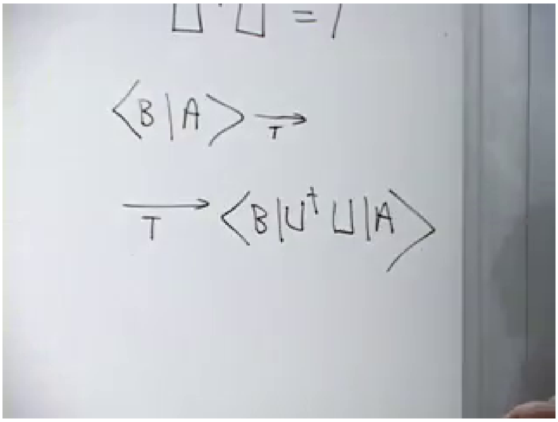

Figure 8.3: The common evolution of two states preserves their inner product and leads to $\hat{U}^\dagger\hat{U} = \mathbf{1}$.
Lecture frame: 00:14:20.

There is a useful geometric image here. The vectors move, but the whole state space moves rigidly: norms and mutual angles remain fixed. In particular, orthogonal states remain orthogonal. For the linear evolution considered here, this preservation law is the governing principle.

Writing the state at an arbitrary initial time t, the evolution law is

$$|\psi(t+T)\rangle = \hat{U}(T)\,|\psi(t)\rangle\,. \tag{8.11}$$

No elapsed time means no change,

$$\hat{U}(0) = \mathbf{1}. \tag{8.12}$$

Successive intervals compose:

$$\hat{U}(T_2)\hat{U}(T_1) = \hat{U}(T_1+T_2) \tag{8.13}$$

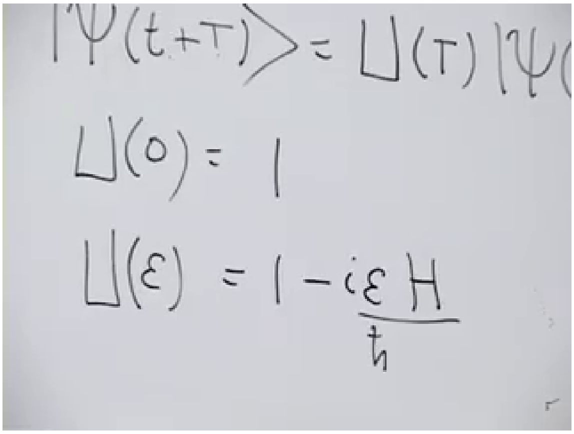

Figure 8.4: A short-time evolution differs from the identity by a term proportional to $-i\epsilon\hat{H}/\hbar$.

Lecture frame: 00:19:54.

when the Hamiltonian has no explicit time dependence. This is why a long evolution can be assembled from many short steps.

8.3 The Infinitesimal Generator

Let ϵ be short enough that terms of order ϵ^2 may be neglected. Since $\hat{U}(0) = \mathbf{1}$, the most general first-order form can be written

$$\hat{U}(\epsilon) = \mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H} + O(\epsilon^2). \tag{8.14}$$

At first, $\hat{H}$ is simply the name of an operator. The factors $-i/\hbar$ are chosen so that $\hat{H}$ has units of energy and so that the classical correspondence will later take its standard form.

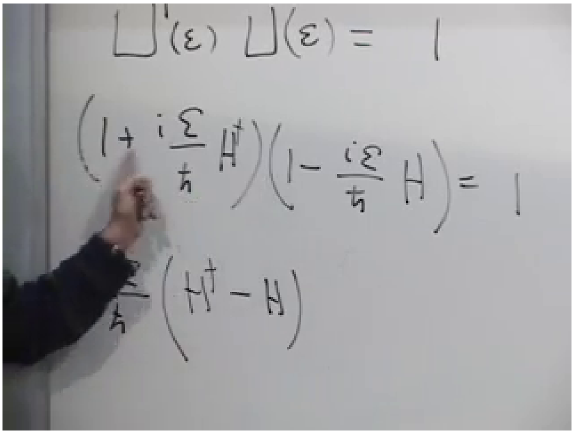

Figure 8.5: Expanding $\hat{U}^\dagger \hat{U} = \mathbf{1}$ to first order leaves the condition $\hat{H}^\dagger - \hat{H} = 0$.

Lecture frame: 00:22:48.

Taking the adjoint gives

$$\hat{U}^\dagger(\epsilon) = \mathbf{1} + \frac{i\epsilon}{\hbar}\hat{H}^\dagger + O(\epsilon^2). \tag{8.15}$$

Unitarity through first order then says

$$\mathbf{1} = \hat{U}^\dagger(\epsilon)\hat{U}(\epsilon) \tag{8.16}$$

$$= \left(\mathbf{1} + \frac{i\epsilon}{\hbar}\hat{H}^\dagger\right)\left(\mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H}\right) + O(\epsilon^2) \tag{8.17}$$

$$= \mathbf{1} + \frac{i\epsilon}{\hbar}\left(\hat{H}^\dagger - \hat{H}\right) + O(\epsilon^2). \tag{8.18}$$

The linear term must vanish, so

$$\boxed{\hat{H}^\dagger = \hat{H}.} \tag{8.19}$$

The generator of continuous unitary evolution is Hermitian. Its eigenvalues are real and, in finite dimensions, its eigenvectors may be chosen as an orthonormal basis. We identify this observable as the Hamiltonian and call its eigenvalues energies. At this point that is an identification within quantum mechanics; recovering familiar classical energy requires a further correspondence argument.

For a time-independent Hamiltonian, compounding the short-time step gives

$$\hat{U}(T) = \exp\left(-\frac{i}{\hbar}\hat{H}T\right). \tag{8.20}$$

The exponential is unitary precisely because $\hat{H}$ is Hermitian.

Question from the audience. If $\hbar$ appears in the denominator, what happens in the classical limit $\hbar \to 0$? [a]

Response. The classical limit is not obtained by setting a denominator to zero inside one fixed microscopic formula. What matters are dimensionless actions measured in units of $\hbar$, such as $\epsilon E/\hbar$. Classical behavior emerges when the relevant actions are large compared with $\hbar$ and quantum phases combine by stationary phase or decoherence. The operator $\hat{H}$ itself has a well-defined classical counterpart even though the phases become rapidly varying.

[a] Lecture timestamp: 00:26:41–00:28:07.

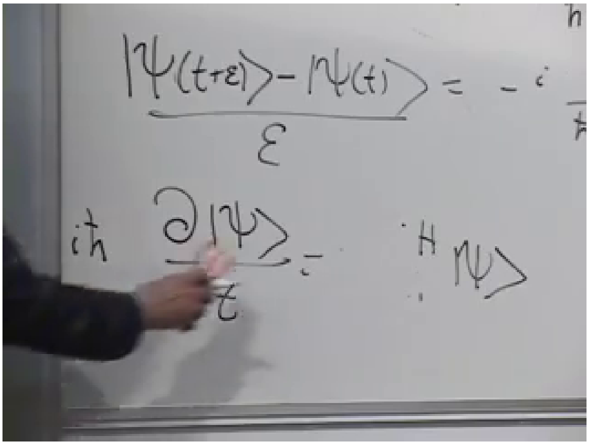

Figure 8.6: The finite difference from t to $t+\epsilon$ becomes the abstract Schrödinger equation.

Lecture frame: 00:32:03.

8.4 The Abstract Schrödinger Equation

Apply Equation (8.14) to a state:

$$|\psi(t+\epsilon)\rangle = \left(\mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H}\right)|\psi(t)\rangle + O(\epsilon^2). \tag{8.21}$$

Subtract $|\psi(t)\rangle$, divide by ϵ, and take $\epsilon \to 0$:

$$\frac{\partial}{\partial t}|\psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\psi(t)\rangle. \tag{8.22}$$

Multiplication by $i\hbar$ gives

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle.} \tag{8.23}$$

This is the general law of quantum time evolution. It is abstract because $|\psi\rangle$ may describe polarization, spin, a particle in space, or any other quantum system. Schrödinger's original wave equation will appear only after we choose a coordinate representation and specify $\hat{H}$.

There is a useful parallel with momentum. In the coordinate representation,

$$\hat{p} = -i\hbar\frac{\partial}{\partial x} \tag{8.24}$$

relates spatial translation to momentum, while Equation (8.23) relates time translation to energy. The sign conventions are conventional; their consistent use is what matters.

8.5 A Polarization Doublet in an Anisotropic Medium

8.5.1 The Energy Eigenstates

Consider a photon of fixed wavelength moving through a material whose two preferred polarization axes have different energies. Choose those material axes as x and y, and suppose

$$\hat{H}\,|x\rangle = E_1\,|x\rangle\,, \qquad \hat{H}\,|y\rangle = E_2\,|y\rangle\,. \tag{8.25}$$

Here x and y label polarization, not position. The assumption is that the material itself distinguishes these two directions.

A general polarization state is

$$|\psi(t)\rangle = \alpha(t)\,|x\rangle + \beta(t)\,|y\rangle = \begin{pmatrix}\alpha(t)\\ \beta(t)\end{pmatrix}. \tag{8.26}$$

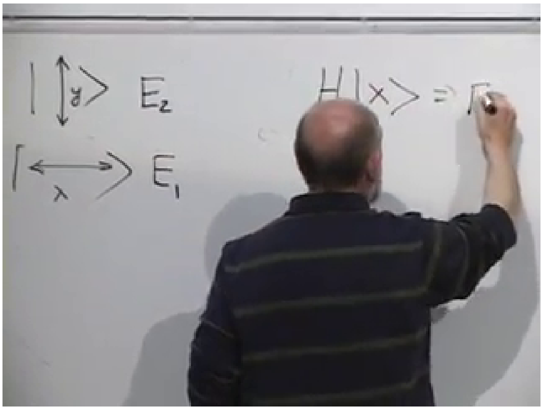

Figure 8.7: The material assigns energies E_1 and E_2 to its two orthogonal polarization eigenstates.
Lecture frame: 00:35:54.

The probabilities of obtaining x and y polarization are $|\alpha(t)|^2$ and $|\beta(t)|^2$.

In the $(|x\rangle, |y\rangle)$ basis,

$$\hat{H} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}. \tag{8.27}$$

The diagonal order follows directly from Equation (8.25).

The Schrödinger equation becomes

$$i\hbar \frac{d}{dt} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{8.28}$$

The two component equations decouple:

$$i\hbar\dot{\alpha} = E_1\alpha, \qquad i\hbar\dot{\beta} = E_2\beta. \tag{8.29}$$

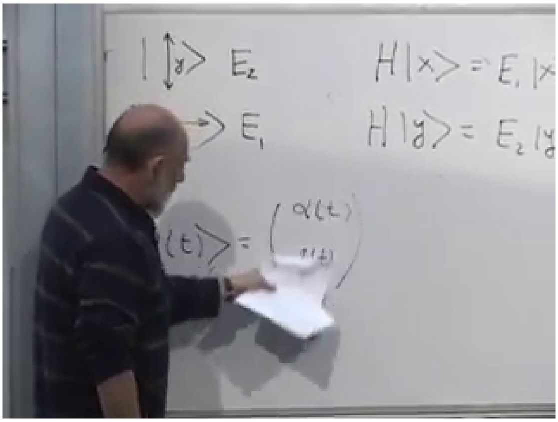

Figure 8.8: The polarization eigenvalue equations used to construct the diagonal Hamiltonian.

Lecture frame: 00:39:31.

Their solutions are

$$\alpha(t) = \alpha(0)e^{-iE_1t/\hbar}, \qquad \beta(t) = \beta(0)e^{-iE_2t/\hbar}. \tag{8.30}$$

Thus

$$|\psi(t)\rangle = \begin{pmatrix} \alpha(0)e^{-iE_1t/\hbar} \\ \beta(0)e^{-iE_2t/\hbar} \end{pmatrix}. \tag{8.31}$$

If we measure in the energy basis itself, nothing seems to change:

$$|\alpha(t)|^2 = |\alpha(0)|^2, \qquad |\beta(t)|^2 = |\beta(0)|^2. \tag{8.32}$$

Each exponential has unit modulus. The relative phase is invisible to an x/y analyzer, but it is not physically irrelevant.

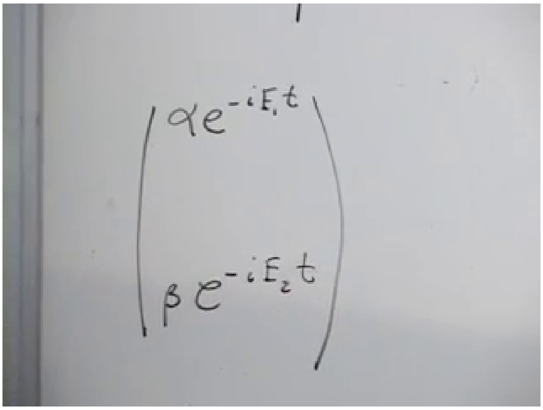

Figure 8.9: Each energy component accumulates its own phase, $e^{-iE_j t/\hbar}$.

Lecture frame: 00:43:55.

8.5.2 Interference in a Rotated Analyzer

Let

$$|d_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |d_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \tag{8.33}$$

be the $+45°$ and $-45°$ linear polarizations. The amplitude to pass a $+45°$ analyzer after time t is

$$\langle d_+ \,|\, \psi(t)\rangle = \frac{1}{\sqrt{2}}\left[\alpha(0)e^{-iE_1 t/\hbar} + \beta(0)e^{-iE_2 t/\hbar}\right]. \tag{8.34}$$

Prepare the photon with a first $+45°$ polarizer. Then

$$\alpha(0) = \beta(0) = \frac{1}{\sqrt{2}}, \tag{8.35}$$

and

$$\mathcal{A}_+(t) = \frac{1}{2}\left(e^{-iE_1 t/\hbar} + e^{-iE_2 t/\hbar}\right). \tag{8.36}$$

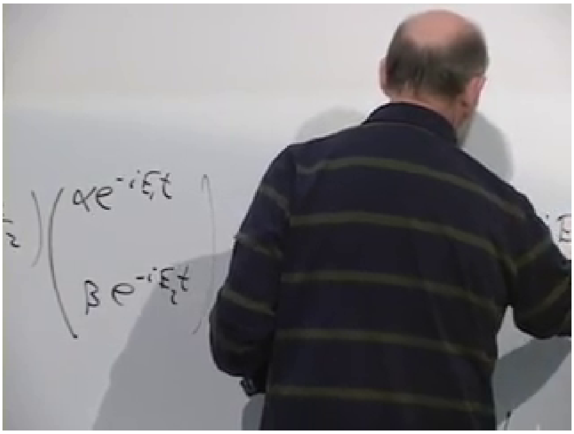

Figure 8.10: Projection of the evolved two-component state onto the 45° analyzer.

Lecture frame: 00:49:24.

The transmission probability is the absolute square:

$$P_+(t) = |\mathcal{A}_+(t)|^2 \tag{8.37}$$

$$= \frac{1}{4}\left(e^{-iE_1 t/\hbar} + e^{-iE_2 t/\hbar}\right)\left(e^{iE_1 t/\hbar} + e^{iE_2 t/\hbar}\right) \tag{8.38}$$

$$= \frac{1}{4}\left[2 + e^{i(E_2-E_1)t/\hbar} + e^{-i(E_2-E_1)t/\hbar}\right]. \tag{8.39}$$

Define the energy splitting

$$\Delta E = E_2 - E_1. \tag{8.40}$$

Using $e^{i\delta} + e^{-i\delta} = 2\cos\delta$, we obtain

$$P_+(t) = \frac{1}{2}\left[1 + \cos\left(\frac{\Delta E\, t}{\hbar}\right)\right] \tag{8.41}$$

$$= \cos^2\left(\frac{\Delta E\, t}{2\hbar}\right). \tag{8.42}$$

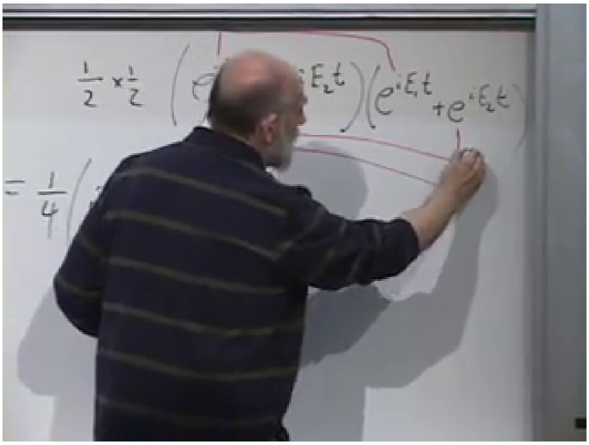

Figure 8.11: The absolute square contains two direct terms and two phase-sensitive cross terms.

Lecture frame: 00:54:54.

The complementary analyzer has

$$P_-(t) = \sin^2\left(\frac{\Delta E\, t}{2\hbar}\right), \qquad P_+(t) + P_-(t) = 1. \tag{8.43}$$

At $t = 0$, two adjacent parallel analyzers transmit with certainty. When $\Delta E\, t/\hbar = \pi$, the state is orthogonal to its initial $+45^\circ$ polarization and passes the -45° analyzer with certainty.

Between those two times the state is generally not linearly polarized. Removing a common phase from Equation (8.31) gives

$$|\psi(t)\rangle \sim \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ e^{-i\Delta E t/\hbar} \end{pmatrix}. \tag{8.44}$$

It begins at $+45^\circ$ linear polarization, becomes circular when the relative phase is $\pm\pi/2$, reaches -45°

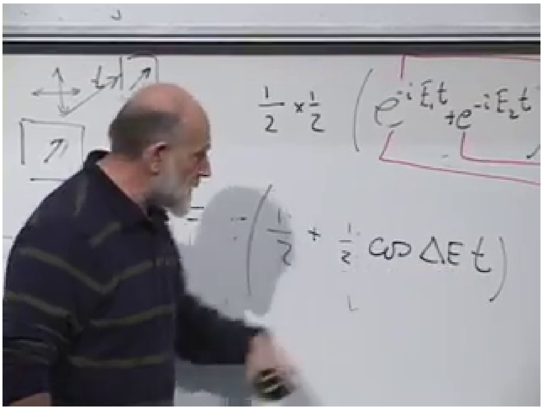

Figure 8.12: The probability for the second parallel analyzer oscillates between one and zero as the relative phase advances.

Lecture frame: 00:58:44.

linear polarization at π, and passes through elliptical states in between. The analyzer oscillation is therefore an interference measurement of a continuously changing polarization state.

8.5.3 Only Energy Differences Are Observable

If a constant C is added to every energy,

$$\hat{H} \longmapsto \hat{H} + C\mathbf{1}, \tag{8.45}$$

then

$$\hat{U}(t) \longmapsto e^{-iCt/\hbar}\hat{U}(t). \tag{8.46}$$

Every state acquires the same overall phase, which changes no transition probability. This is why the interference depends on ΔE, not on the arbitrary zero of energy.

Question from the audience. Must the material fill the entire region between the two polarizers?[a]

Response. The idealized calculation assumes that the photon spends a known time in a uniform anisotropic region. A finite slab works just as well: replace t by the transit time through that slab. Outside it, any common vacuum phase does not alter the polarization probabilities.

[a] Lecture timestamp: 01:00:52–01:01:27.

Question from the audience. The graph uses time on its horizontal axis. How does that become a downstream position in the experiment? [a]

Response. If the photon travels through the material with group velocity v, a slab length L corresponds to an interaction time $t = L/v$. Moving the second analyzer downstream, or changing the material thickness, therefore changes the accumulated relative phase $\Delta EL/(\hbar v)$.

[a] Lecture timestamp: 01:01:29–01:02:43.

Question from the audience. What happens if the incident photon is circularly polarized instead of linearly polarized?[a]

Response. A circular state is an equal-amplitude x/y superposition with relative phase $\pm\pi/2$. The material adds $-\Delta E t/\hbar$ to that phase, so the state moves through elliptical polarizations, becomes linearly polarized at appropriate times, and later reaches the opposite circular handedness. The same interference calculation applies after changing the initial coefficients.

[a]Lecture timestamp: 01:02:44–01:03:40.

Question from the audience. If the labels x and y are arbitrary, why should a different choice change the prediction? [a]

Response. The labels are conventional, but the axes are not. The material has two physical principal axes. We chose coordinates aligned with them so that $\hat{H}$ is diagonal. Rotating only the coordinate labels changes no prediction; rotating the polarizers relative to the material changes the preparation and measurement basis and therefore reveals the relative phase.

[a]Lecture timestamp: 01:03:43–01:05:15.

8.6 Expectation Values and Commutators

Let $\hat{K}$ be a fixed observable with no explicit time dependence. Its expectation value in the evolving

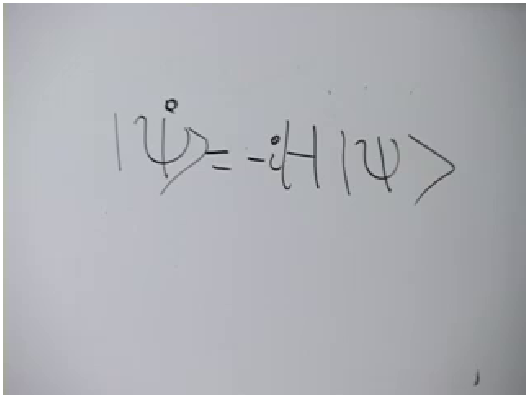

Figure 8.13: The ket Schrödinger equation and the corresponding bra equation carry opposite signs.
Lecture frame: 01:11:45.

state is

$$\left\langle \hat{K} \right\rangle = \langle \psi(t) | \, \hat{K} \, | \psi(t) \rangle \,. \tag{8.47}$$

Only the bra and ket vary. The product rule gives

$$\frac{d}{dt} \left\langle \hat{K} \right\rangle = \left\langle \dot{\psi} \right| \hat{K} \left| \psi \right\rangle + \left\langle \psi \right| \hat{K} \left| \dot{\psi} \right\rangle . \tag{8.48}$$

The ket equation and its adjoint are

$$\left| \dot{\psi} \right\rangle = -\frac{i}{\hbar} \hat{H} \left| \psi \right\rangle , \qquad \left\langle \dot{\psi} \right| = \frac{i}{\hbar} \left\langle \psi \right| \hat{H}, \tag{8.49}$$

where Hermiticity of $\hat{H}$ has been used.

Substitution gives

$$\frac{d}{dt} \left\langle \hat{K} \right\rangle = \frac{i}{\hbar} \left\langle \psi \right| \hat{H} \hat{K} \left| \psi \right\rangle - \frac{i}{\hbar} \left\langle \psi \right| \hat{K} \hat{H} \left| \psi \right\rangle \tag{8.50}$$

$$= \frac{i}{\hbar} \left\langle [\hat{H}, \hat{K}] \right\rangle \tag{8.51}$$

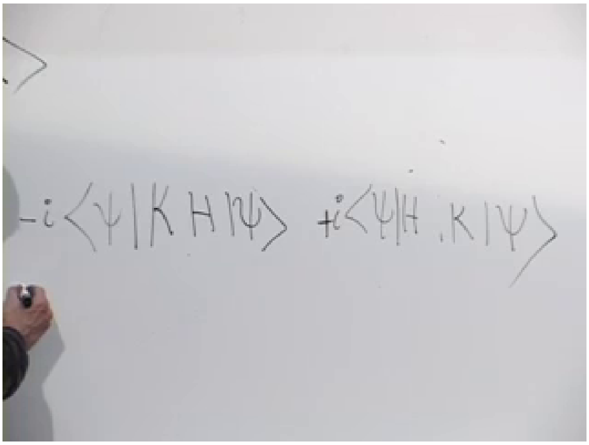

Figure 8.14: Differentiating the expectation-value sandwich produces the two operator orderings $\hat{H}\hat{K}$ and $\hat{K}\hat{H}$.

Lecture frame: 01:13:20.

$$= -\frac{i}{\hbar}\left\langle [\hat{K}, \hat{H}] \right\rangle . \tag{8.52}$$

Question from the audience. Is the result written with $[\hat{K}, \hat{H}]$ or $[\hat{H}, \hat{K}]$, and which sign belongs in front? [a]

Response. Both forms are equivalent because $[\hat{H}, \hat{K}] = -[\hat{K}, \hat{H}]$. With $[\hat{K}, \hat{H}] = \hat{K}\hat{H} - \hat{H}\hat{K}$,

$$\frac{d}{dt}\left\langle \hat{K} \right\rangle = -\frac{i}{\hbar}\left\langle [\hat{K}, \hat{H}] \right\rangle .$$

Writing the commutator in the opposite order changes the prefactor to $+i/\hbar$.

[a] Lecture timestamp: 01:14:46–01:15:18.

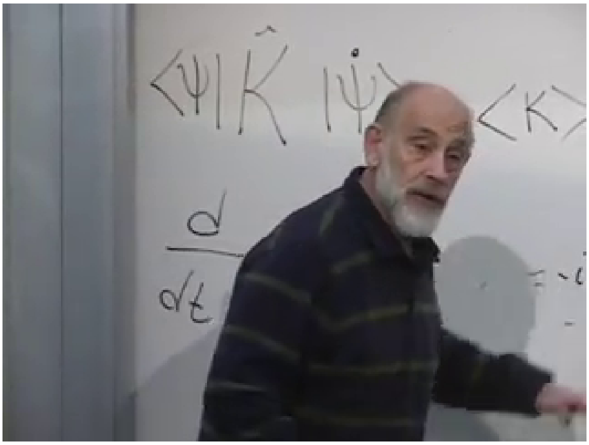

Figure 8.15: The completed commutator form of the expectation-value evolution law.

Lecture frame: 01:16:37.

This is the quantum counterpart of Equation (8.6). In classical mechanics, take a Poisson bracket with H; in quantum mechanics, take a commutator with $\hat{H}$. The precise correspondence is

$$\{F, H\}_{\mathrm{PB}} \quad \longleftrightarrow \quad \frac{1}{i\hbar}[\hat{F}, \hat{H}] \tag{8.53}$$

in the appropriate classical limit.

8.7 Conserved Quantities

If

$$[\hat{K}, \hat{H}] = 0, \tag{8.54}$$

then Equation (8.52) gives

$$\frac{d}{dt}\left\langle \hat{K} \right\rangle = 0 \tag{8.55}$$

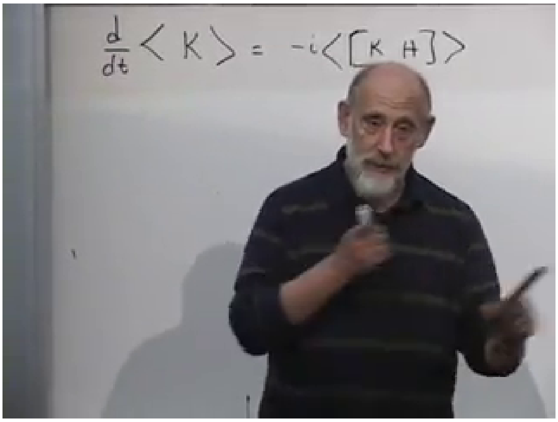

Figure 8.16: A vanishing commutator with the Hamiltonian is the criterion for a time-independent observable distribution.

Lecture frame: 01:18:49.

in every state. More strongly, the spectral projectors of $\hat{K}$ commute with $\hat{H}$, so the full probability distribution of K is time-independent.

The converse requires the same scope: if the expectation value is constant for every state, and $\hat{K}$ has no explicit time dependence, then $[\hat{K}, \hat{H}] = 0$. Constancy in one specially chosen state alone would not be enough.

The Hamiltonian commutes with itself:

$$[\hat{H}, \hat{H}] = 0. \tag{8.56}$$

Therefore a time-independent Hamiltonian is conserved. Since it represents energy, this is energy conservation. The corresponding classical statement is

$$\{H, H\}_{\mathrm{PB}} = 0. \tag{8.57}$$

If the Hamiltonian depends explicitly on time, an additional term $\left\langle \partial \hat{H}/\partial t \right\rangle$ appears and energy need not be conserved. The lecture's argument assumes no such explicit dependence.

8.8 A Free Particle on a Line

8.8.1 From an Abstract Ket to a Wave Function

For a particle on a line, the state may be represented by $\psi(x,t)$. Its position probability density is

$$\rho(x,t) = |\psi(x,t)|^2. \tag{8.58}$$

The Fourier transform $\widetilde{\psi}(p,t)$ gives the momentum representation, with $|\widetilde{\psi}(p,t)|^2$ as the momentum probability density. These are not new kinds of state; they are coordinate representations of the same abstract vector.

Multiplication by x and differentiation with respect to x are linear operators on the vector space of wave functions:

$$\hat{x}\psi = x\psi, \qquad \hat{p}\psi = -i\hbar\frac{\partial \psi}{\partial x}. \tag{8.59}$$

For a force-free particle, classical mechanics suggests

$$H = \frac{p^2}{2m}. \tag{8.60}$$

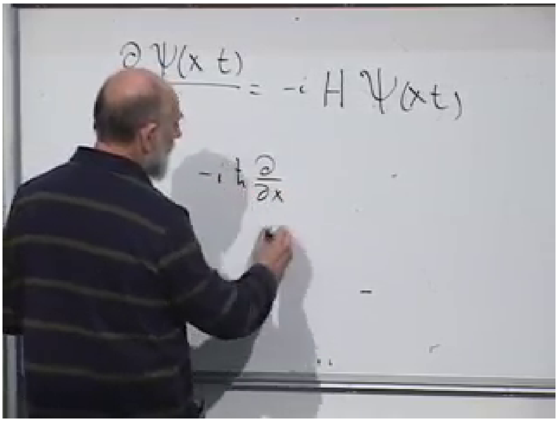

Figure 8.17: The wave-function Schrödinger equation is paired with the differential representation of momentum.

Lecture frame: 01:24:03.

This does not prove the quantum rule, but it gives the correspondence postulate

$$\hat{H} = \frac{\hat{p}^2}{2m}. \tag{8.61}$$

Acting twice with momentum,

$$\hat{p}^2\psi = \left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial\psi}{\partial x}\right) \tag{8.62}$$

$$= -\hbar^2\frac{\partial^2\psi}{\partial x^2}. \tag{8.63}$$

Consequently

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}. \tag{8.64}$$

Question from the audience. Where do the minus sign and the powers of $\hbar$ go in the free-particle equation? [a]

Response. Keep the abstract equation and momentum operator visible at the same time:

$$i\hbar\,\partial_t\psi = \hat{H}\psi, \qquad \hat{H} = \frac{\hat{p}^2}{2m}, \qquad \hat{p} = -i\hbar\,\partial_x.$$

Squaring $\hat{p}$ produces $(-i)^2\hbar^2\partial_x^2 = -\hbar^2\partial_x^2$. No additional factor of i belongs to $\hat{H}$.

[a] Lecture timestamp: 01:27:15–01:29:09.

Substitution into the abstract equation yields Schrödinger's free-particle wave equation:

$$\boxed{i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2}.} \tag{8.65}$$

The Hamiltonian is Hermitian on an appropriate domain because $\hat{p}$ is Hermitian there and the square of a Hermitian operator is Hermitian. Boundary conditions and domains matter in a full treatment; on the line, wave functions are taken to decay sufficiently or are handled with the standard self-adjoint momentum domain.

8.8.2 A Momentum Eigenstate

Look for a solution that remains a momentum eigenstate:

$$\psi_p(x,t) = f(t)e^{ipx/\hbar}. \tag{8.66}$$

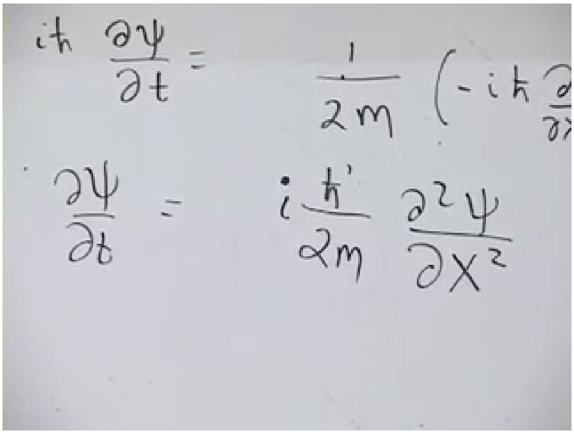

Figure 8.18: The complete free-particle Schrödinger equation with its $\hbar$, i, and minus-sign factors restored.

Lecture frame: 01:30:44.

The number p is the momentum eigenvalue, because

$$-i\hbar\frac{\partial}{\partial x}e^{ipx/\hbar} = p\, e^{ipx/\hbar}. \tag{8.67}$$

The function $f(t)$ is a spectator under spatial differentiation and contains all time dependence.

Substituting Equation (8.66) into Equation (8.65) gives

$$i\hbar\dot{f}(t) = \frac{p^2}{2m}f(t). \tag{8.68}$$

Thus

$$f(t) = f(0)\exp\left(-\frac{i}{\hbar}\frac{p^2}{2m}t\right), \tag{8.69}$$

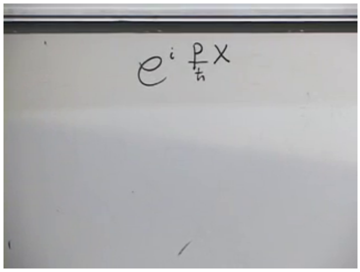

Figure 8.19: The spatial factor $e^{ipx/\hbar}$ is an eigenfunction of the momentum operator.

Lecture frame: 01:32:24.

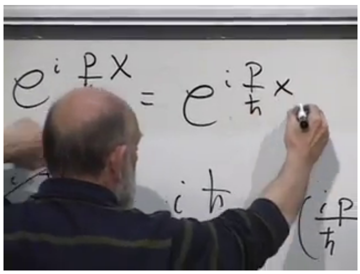

Figure 8.20: The time factor and spatial momentum eigenfunction combine into the free plane wave.

Lecture frame: 01:36:59.

and the full solution is

$$\boxed{\psi_p(x,t) = C \exp\left[\frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)\right].} \qquad (8.70)$$

The energy is $E = p^2/(2m)$, so the familiar phase is $(px - Et)/\hbar$. A single ideal momentum eigenstate changes only by an energy phase and retains definite momentum. Such a plane wave is delta-normalized rather than square-integrable; a physical localized state is a superposition of these waves. Its different momentum components acquire different phases, producing motion and spreading, which is the next step in the analysis.

Finally,

$$[\hat{p}, \hat{H}] = \left[\hat{p}, \frac{\hat{p}^2}{2m}\right] = 0. \tag{8.71}$$

Momentum is therefore conserved for a free particle. The classical Hamiltonian flow, unitary state evolution, polarization interference, commutator conservation law, and free-particle wave equation are all different expressions of the same organizing principle: the Hamiltonian generates change with time.

CHAPTER 9

FREE PARTICLES, FOURIER EVOLUTION, AND THE CLASSICAL LIMIT

9.1 When Classical Repair Was No Longer Enough

There are two ways to respond when an old theory fails. One can try to repair the mechanism, adding enough machinery to preserve the familiar picture, or one can search for a principle that changes the picture itself. The Michelson–Morley experiment encouraged both reactions. Elaborate mechanical models of the ether could be made to imitate the observed result, but Einstein's relativity began from a sharper principle and made the old mechanism unnecessary. Quantum theory followed a similarly reluctant path from repair to reconstruction.

For the present discussion, the decisive classical failure is blackbody radiation. Electromagnetic radiation in a cavity can be decomposed into normal modes. Each mode behaves mathematically like a harmonic oscillator, and classical equipartition assigns to it an average thermal energy of order

$$E_{\text{mode}} = kT. \tag{9.1}$$

The cavity contains infinitely many modes, extending to arbitrarily high frequency. If every one carries

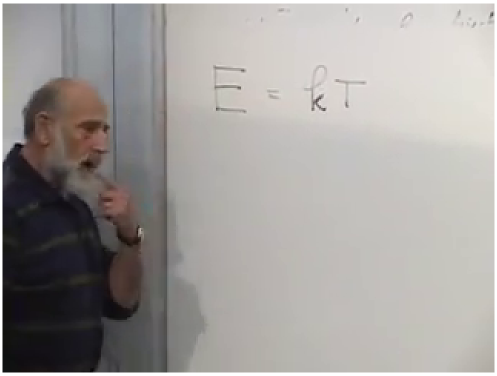

Figure 9.1: The classical equipartition assignment for each electromagnetic normal mode of a thermal cavity.

Lecture frame: 00:03:05.

the nonzero energy in Equation (9.1), their sum diverges. The prediction is not merely somewhat inaccurate in the ultraviolet; it is infinite. This is the ultraviolet catastrophe.

Planck found the rule that cuts off the divergence:

$$E_n = nh\nu, \qquad n = 0, 1, 2, \ldots. \tag{9.2}$$

At first, however, the rule was attached to oscillators in the cavity walls. The radiation itself was still imagined as an ordinary classical wave. Einstein took the more radical step and attributed the quanta to the radiation. If $h\nu \gg kT$, thermal agitation ordinarily cannot supply even one quantum at that frequency. A classical fraction of a quantum is not available; the most probable occupation is zero. High-frequency modes are therefore suppressed.

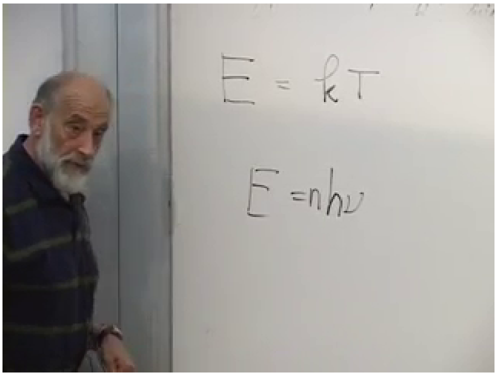

Figure 9.2: Planck's discrete energy rule, whose interpretation Einstein extended from material emitters to radiation itself.

Lecture frame: 00:05:15.

This did not make photons immediately welcome. A wave theory had explained interference and diffraction too well to be discarded casually. The right conclusion was not that waves were simply false, but that the classical wave description was incomplete. Compton scattering eventually made the particle-like transfer of energy and momentum especially difficult to evade.

Question from the audience. Why was the photoelectric effect not already decisive evidence for photons? [a]

Response. It was important evidence, and Einstein's explanation was one of the earliest direct uses of light quanta. Historically, however, it did not force everyone to abandon a classical field interacting with quantized matter. Compton scattering displayed photon-like energy and momentum conservation so transparently that the remaining classical interpretations became much harder to sustain.

[a] Lecture timestamp: 00:10:09–00:10:19.

Question from the audience. Where does Bohr's 1913 atomic model belong in this transition? [a]

Response. It is one of several sharp breaks rather than the final theory. Bohr imposed discrete stationary orbits and quantum jumps on a largely classical orbital picture. The complete conceptual transition came later, especially through Heisenberg's matrix mechanics, Schrödinger's wave mechanics, and Dirac's unifying operator language.

[a] Lecture timestamp: 00:10:20–00:11:13.

Einstein had also recognized very early that atomic and radioactive events carry an irreducibly statistical appearance. An excited atom can remain undecayed for an unpredictable interval and then emit spontaneously. A radioactive nucleus behaves similarly.

One can predict distributions and lifetimes with great precision without predicting the moment of an individual event.

The old orbital models tried to preserve familiar mechanics by adding special rules: only certain orbits were allowed, and radiation somehow occurred only in jumps between them. Such models could organize data, but they did not provide a consistent mechanics. Heisenberg's decisive move was to stop describing unobservable electron orbits and build the theory from observable transition quantities.

Question from the audience. Why did the old orbit models persist if they were already in tension with classical mechanics? [a]

Response. They retained a familiar picture and reproduced important spectral regularities. That made them useful transitional models. Their special rules were not a coherent dynamics, however, and increasingly complicated atoms exposed the weakness. Heisenberg broke with the orbit picture rather than adding another repair.

[a] Lecture timestamp: 00:12:50–00:13:22.

Question from the audience. If every observation of a photon uses a quantized charged particle, could quantizing only the matter be enough? [a]

Response. Not in a consistent interaction theory. A quantum system entangles with whatever it exchanges energy and momentum with. Leaving the electromagnetic field classical while matter is quantum can imitate limited experiments, but it cannot consistently describe emission, absorption, fluctuations, and photon statistics together. Both sides of the interaction must ultimately be treated quantum mechanically.

[a] Lecture timestamp: 00:14:06–00:14:55.

Question from the audience. Why did Einstein later resist quantum mechanics after recognizing spontaneous probability so early? [a]

Response. He recognized the empirical need for probabilistic transition laws, but he remained dissatisfied with probability as the final description of an individual physical system. The issue was not whether the statistical predictions worked; it was what they meant and whether a deeper state of affairs had been omitted. His later objections were attempts to show that the formalism, though successful, might not be complete.

[a] Lecture timestamp: 00:14:59–00:17:08.

Question from the audience. Could probability merely reflect ignorance, as it does when a classical coin is tossed? [a]

Response. That possibility was precisely the source of the dispute. Quantum mechanics assigns probabilities even when the state has been specified as completely as the theory allows. The formalism predicts not only uncertainty but interference between alternatives. Treating those probabilities as ordinary ignorance does not reproduce the full structure without adding a much more subtle underlying theory.

[a] Lecture timestamp: 00:17:09–00:18:19.

The historical prelude fixes the direction of the argument that follows. Quantum mechanics is the primary framework. Classical mechanics is not used to derive it; at most, classical expressions suggest a Hamiltonian whose quantum consequences we then work out. Finally we must recover classical motion as an approximation to the quantum theory.

9.2 The Free Hamiltonian as a Quantum Guess

The abstract Schrödinger equation is

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}\left|\psi(t)\right\rangle. \tag{9.3}$$

In the position representation, momentum is

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}. \tag{9.4}$$

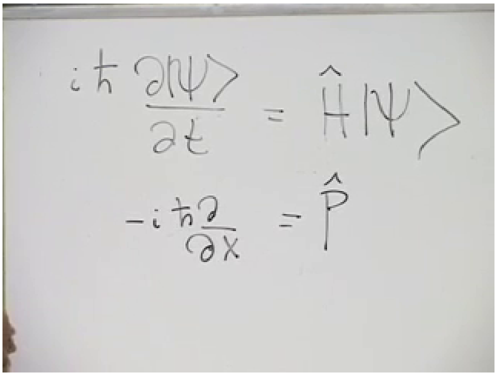

Figure 9.3: Time and space derivatives paired with the Hamiltonian and momentum operators.
Lecture frame: 00:21:40.

The two equations have a revealing parallel. A derivative with respect to time is generated by the Hamiltonian; a derivative with respect to position is generated by momentum. The sign in Equation (9.4) is conventional, but once chosen it must be used consistently.

For a nonrelativistic free particle on a line, we guess

$$\boxed{\hat{H} = \frac{\hat{p}^2}{2m}.} \tag{9.5}$$

This is a correspondence-guided guess, not a derivation of quantum mechanics from classical mechanics. Free space has no preferred direction, so the energy should be unchanged under $p \mapsto -p$. The simplest even function with the familiar low-speed behavior is $p^2/(2m)$. The constant m earns its interpretation as inertial mass through the resulting relation between momentum and velocity.

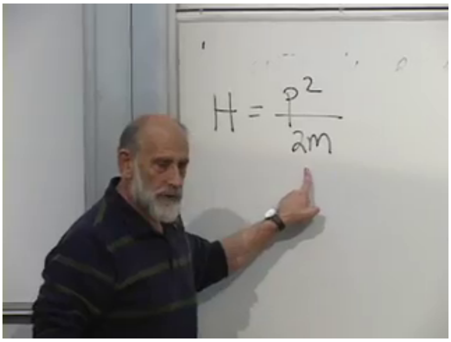

Figure 9.4: The nonrelativistic free-particle Hamiltonian chosen as an even function of momentum.

Lecture frame: 00:24:00.

The square of an operator means composition: $\hat{p}^2\psi$ is $\hat{p}(\hat{p}\psi)$. Writing $\psi(x,t) = \langle x \,|\, \psi(t)\rangle$, the probability of finding the particle in an interval dx at a specified time is

$$\Pr(x \in [x, x+dx]; t) = |\psi(x,t)|^2 \, dx. \tag{9.6}$$

Position labels the possible outcomes of an observable. Time labels the evolution of the state; in this elementary formulation it is not another coordinate whose value is measured with probability $|\psi|^2$.

Substituting Equations (9.4) and (9.5) into Equation (9.3) gives

$$i\hbar\frac{\partial \psi}{\partial t} = \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)^2 \psi \tag{9.7}$$

$$= -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2}. \tag{9.8}$$

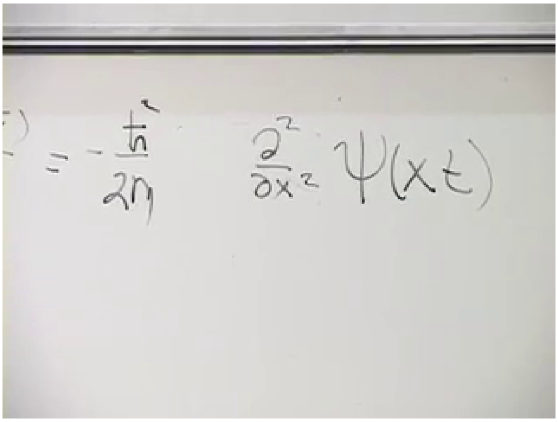

Figure 9.5: The free Schrödinger equation obtained by applying the momentum operator twice.

Lecture frame: 00:28:02.

Thus the free-particle equation is

$$\boxed{i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2}.} \tag{9.9}$$

9.3 Plane Waves and the Two Directions of Momentum

We first seek a solution with definite momentum. Following the sign convention used at the board, take

$$\psi(x,t) = e^{ipx/\hbar}e^{i\omega t}. \tag{9.10}$$

Question from the audience. Should the frequency ω also appear in the spatial exponent? [a]

Response. No. The spatial exponent is fixed by the momentum eigenvalue: $-i\hbar\partial_x e^{ipx/\hbar} = pe^{ipx/\hbar}$. The independent time factor is then determined by the Schrödinger equation. If one writes the wave as $e^{i(kx-\omega t)}$, the relation is $p = \hbar k$ and $E = \hbar\omega$.

[a] Lecture timestamp: 00:29:05–00:29:20.

The time derivative of Equation (9.10) is $\partial_t\psi = i\omega\psi$, while

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{p^2}{\hbar^2}\psi. \tag{9.11}$$

The free Schrödinger equation therefore requires

$$-\hbar\omega = \frac{p^2}{2m}, \qquad \omega = -\frac{p^2}{2m\hbar}. \tag{9.12}$$

The negative value is a consequence of having written the time factor as $e^{+i\omega t}$. In the conventional positive-frequency notation,

$$\boxed{\psi_p(x,t) = \exp\left[\frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)\right],} \tag{9.13}$$

so $E = p^2/(2m)$ and the time factor is $e^{-iEt/\hbar}$.

The same energy belongs to two momentum eigenstates:

$$p_E = \pm\sqrt{2mE}. \tag{9.14}$$

The right-moving and left-moving waves differ in their spatial phases but share the same time phase.

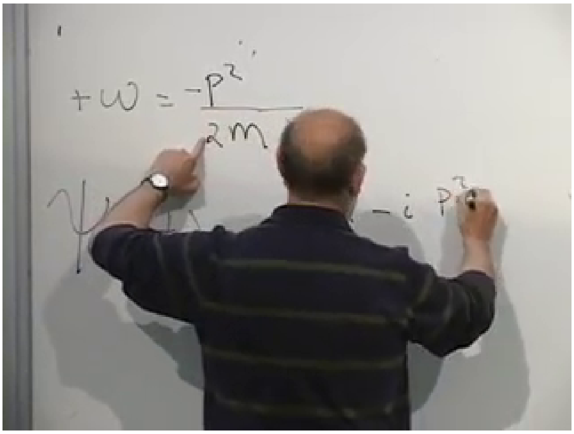

Figure 9.6: The plane-wave substitution fixes the time frequency through $E = p^2/(2m)$.

Lecture frame: 00:34:45.

Any superposition

$$\psi(x,t) = \alpha\psi_p(x,t) + \beta\psi_{-p}(x,t) \tag{9.15}$$

therefore remains in the same energy eigenspace. In a two-state, box-normalized version of the problem, the probabilities for momenta $\pm p$ are $|\alpha|^2$ and $|\beta|^2$, while the probability of the common energy is their sum.

Ideal plane waves are not square-integrable on the whole line. They are generalized, delta-normalized eigenstates. Normalizable particle states are wave packets assembled from them.

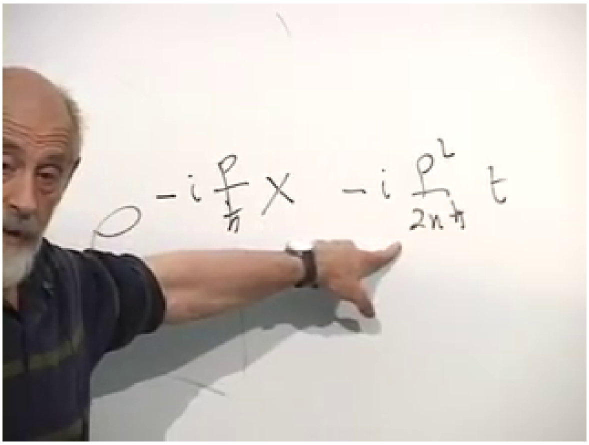

Figure 9.7: The $+p$ and $-p$ plane waves have a common energy because the Hamiltonian depends on p^2.

Lecture frame: 00:37:40.

9.4 The General Free State in Momentum Space

Linearity now gives the general free evolution as a Fourier superposition:

$$\boxed{\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp\, e^{ipx/\hbar} e^{-ip^2 t/(2m\hbar)} \widetilde{\psi}(p).} \tag{9.16}$$

The coefficient $\widetilde{\psi}(p)$ is the momentum-space amplitude at $t = 0$. Every value of momentum evolves independently by the phase dictated by its energy.

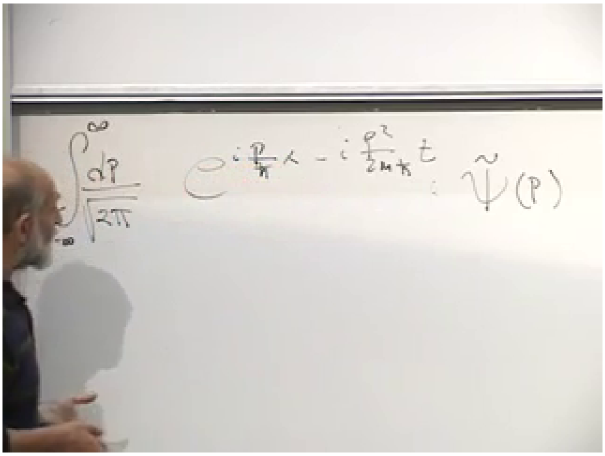

Figure 9.8: The general free state as a continuous superposition of momentum eigenwaves.

Lecture frame: 00:41:10.

Question from the audience. In the integral, is p a particular measured momentum or only a dummy variable? [a]

Response. Inside the integral it is a dummy variable ranging over all real momenta. A particular outcome may instead be denoted q. Its probability density is $|\widetilde{\psi}(q)|^2$, independent of what symbol was used for the integration variable.

[a] Lecture timestamp: 00:41:55–00:42:13.

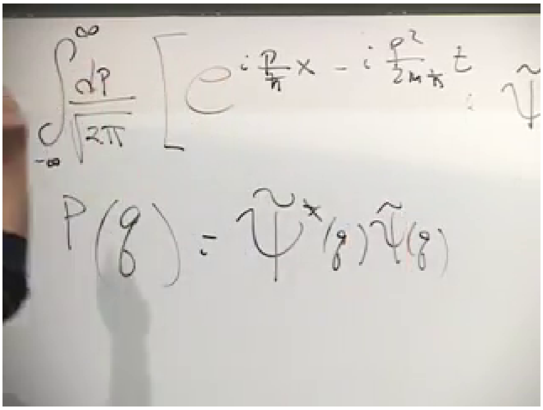

Figure 9.9: The momentum-space Born rule: $P(q) = \widetilde{\psi}^*(q)\widetilde{\psi}(q)$.

Lecture frame: 00:44:24.

Question from the audience. What is held fixed while the momentum integral is performed? [a]

Response. The spacetime point x, t is fixed. We sum the complex contribution made at that point by every momentum component. Constants such as m and $\hbar$ are fixed as well; only the integration variable runs.

[a] Lecture timestamp: 00:42:14–00:42:57.

With the normalization of Equation (9.16),

$$\Pr(p \in [q, q + dq]) = |\widetilde{\psi}(q)|^2 \, dq. \tag{9.17}$$

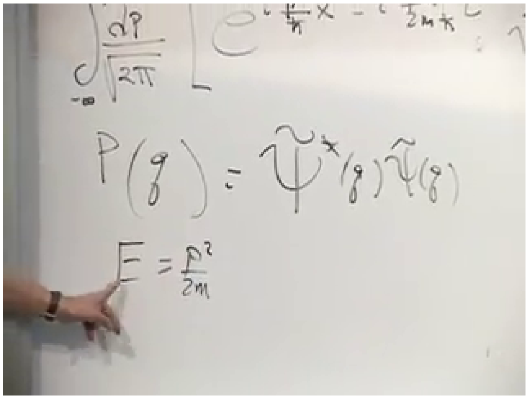

Figure 9.10: For a given positive energy, the free Hamiltonian admits the two momentum roots $p = \pm\sqrt{2mE}$.

Lecture frame: 00:45:44.

Question from the audience. Does the symbol q in $P(q)$ denote a position? [a]

Response. No. Here q is merely a fresh name for a possible momentum outcome. Position is still denoted x. The change of letter prevents confusion with the dummy momentum p already used inside the integral.

[a]Lecture timestamp: 00:44:26–00:45:05.

Energy is a coarser label than momentum because the map

$$E = \frac{p^2}{2m} \tag{9.18}$$

identifies p and $-p$.

For discrete or box-normalized momentum states, the weight assigned to energy E is the sum of the

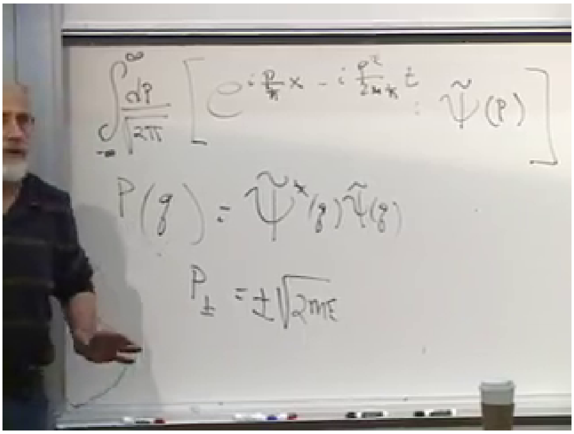

Figure 9.11: The board collects both momentum amplitudes associated with one free-particle energy.
Lecture frame: 00:48:20.

weights at its two roots. For a continuum *density* in energy, the change of measure must also be included:

$$p_E = \sqrt{2mE}, \tag{9.19}$$

$$\rho_E(E) = \sum_{p_i:\, p_i^2/(2m)=E} \frac{|\tilde{\psi}(p_i)|^2}{|dE/dp|_{p_i}} \tag{9.20}$$

$$= \frac{m}{p_E}\left[|\tilde{\psi}(p_E)|^2 + |\tilde{\psi}(-p_E)|^2\right], \qquad E > 0. \tag{9.21}$$

The lecture's sum over the two momentum probabilities expresses the essential degeneracy. Equation (9.21) adds the Jacobian needed when probability is quoted per unit energy rather than for an energy bin.

Question from the audience. Should $\hbar$ appear in the energy relation $E = p^2/(2m)$? [a]

Response. No. Momentum and energy retain the familiar nonrelativistic relation. Planck's constant converts those quantities into spatial and temporal phase rates: $k = p/\hbar$ and $\omega = E/\hbar$. It belongs in the wave, not as an extra factor in the dispersion relation.

[a]Lecture timestamp: 00:47:58–00:48:24.

The physically admissible free states are precisely the suitable superpositions of these modes. For ordinary one-particle wave packets, $\widetilde{\psi}$ belongs to $L^2(\mathbb{R})$, so the norm is finite. Solutions that grow without bound and cannot represent finite probability are not admitted as normalized states, although generalized eigenfunctions remain useful as a basis.

Question from the audience. Does this claim of generality assume that every solution is separable into a function of x times a function of t? [a]

Response. No. Individual energy–momentum eigenmodes are separable, but a general wave packet is their superposition and is usually not separable. Completeness of the Fourier modes, not separability of the final state, is what makes the integral general.

[a]Lecture timestamp: 00:49:12–00:50:09.

Question from the audience. May $\widetilde{\psi}(p)$ contain an arbitrary momentum-dependent phase? [a]

Response. Yes, and it matters. Its magnitude fixes the momentum probability density, but its phase controls interference among momentum components in position space. Two states with the same $|\widetilde{\psi}(p)|^2$ can therefore have very different spatial wave packets.

[a] Lecture timestamp: 00:50:09–00:50:51.

9.5 Initial Data and Dispersive Evolution

At $t = 0$, the time phase in Equation (9.16) disappears:

$$\psi(x,0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp\, e^{ipx/\hbar} \widetilde{\psi}(p). \tag{9.22}$$

Fourier inversion then gives

$$\boxed{\widetilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\, e^{-ipx/\hbar} \psi(x,0).} \tag{9.23}$$

The factors of $\hbar$ have been kept explicit so that $px/\hbar$ is dimensionless and the two representations have compatible normalization.

The initial-value problem is now constructive:

1. begin with $\psi(x,0)$;
2. Fourier transform it to obtain $\widetilde{\psi}(p,0)$;
3. multiply each momentum component by its free-evolution phase;

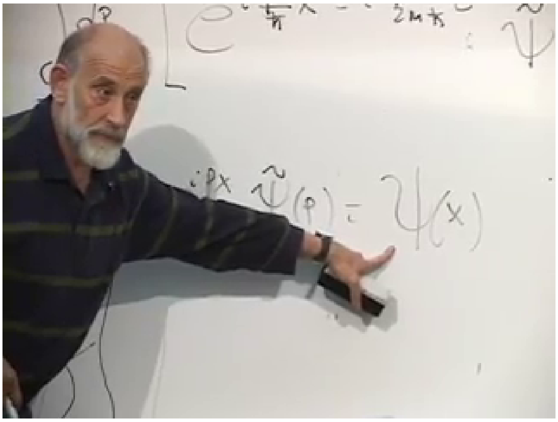

Figure 9.12: Setting $t = 0$ identifies the momentum superposition with the initial position-space wave function.

Lecture frame: 00:52:02.

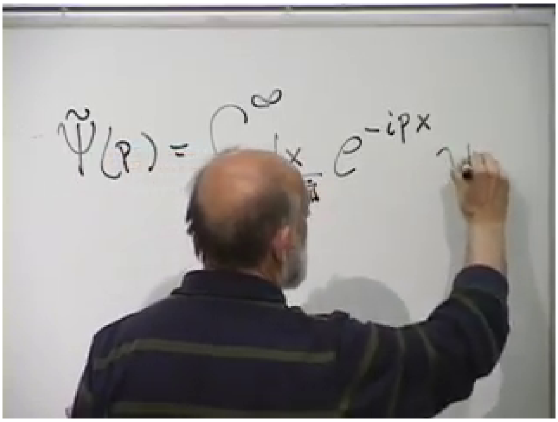

Figure 9.13: The inverse Fourier transform recovers the momentum amplitude from the initial wave function.

Lecture frame: 00:53:05.

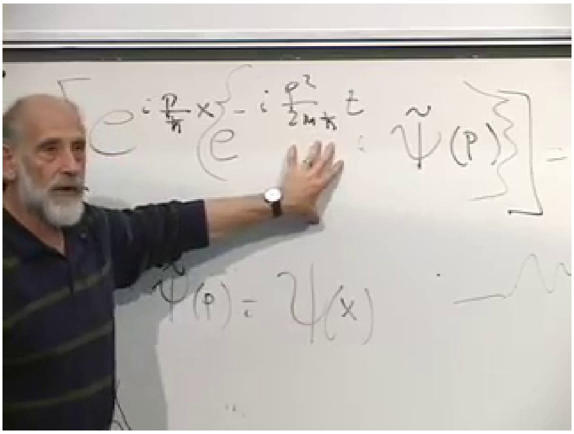

Figure 9.14: Free evolution in momentum space consists of an energy-dependent phase for each Fourier component.

Lecture frame: 00:57:06.

4. inverse transform to recover $\psi(x,t)$.

Equivalently,

$$\tilde{\psi}(p,t) = e^{-ip^2 t/(2m\hbar)}\tilde{\psi}(p,0). \tag{9.24}$$

Question from the audience. Can the time dependence simply be included in a time-dependent momentum wave function? [a]

Response. Yes. Equation (9.24) is exactly that description. Its magnitude remains fixed for a free particle, so momentum probabilities are conserved, while its phase changes at a rate set by $p^2/(2m\hbar)$.

[a] Lecture timestamp: 00:56:32–00:57:34.

Because the phase is quadratic in p, different components slip relative to one another. A localized packet generally spreads. With $p = \hbar k$ and the conventional positive frequency

$$\omega(k) = \frac{\hbar k^2}{2m}, \tag{9.25}$$

the phase velocity and group velocity are

$$v_{\mathrm{ph}} = \frac{\omega}{k} = \frac{p}{2m}, \qquad v_{\mathrm{g}} = \frac{d\omega}{dk} = \frac{p}{m}. \tag{9.26}$$

The group velocity agrees with the classical velocity associated with the central momentum of a narrow packet. The variation of v_{g} over the packet's momentum width is what produces dispersion.

Question from the audience. Is this the distinction between phase velocity and group velocity? [a]

Response. Yes. The crests of an individual Fourier component move with v_{ph}, while the envelope of a narrow packet moves with v_{g}. For the nonrelativistic quadratic dispersion relation they are not equal, and components with different momenta have different group velocities, so a general packet spreads.

[a] Lecture timestamp: 00:57:35–00:58:17.

Question from the audience. Can the same free-particle equation be used for a photon? [a]

Response. No. Equation (9.5) is the nonrelativistic massive-particle Hamiltonian. A photon obeys $E = c|p|$, not $E = p^2/(2m)$, and has no rest-mass parameter of this kind. Its propagation and polarization require a relativistic field description.

[a] Lecture timestamp: 01:00:05–01:00:55.

9.6 Commutators and Poisson Brackets

We now step back from the explicit free solution. For an operator $\hat{K}$ that may itself depend on time,

$$\frac{d}{dt}\langle K\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{K}]\rangle + \left\langle\frac{\partial\hat{K}}{\partial t}\right\rangle. \tag{9.27}$$

The lecture considers operators with no explicit time dependence, so the last term vanishes. The remaining formula follows by differentiating $\langle\psi|\,\hat{K}\,|\psi\rangle$ and applying the Schrödinger equation to both the ket and the bra.

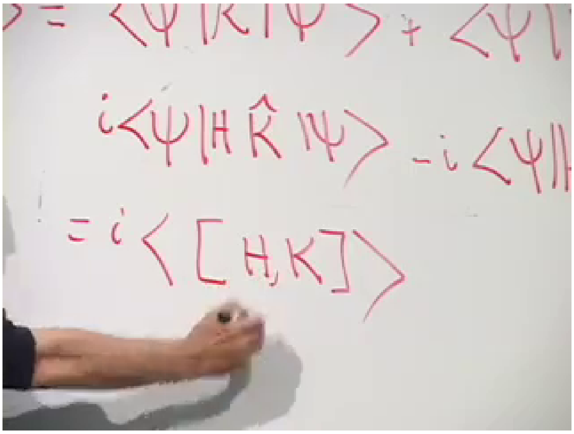

Figure 9.15: The two differentiated state vectors combine into the expectation value of a commutator.
Lecture frame: 01:04:50.

Question from the audience. Does the sign depend on whether the commutator is written as $[\hat{H}, \hat{K}]$ or $[\hat{K}, \hat{H}]$? [a]

Response. Yes, but the two standard forms are identical:

$$\frac{d}{dt}\langle K\rangle = \frac{i}{\hbar}\langle[\hat{H}, \hat{K}]\rangle = \frac{1}{i\hbar}\langle[\hat{K}, \hat{H}]\rangle.$$

Reversing the order changes the sign of the commutator, and $1/i = -i$ changes it back.

[a] Lecture timestamp: 01:03:11–01:05:17.

The classical law for a phase-space function $K(x, p)$ is

$$\dot{K} = \{K, H\}_{\text{PB}}, \tag{9.28}$$

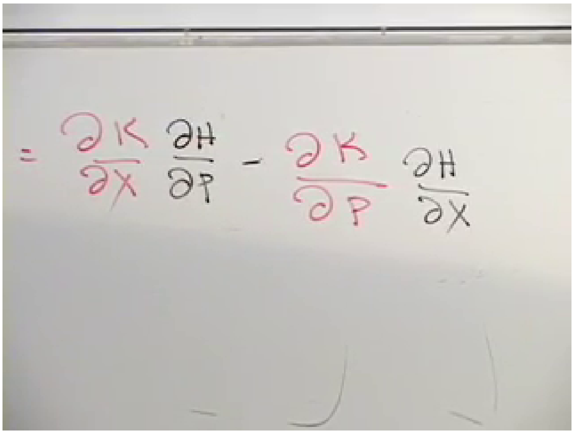

Figure 9.16: Hamilton's equations lead directly to $\dot{K} = \{K, H\}_{\text{PB}}$.

Lecture frame: 01:09:31.

where

$$\{A, B\}_{\text{PB}} = \frac{\partial A}{\partial x}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial x}. \tag{9.29}$$

The order in Equation (9.28) follows directly from the chain rule and Hamilton's equations:

$$\dot{K} = \frac{\partial K}{\partial x}\dot{x} + \frac{\partial K}{\partial p}\dot{p} \tag{9.30}$$

$$= \frac{\partial K}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial K}{\partial p}\frac{\partial H}{\partial x}. \tag{9.31}$$

The elementary classical identity

$$\{f(x), p\}_{\text{PB}} = f'(x) \tag{9.32}$$

has an exact quantum counterpart. Act with the

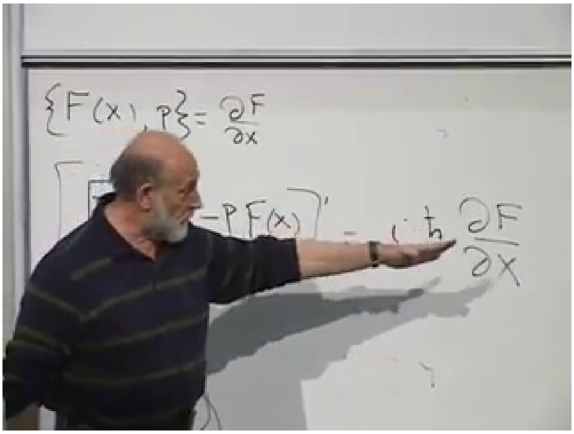

Figure 9.17: Applying the commutator to an arbitrary wave function exposes the derivative $f'(\hat{x})$.
Lecture frame: 01:19:12.

commutator on an arbitrary test wave function:

$$[f(\hat{x}), \hat{p}]\psi = f(x)\left(-i\hbar\frac{d\psi}{dx}\right) + i\hbar\frac{d}{dx}\big(f(x)\psi(x)\big) \tag{9.33}$$

$$= i\hbar f'(x)\psi(x). \tag{9.34}$$

Because this holds for every wave function in the common operator domain,

$$\boxed{[f(\hat{x}), \hat{p}] = i\hbar f'(\hat{x}).} \tag{9.35}$$

In the momentum representation the reciprocal identity is

$$[\hat{x}, g(\hat{p})] = i\hbar g'(\hat{p}), \tag{9.36}$$

and the special case $f(x) = x$ yields

$$[\hat{x}, \hat{p}] = i\hbar. \tag{9.37}$$

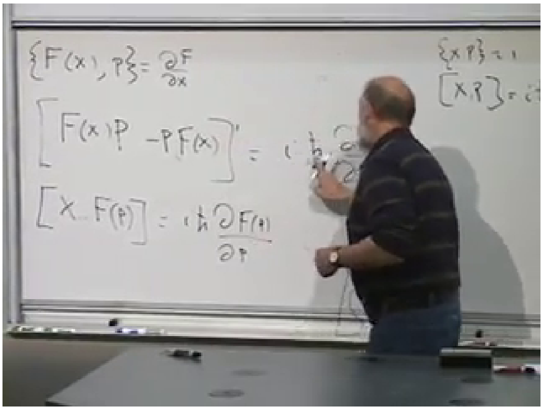

Figure 9.18: The canonical commutator and Poisson bracket display the basic quantum–classical correspondence.

Lecture frame: 01:21:52.

These examples motivate the correspondence

$$[\hat{A}, \hat{B}] \longleftrightarrow i\hbar\,\widehat{\{A, B\}}_{\mathrm{PB}}. \tag{9.38}$$

Both brackets are antisymmetric,

$$[A, B] = -[B, A], \qquad \{A, B\}_{\mathrm{PB}} = -\{B, A\}_{\mathrm{PB}}, \tag{9.39}$$

and both act as derivations on products:

$$[AB, C] = A[B, C] + [A, C]B, \tag{9.40}$$

$$\{AB, C\}_{\mathrm{PB}} = A\{B, C\}_{\mathrm{PB}} + \{A, C\}_{\mathrm{PB}}B. \tag{9.41}$$

For classical functions the order of the ordinary factors is immaterial. For operators it is not, which is why the placement of B in Equation (9.40) must be preserved.

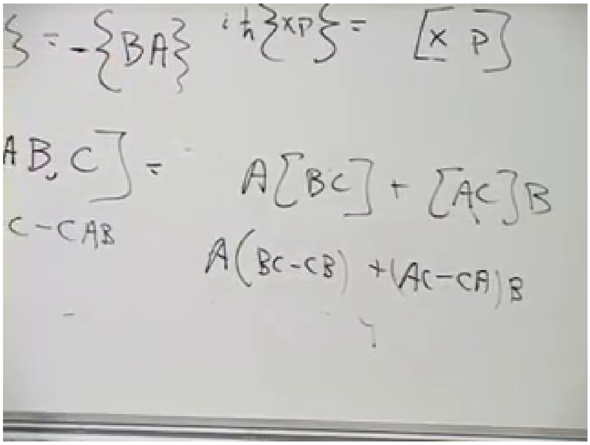

Figure 9.19: The product rules for Poisson brackets and commutators, with operator order retained in the quantum identity.

Lecture frame: 01:29:55.

Question from the audience. In the commutator product rule, should the second term be $B[A,C]$ or $[A,C]B$? [a]

Response. It is $[A,C]B$. Expanding $ABC - CAB$ and adding and subtracting ACB gives

$$ABC - ACB + ACB - CAB = A[B,C] + [A,C]B.$$

The classroom correction matters because the factors need not commute.

[a] Lecture timestamp: 01:29:31–01:30:55.

The correspondence in Equation (9.38) is exact for the canonical elementary identities just derived, but it is not a universal replacement rule for arbitrary functions. A classical monomial such as x^3p has several inequivalent operator orderings. Symmetrization often provides a natural choice, yet higher-order $\hbar$ corrections remain in general. The classical bracket is the leading correspondence limit, not a license to reverse or rearrange arbitrary operator strings.

9.7 A Potential and Ehrenfest's Theorem

Add a position-dependent potential:

$$\hat{H} = \frac{\hat{p}^2}{2m} + U(\hat{x}). \tag{9.42}$$

In position space, $U(\hat{x})$ acts by multiplication, and the Schrödinger equation becomes

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U(x)\right]\psi(x,t). \tag{9.43}$$

The associated force is

$$F(x) = -U'(x). \tag{9.44}$$

Question from the audience. Does the momentum operator itself now depend on time? [a]

Response. In the Schrödinger picture, $\hat{p} = -i\hbar\partial_x$ has no explicit time dependence. The state changes, and therefore its momentum distribution and expectation value can change. One may instead transfer time dependence to operators in the Heisenberg picture, but that is a different representation of the same physics.

[a] Lecture timestamp: 01:35:01–01:35:40.

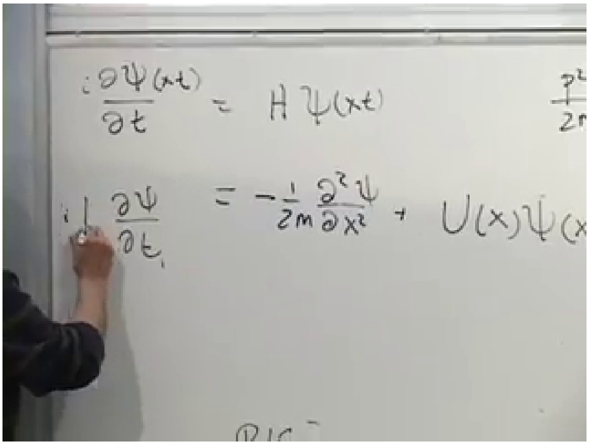

Figure 9.20: The Schrödinger equation for kinetic energy plus a position-dependent potential.

Lecture frame: 01:34:48.

We now ask how the center of a wave packet moves. First set $\hat{K} = \hat{x}$ in Equation (9.27). Since $[U(\hat{x}), \hat{x}] = 0$,

$$\frac{d}{dt}\langle x\rangle = \frac{i}{\hbar}\left\langle \left[\frac{\hat{p}^2}{2m}, \hat{x}\right]\right\rangle . \tag{9.45}$$

The product rule and $[\hat{p}, \hat{x}] = -i\hbar$ give

$$\begin{aligned} [\hat{p}^2, \hat{x}] &= \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p} && (9.46)\\ &= -2i\hbar\hat{p}. && (9.47) \end{aligned}$$

Consequently,

$$\boxed{\frac{d}{dt}\langle x\rangle = \frac{\langle p\rangle}{m}.} \tag{9.48}$$

Next choose $\hat{K} = \hat{p}$. The kinetic term commutes with momentum, while Equation (9.35) gives $[U(\hat{x}), \hat{p}] =$

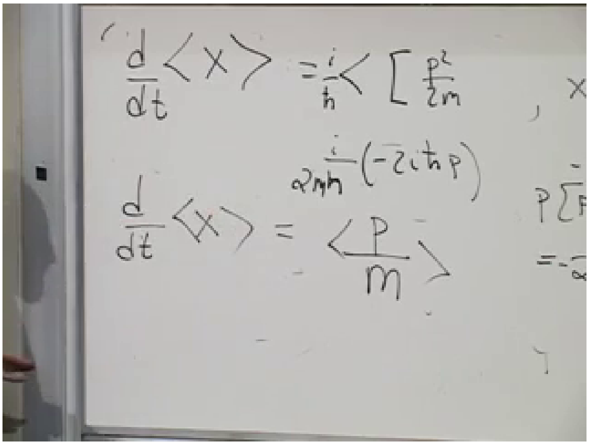

Figure 9.21: The first Ehrenfest equation: average velocity equals average momentum divided by mass.
Lecture frame: 01:42:18.

$i\hbar U'(\hat{x})$. Therefore

$$\frac{d}{dt}\langle p\rangle = \frac{i}{\hbar}\,\langle[U(\hat{x}),\hat{p}]\rangle \tag{9.49}$$

$$= -\langle U'(\hat{x})\rangle = \langle F(\hat{x})\rangle. \tag{9.50}$$

Combining the two exact identities,

$$m\frac{d^2}{dt^2}\langle x\rangle = -\langle U'(\hat{x})\rangle. \tag{9.51}$$

This resembles Newton's equation, but an important distinction remains:

$$\langle U'(\hat{x})\rangle \neq U'(\langle x\rangle) \quad \text{in general.} \tag{9.52}$$

If the packet is narrow compared with the length scale over which the force changes, then

$$\langle U'(\hat{x})\rangle \approx U'(\langle x\rangle), \tag{9.53}$$

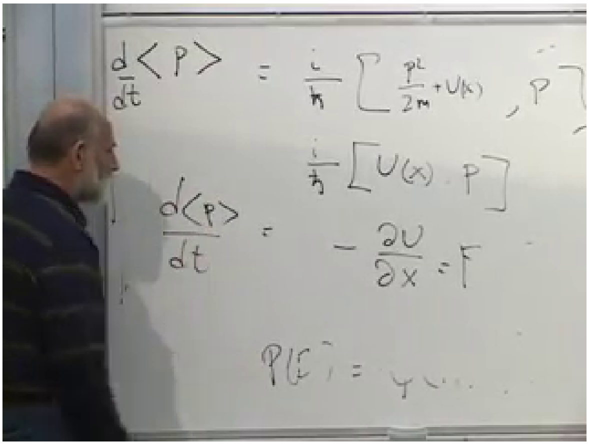

Figure 9.22: The second Ehrenfest equation: the mean momentum changes by the expectation value of the force.

Lecture frame: 01:45:39.

and

$$m\frac{d^2}{dt^2}\langle x\rangle \approx F(\langle x\rangle). \tag{9.54}$$

For potentials of at most quadratic degree, $U'(x)$ is at most linear, so the replacement in Equation (9.53) is exact for the first moments even when the packet is broad. For a general nonlinear potential, packet width and higher moments matter.

Question from the audience. Are all the apparent Poisson-bracket and commutator analogies exact, and is $\langle F \rangle$ the quantum version of the resultant force? [a]

Response. The elementary canonical relations and the two Ehrenfest equations above are exact. A general replacement of every Poisson bracket by a commutator is only a correspondence rule because operator ordering can matter. In one dimension, $\langle F(\hat{x}) \rangle$ is the probability weighted average force acting across the packet. It equals the force at the average position only under the additional conditions just stated.

[a] Lecture timestamp: 01:46:36–01:47:35.

This is the promised order of explanation. Quantum mechanics supplies the fundamental evolution. Classical motion emerges when the observables of interest vary little across the state, so that averages close approximately on themselves. We do not first assume a classical particle and then decorate it with a wave; we derive the motion of a classical-looking center from the quantum wave packet.

9.8 What Has Been Established

The historical crisis taught us not to confuse a successful approximation with a fundamental mechanism. With that lesson in place, the free Hamiltonian $\hat{p}^2/(2m)$ led to the Schrödinger equation, plane waves, and the dispersion relation $E = p^2/(2m)$. Their Fourier superpositions gave every normalizable free state and a direct procedure for evolving arbitrary initial data.

The second half of the lecture exposed the algebra beneath the quantum–classical relation. Commutators govern expectation values in the same structural position that Poisson brackets govern classical observables. For a particle in a potential, that parallel produces the exact Ehrenfest equations

$$\frac{d}{dt}\langle x\rangle = \frac{\langle p\rangle}{m}, \qquad \frac{d}{dt}\langle p\rangle = -\langle U'(\hat{x})\rangle. \tag{9.55}$$

Only after these quantum statements are established do we take the narrow-packet, smooth-force limit and recover Newton's equation for the packet's center.

CHAPTER 10

THE QUANTUM HARMONIC OSCILLATOR AND ITS LADDER OF STATES

10.1 The Universal Model Near Stable Equilibrium

This is the last lecture in the present sequence, though not the last word on quantum mechanics. Angular momentum is the other natural closing subject, but the harmonic oscillator fills the available time and angular momentum will have to wait for the later discussion of quantum fields.

The oscillator deserves that time because it is not merely one special spring problem. Whenever a system has a stable equilibrium and is displaced slightly from it, the first nontrivial approximation to its motion is usually harmonic. A weight suspended from a spring is the obvious model. Molecular vibrations, sound modes, electromagnetic modes, and small disturbances of many more complicated systems share the same mathematics.

Consider the hanging mass. Gravity pulls downward with an approximately constant force, while the spring force grows with extension. At one position the two balance. If x measures displacement from that equilibrium rather than from the spring's unstretched length, gravity disappears from the

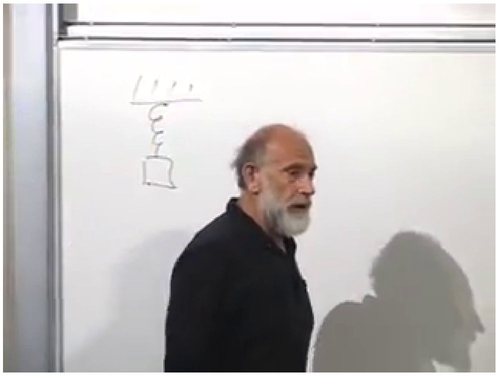

Figure 10.1: A suspended mass and spring provide the concrete oscillator model; the coordinate is measured from the force-balanced equilibrium.

Lecture frame: 00:02:02.

equation of small motion and the restoring force is

$$F(x) = -kx. \tag{10.1}$$

The minus sign says that either displacement produces a force back toward $x = 0$.

The same conclusion follows for a general potential $V(q)$ with a nondegenerate minimum at $q = q_0$. Put $x = q - q_0$. Taylor expansion gives

$$V(q_0 + x) = V(q_0) + \underbrace{V'(q_0)}_{0}\, x + \frac{1}{2}V''(q_0)x^2 + O(x^3). \tag{10.2}$$

The constant does not affect the force, the linear term vanishes at the minimum, and the positive quadratic coefficient is the spring constant: $k = V''(q_0) > 0$. Harmonic motion is therefore the local language of stable equilibrium.

10.2 Classical Preparation

Before quantizing, let us put the classical system into a convenient form. For mass m and angular frequency $\omega = \sqrt{k/m}$,

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2. \tag{10.3}$$

The rescaling $x = \sqrt{m}\,q$ absorbs the mass:

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \tag{10.4}$$

We shall use this unit-mass coordinate through most of the lecture and restore m whenever its physical role matters.

The Euler–Lagrange equation gives

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}, \tag{10.5}$$

or

$$\boxed{\ddot{x} = -\omega^2 x.} \tag{10.6}$$

Its general real solution is

$$x(t) = A\cos\omega t + B\sin\omega t. \tag{10.7}$$

Twice differentiating either sinusoid returns the function multiplied by $-\omega^2$, so the frequency in the potential is exactly the frequency of the motion.

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x}, \tag{10.8}$$

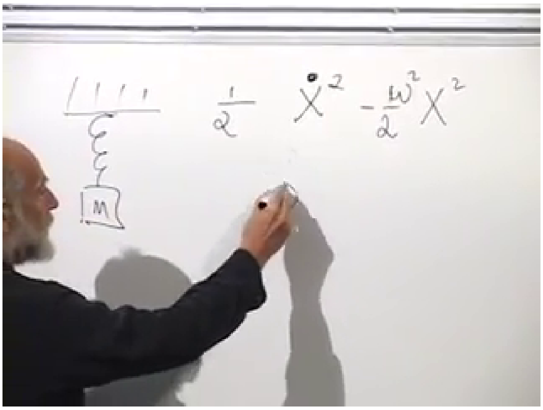

Figure 10.2: The spring model and its unit-mass Lagrangian, with a quadratic potential characterized by ω^2.

Lecture frame: 00:10:57.

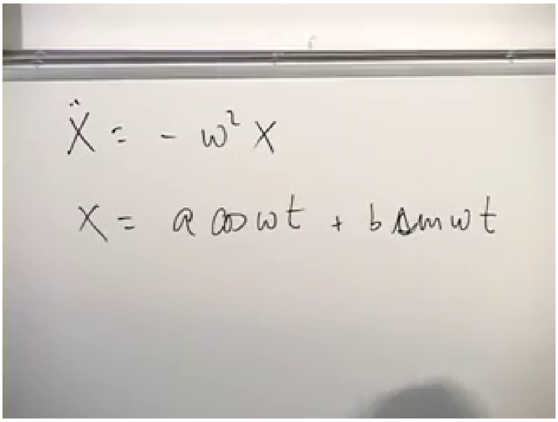

Figure 10.3: The restoring-force equation and its sinusoidal classical solution.

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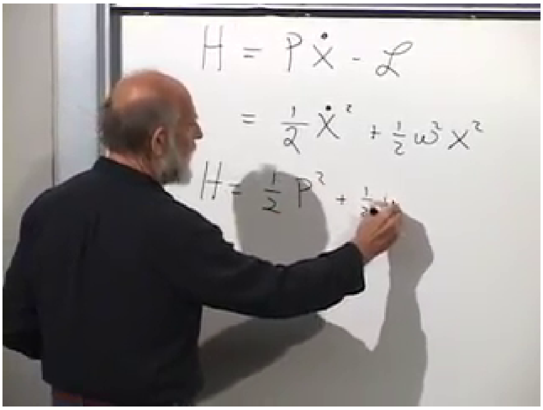

Figure 10.4: The Legendre transform rewrites the oscillator energy in the canonical variables x and p.
Lecture frame: 00:17:18.

and the Legendre transform produces

$$H = p\dot{x} - L \tag{10.9}$$

$$= \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \tag{10.10}$$

The Lagrangian is kinetic minus potential energy; the Hamiltonian is their sum.

10.3 Quantization and the Two Jobs of the Hamiltonian

The quantum state space consists of square-integrable wave functions $\psi(x)$. At a specified time,

$$|\psi(x)|^2\,dx \tag{10.11}$$

is the probability of finding the oscillator coordinate in an interval dx, and normalization requires

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1. \tag{10.12}$$

In the position representation,

$$(\hat{x}\psi)(x) = x\psi(x), \qquad (\hat{p}\psi)(x) = -i\hbar\frac{\partial\psi}{\partial x}. \tag{10.13}$$

Writing $\hat{x}\,|\psi\rangle$ and writing multiplication by x are not two different operations; the second is the position-space representation of the first. Similarly, $\hat{p}^{\,2}$ means applying the momentum operator twice.

Promoting the classical variables in Equation (10.10) to operators gives

$$\hat{H} = \frac{1}{2}\hat{p}^{\,2} + \frac{1}{2}\omega^2\hat{x}^{\,2}. \tag{10.14}$$

Its action on a wave function is

$$(\hat{H}\psi)(x) = -\frac{\hbar^2}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}\omega^2 x^2\psi(x). \tag{10.15}$$

The Hamiltonian has two closely related jobs. First, it generates time evolution:

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2}\frac{\partial^2}{\partial x^2} + \frac{1}{2}\omega^2 x^2\right]\psi(x,t). \tag{10.16}$$

Even if $\psi(x,0)$ happens to be real, the factor i generally produces an imaginary component immediately. A nonstationary wave function is therefore intrinsically complex.

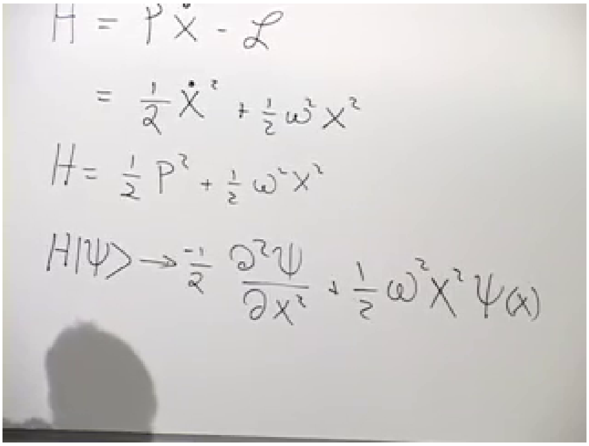

Figure 10.5: The classical Hamiltonian and its action as a differential operator on the wave function.

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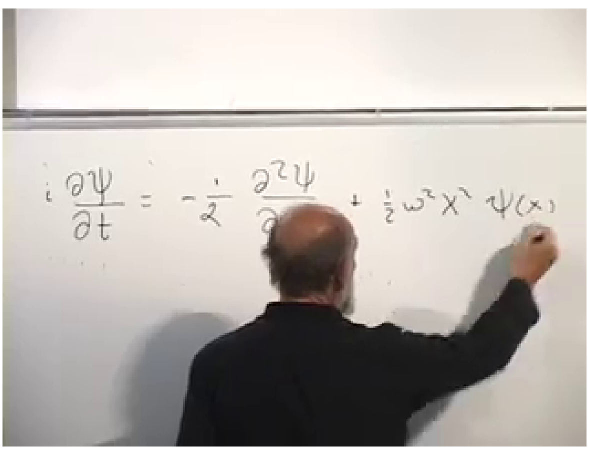

Figure 10.6: The time-dependent Schrödinger equation for the oscillator.

Lecture frame: 00:24:24.

Question from the audience. Why are the derivatives partial derivatives rather than ordinary derivatives? [a]

Response. The wave function depends independently on x and t. The derivative $\partial_t\psi$ changes time while holding position fixed, whereas $\partial_x\psi$ changes position while holding time fixed. Time labels evolution; it does not have the same operator status as position in this formulation.

[a] Lecture timestamp: 00:21:24–00:24:07.

Given an initial wave function, one could integrate Equation (10.16) forward in small time steps. The second job of $\hat{H}$, however, gives a more revealing route. Solve the stationary eigenvalue problem

$$\hat{H}\psi_E = E\psi_E, \tag{10.17}$$

or explicitly,

$$-\frac{\hbar^2}{2}\frac{d^2\psi_E}{dx^2} + \frac{1}{2}\omega^2 x^2\psi_E = E\psi_E. \tag{10.18}$$

Once the eigenfunctions are known, arbitrary initial data may be expanded as

$$\psi(x,0) = \sum_n c_n\psi_n(x), \tag{10.19}$$

and its time evolution is immediate:

$$\psi(x,t) = \sum_n c_n e^{-iE_nt/\hbar}\psi_n(x). \tag{10.20}$$

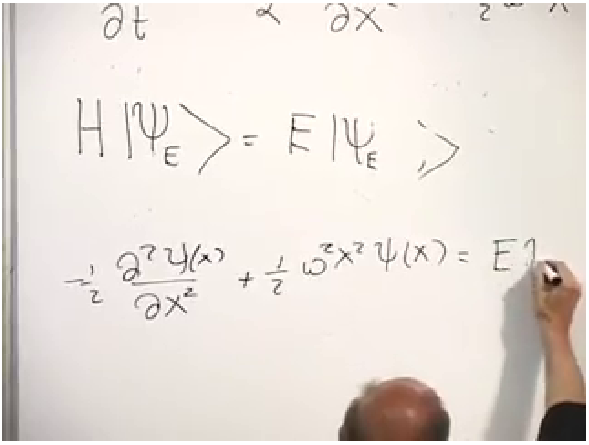

Figure 10.7: The stationary Schrödinger equation simultaneously determines the oscillator eigenfunctions and their allowed energies.

Lecture frame: 00:29:24.

10.4 Why the Bound-State Energies Are Discrete

Equation (10.18) is second order, so for a fixed E it has two linearly independent local solutions. Far from the origin, the quadratic potential dominates. The two asymptotic behaviors are of the form

$$\psi(x) \sim \exp\left(\pm\frac{\omega x^2}{2\hbar}\right). \tag{10.21}$$

One decays and one grows. For a generic trial energy, a solution adjusted to decay at $+\infty$ contains a growing component at $-\infty$, or vice versa. Only selected energies allow the growing component to vanish at both ends.

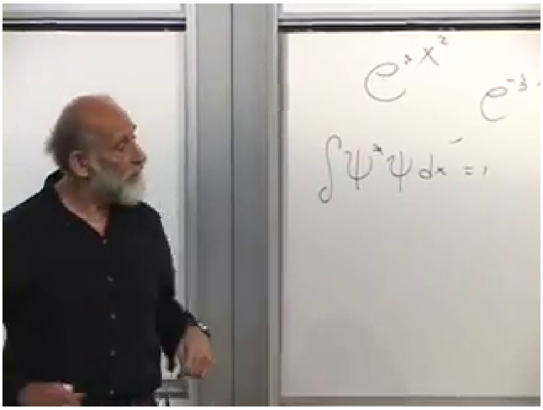

Figure 10.8: Normalizability rejects the exponentially growing asymptotic branch and selects isolated energies that decay at both ends.

Lecture frame: 00:34:20.

This is the origin of the discrete bound-state spectrum. The differential equation exists for arbitrary E, but a physical state must belong to the Hilbert space. A function that grows exponentially cannot satisfy Equation (10.12); in the oscillator potential it also has divergent potential-energy expectation.

There is a second way to anticipate the lowest energy. Classically one minimizes

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2 \qquad (10.22)$$

by setting $x = p = 0$. Quantum mechanically,

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}, \qquad (10.23)$$

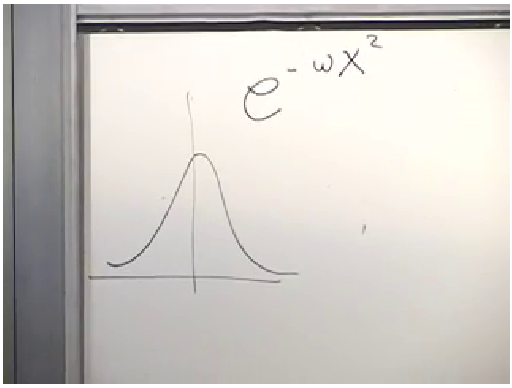

Figure 10.9: The Gaussian trial state is concentrated near the stable equilibrium and decays in both directions.

Lecture frame: 00:38:18.

so no normalized state can make both contributions vanish. The ground-state energy must be positive.

10.5 Finding the Gaussian Ground State

The asymptotic form suggests a Gaussian:

$$\psi_0(x) = C \exp\left(-\frac{\omega x^2}{2\hbar}\right), \tag{10.24}$$

where $\omega > 0$ denotes the positive oscillator frequency. Its first two derivatives are

$$\psi_0' = -\frac{\omega x}{\hbar}\psi_0, \tag{10.25}$$

$$\psi_0'' = \left(\frac{\omega^2 x^2}{\hbar^2} - \frac{\omega}{\hbar}\right)\psi_0. \tag{10.26}$$

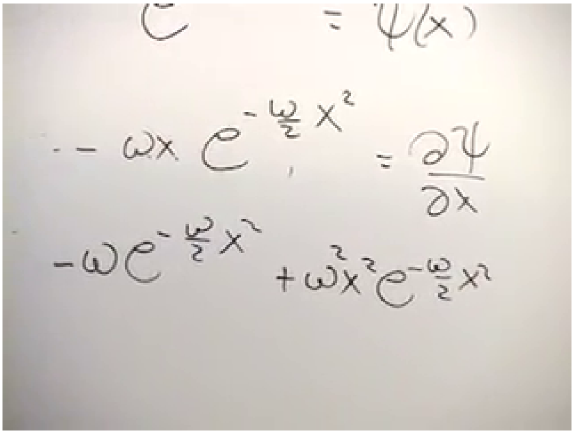

Figure 10.10: Differentiating the Gaussian twice produces a constant term and a quadratic term.
Lecture frame: 00:41:05.

Applying the Hamiltonian,

$$\hat{H}\psi_0 = -\frac{\hbar^2}{2}\left(\frac{\omega^2 x^2}{\hbar^2} - \frac{\omega}{\hbar}\right)\psi_0 + \frac{1}{2}\omega^2 x^2 \psi_0 \tag{10.27}$$

$$= \frac{1}{2}\hbar\omega\,\psi_0. \tag{10.28}$$

The two terms proportional to x^2 cancel, leaving

$$\boxed{E_0 = \frac{1}{2}\hbar\omega.} \tag{10.29}$$

Normalization fixes

$$C = \left(\frac{\omega}{\pi\hbar}\right)^{1/4}, \tag{10.30}$$

up to an irrelevant overall phase. For a physical coordinate q with mass m, the replacements $x =$

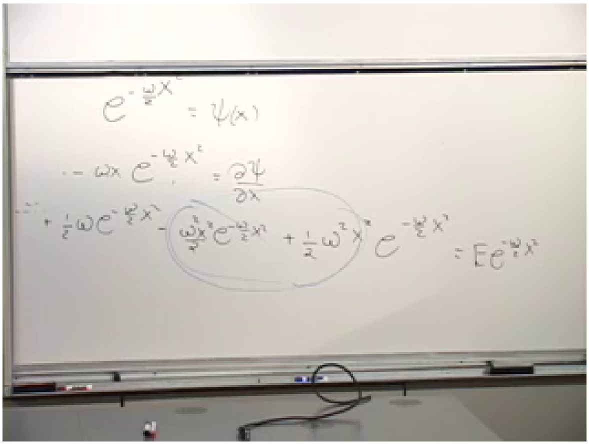

Figure 10.11: The Gaussian eigenfunction calculation leaves the zero-point energy after the quadratic terms cancel.

Lecture frame: 00:44:34.

$\sqrt{m}\, q$ give

$$\psi_0(q) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega q^2}{2\hbar}\right). \tag{10.31}$$

Question from the audience. Does the uncertainty principle mean that the oscillator can never have zero energy? [a]

Response. Yes. A zero-energy state would require both positive terms $\langle p^2\rangle/2$ and $\omega^2\langle x^2\rangle/2$ to vanish. That would make both position and momentum sharp at zero, contradicting Equation (10.23). The lowest allowed value is $\hbar\omega/2$.

[a] Lecture timestamp: 00:45:23–00:46:13.

Question from the audience. Why not take $\psi(x) = 1$ as a zero-energy state? [a]

Response. A constant wave function is a formal zero-momentum eigenfunction, so the kinetic term vanishes, but it is neither normalizable on the line nor localized at $x = 0$. The potential term produces $\frac{1}{2}\omega^2 x^2 \psi$, which is not zero and is not proportional to the constant state. It therefore does not solve the oscillator eigenvalue equation.

[a] Lecture timestamp: 00:46:13–00:47:49.

Question from the audience. For $E = 0$, can some linear combination of the two second-order solutions remain bounded at both ends? [a]

Response. No. Initial data can be tuned so the solution decays in one direction, but at $E = 0$ it grows in the other. The exceptional tuning that removes the growing branch at both ends occurs only at the discrete eigenvalues, the lowest of which is $\hbar\omega/2$.

[a] Lecture timestamp: 00:48:02–00:50:27.

Question from the audience. Why must the ω in the Gaussian exponent be the positive root? [a]

Response. The classical equation contains only ω^2, so its sign is irrelevant there. In the wave function, $\exp[-\omega x^2/(2\hbar)]$ decays only for $\omega > 0$. Choosing the positive frequency is therefore the normalizable branch.

[a] Lecture timestamp: 00:50:33–00:51:15.

Restoring $\hbar$ is essential. Frequency has units of inverse time, while energy requires $\hbar\omega$. For a macroscopic spring the spacing is usually far too small to notice, and the zero-point energy disappears relative to ordinary classical scales as $\hbar \to 0$.

10.6 Factorizing the Hamiltonian

The Gaussian gives one eigenstate. To find all of them, factor the sum of squares. Define the unnormalized operators

$$\hat{b}_- = \hat{p} - i\omega\hat{x}, \qquad \hat{b}_+ = \hat{p} + i\omega\hat{x} = \hat{b}_-^\dagger. \tag{10.32}$$

Classically, $(p + i\omega x)(p - i\omega x) = p^2 + \omega^2 x^2$. For noncommuting quantum operators, the chosen order contributes an additional term:

$$\begin{aligned} \hat{b}_+\hat{b}_- &= (\hat{p} + i\omega\hat{x})(\hat{p} - i\omega\hat{x}) && (10.33)\\ &= \hat{p}^2 + \omega^2\hat{x}^2 + i\omega[\hat{x}, \hat{p}] && (10.34)\\ &= \hat{p}^2 + \omega^2\hat{x}^2 - \hbar\omega. && (10.35) \end{aligned}$$

Hence

$$\boxed{\hat{H} = \frac{1}{2}\hat{b}_+\hat{b}_- + \frac{1}{2}\hbar\omega.} \tag{10.36}$$

The first-order operator $\hat{b}_-$ annihilates the Gaussian:

$$\begin{aligned} \hat{b}_-\psi_0 &= \left(-i\hbar\frac{d}{dx} - i\omega x\right)\psi_0 && (10.37)\\ &= (i\omega x - i\omega x)\,\psi_0 = 0. && (10.38) \end{aligned}$$

Therefore the product $\hat{b}_+\hat{b}_-$ contributes zero on the ground state, and Equation (10.36) immediately returns $E_0 = \hbar\omega/2$.

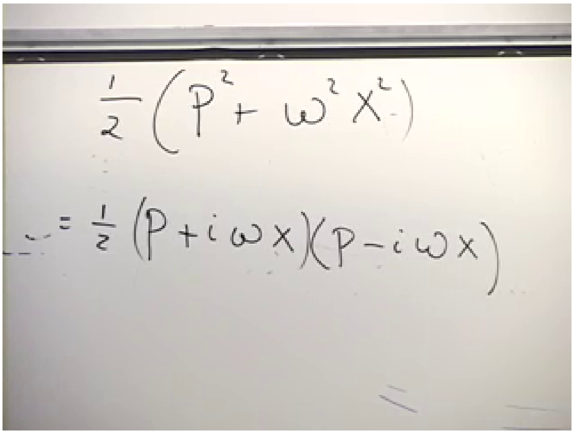

Figure 10.12: The classical-looking factorization must retain operator order in quantum mechanics.

Lecture frame: 00:55:24.

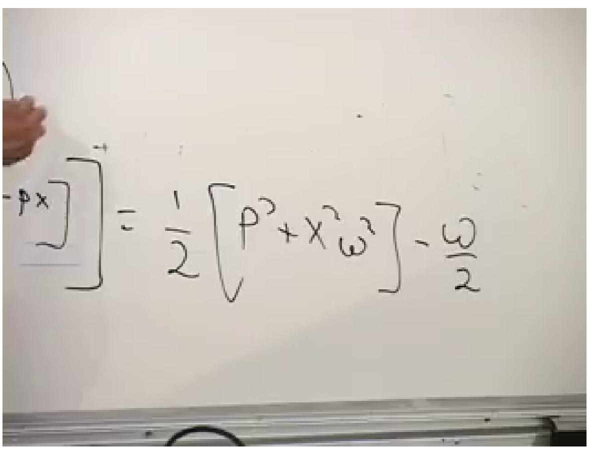

Figure 10.13: The noncommuting cross terms generate the additive zero-point contribution in the factorized Hamiltonian.

Lecture frame: 00:59:31.

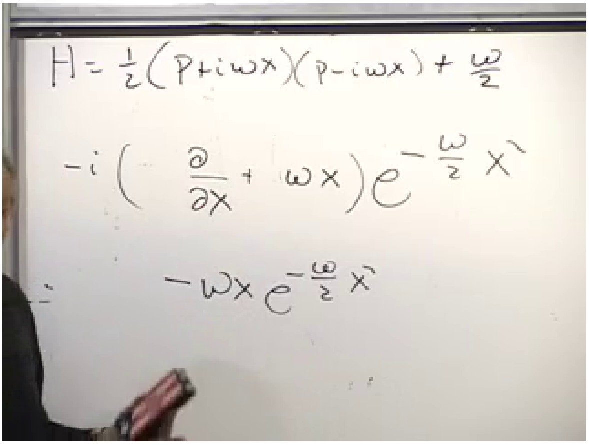

Figure 10.14: The lower first-order factor annihilates the Gaussian ground state.

Lecture frame: 01:03:56.

Normalize the operators:

$$\hat{a} = \frac{\hat{b}_-}{\sqrt{2\hbar\omega}}, \qquad \hat{a}^\dagger = \frac{\hat{b}_+}{\sqrt{2\hbar\omega}}. \tag{10.39}$$

Since

$$[\hat{b}_-, \hat{b}_+] = 2\hbar\omega, \tag{10.40}$$

we obtain

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \tag{10.41}$$

Equivalently, $[\hat{a}^\dagger, \hat{a}] = -1$. This resolves the sign uncertainty that arises if the order of the two operators is changed while they are being named.

The Hamiltonian is now

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).} \tag{10.42}$$

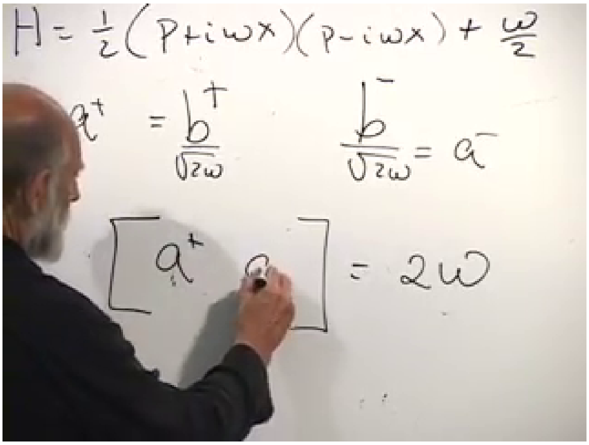

Figure 10.15: Rescaling the first-order factors produces dimensionless ladder operators with a unit commutator.

Lecture frame: 01:08:22.

The operator

$$\hat{N} = \hat{a}^\dagger \hat{a} \tag{10.43}$$

is positive:

$$\langle\psi|\,\hat{N}\,|\psi\rangle = \|\hat{a}\,|\psi\rangle\,\|^2 \geq 0. \tag{10.44}$$

This factorization proves, rather than merely guesses, that no state lies below $\hbar\omega/2$.

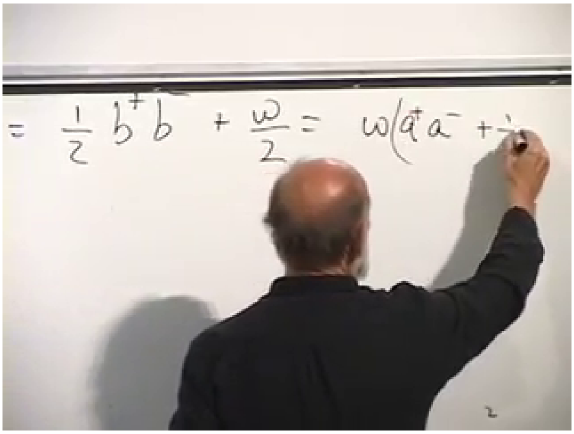

Figure 10.16: The oscillator Hamiltonian reduced to the number operator plus the zero-point half.

Lecture frame: 01:09:37.

Question from the audience. Which commutator has sign +1, and which operator really annihilates the ground state? [a]

Response. With the consistent definitions in Equation (10.39), $\hat{a} \propto \hat{p} - i\omega\hat{x}$ annihilates the Gaussian, $[\hat{a}, \hat{a}^\dagger] = 1$, and therefore $[\hat{a}^\dagger, \hat{a}] = -1$. The extended board discussion mixed the naming order, not the physics. Testing the operator directly on ψ_0 fixes the convention unambiguously.

[a]Lecture timestamp: 01:11:05–01:14:09.

Writing the ground state as $|0\rangle$,

$$\hat{a}\,|0\rangle = 0, \qquad \hat{N}\,|0\rangle = 0, \qquad \hat{H}\,|0\rangle = \frac{1}{2}\hbar\omega\,|0\rangle\,. \tag{10.45}$$

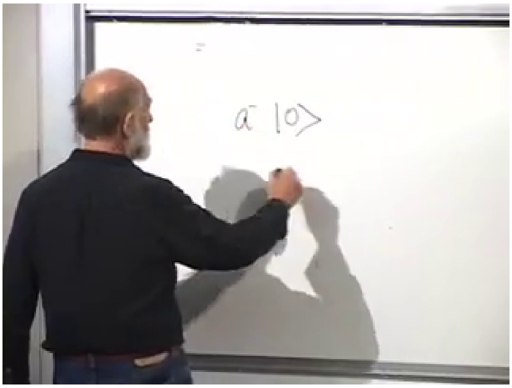

Figure 10.17: The algebraic definition of the ground state: $\hat{a}\,|0\rangle = 0$.

Lecture frame: 01:14:47.

10.7 Climbing the Energy Ladder

Suppose

$$\hat{N}\,|n\rangle = n\,|n\rangle\,. \tag{10.46}$$

Use the commutator to move $\hat{N}$ past the raising operator:

$$\begin{aligned}\hat{N}\hat{a}^\dagger &= \hat{a}^\dagger\hat{a}\hat{a}^\dagger && (10.47)\\ &= \hat{a}^\dagger\left(\hat{a}^\dagger\hat{a}+1\right) && (10.48)\\ &= \hat{a}^\dagger(\hat{N}+1). && (10.49)\end{aligned}$$

Acting on $|n\rangle$,

$$\hat{N}(\hat{a}^\dagger\,|n\rangle) = (n+1)\hat{a}^\dagger\,|n\rangle\,. \tag{10.50}$$

Thus $\hat{a}^\dagger\,|n\rangle$ is an eigenstate with number eigenvalue $n+1$, unless it vanishes.

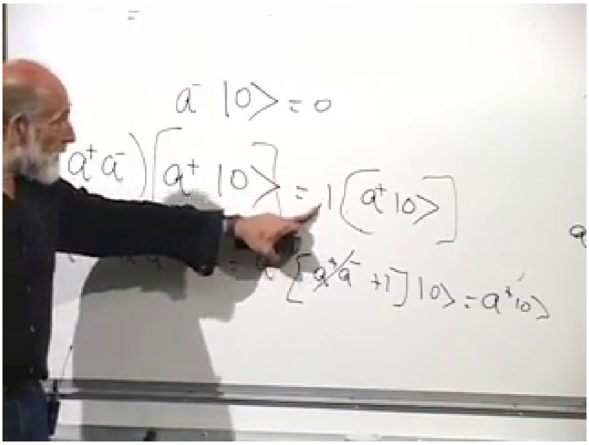

Figure 10.18: Commuting the number operator through $\hat{a}^\dagger$ raises its eigenvalue by one.

Lecture frame: 01:19:29.

The first excited state is proportional to $\hat{a}^\dagger |0\rangle$ and has energy

$$E_1 = \hbar\omega \left(1 + \frac{1}{2}\right) = \frac{3}{2}\hbar\omega. \qquad (10.51)$$

Repeating the argument generates

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle , \qquad n = 0, 1, 2, \ldots , \qquad (10.52)$$

with the normalized ladder actions

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1}\, |n+1\rangle , \qquad \hat{a}\, |n\rangle = \sqrt{n}\, |n-1\rangle . \qquad (10.53)$$

At the bottom, $\hat{a}\,|0\rangle = 0$, so the ladder cannot continue to negative n.

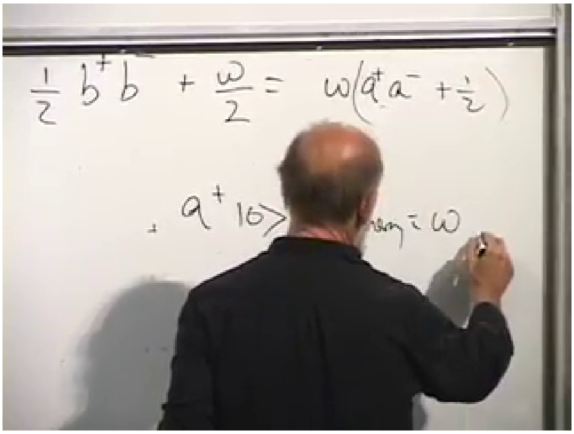

Figure 10.19: One application of the raising operator adds $\hbar\omega$ to the ground-state energy.

Lecture frame: 01:20:00.

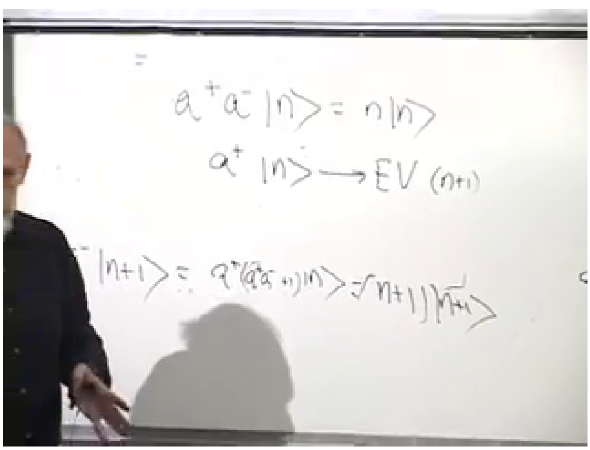

Figure 10.20: The raising theorem maps a number eigenstate to the next eigenstate.

Lecture frame: 01:24:52.

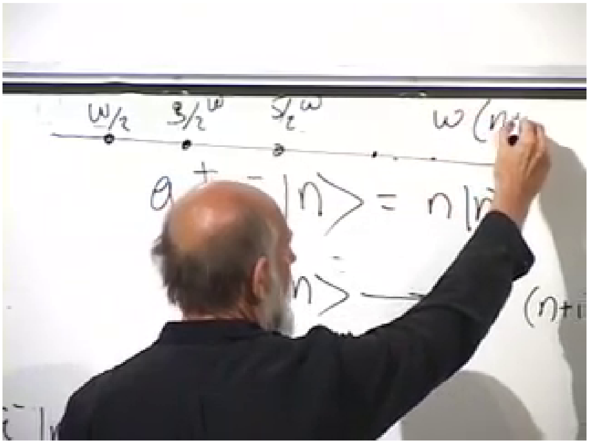

Figure 10.21: The equally spaced oscillator spectrum beginning at the zero-point energy.

Lecture frame: 01:25:38.

The complete spectrum is

$$\boxed{E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots.} \tag{10.54}$$

Successive levels are separated by the fixed quantum $\hbar\omega$:

$$E_{n+1} - E_n = \hbar\omega. \tag{10.55}$$

Question from the audience. Are there other Hamiltonians with an equally spaced spectrum? [a]

Response. One can construct them. For example, on a particle-on-a-ring Hilbert space, an operator proportional to angular momentum has equally spaced eigenvalues $m\hbar$ with positive and negative integers m. Used directly as a Hamiltonian, however, that spectrum is unbounded below and has no stable ground state. The oscillator is distinguished by its equally spaced ladder together with a bottom.

[a]Lecture timestamp: 01:27:35–01:28:54.

10.8 What the Eigenfunctions Mean

Question from the audience. What physical system does a quantum harmonic oscillator represent? Is it an electron orbiting an atom? [a]

Response. A Coulomb orbit is not itself a harmonic oscillator. The general setting is any stable configuration examined through small displacements. A diatomic molecule vibrating about its equilibrium bond length is a good example. More broadly, Equation (10.2) shows why every ordinary nondegenerate minimum is approximately quadratic. A quantum system is characterized by measurable probability distributions and spectra, not by a literal photograph of a microscopic trajectory.

[a]Lecture timestamp: 01:29:02–01:32:31.

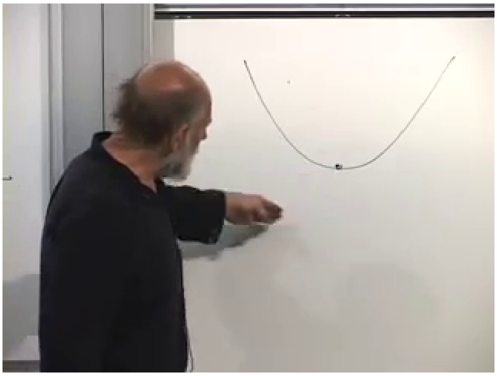

Figure 10.22: Near a stable minimum, a general potential is locally parabolic and therefore harmonic to leading order.

Lecture frame: 01:30:41.

For the ground state,

$$|\psi_0(x)|^2 = \sqrt{\frac{\omega}{\pi\hbar}} \exp\left(-\frac{\omega x^2}{\hbar}\right). \tag{10.56}$$

Its variance is

$$(\Delta x)^2 = \frac{\hbar}{2\omega} \tag{10.57}$$

in the unit-mass coordinate. Increasing ω stiffens the restoring force and narrows the spatial distribution.

Apply the raising operator once:

$$\psi_1(x) \propto \left(-\hbar\frac{d}{dx} + \omega x\right) e^{-\omega x^2/(2\hbar)} \tag{10.58}$$

$$\propto x\, e^{-\omega x^2/(2\hbar)}. \tag{10.59}$$

It is odd, changes sign at the origin, and has one node. Multiplying a state by an overall constant or

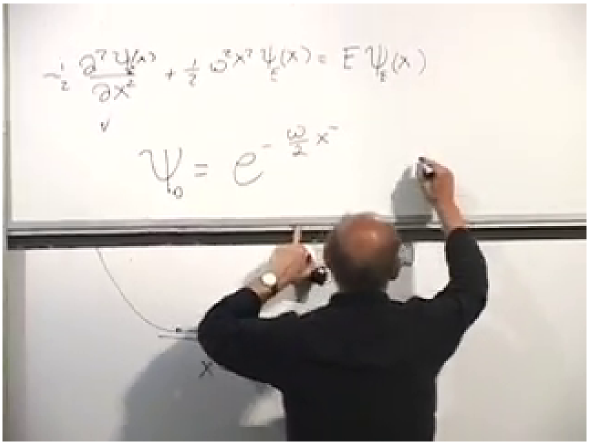

Figure 10.23: The Gaussian ground-state probability is more tightly localized for a stiffer oscillator.

Lecture frame: 01:32:38.

phase, including a minus sign, does not change its physical predictions.

Repeated raising produces a polynomial times the same Gaussian:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{\omega}{\hbar}}x\right) e^{-\omega x^2/(2\hbar)}, \tag{10.60}$$

where H_n is the nth Hermite polynomial. The parity alternates,

$$\psi_n(-x) = (-1)^n \psi_n(x), \tag{10.61}$$

and ψ_n has n real nodes. Higher states oscillate more rapidly, carry larger characteristic momentum, and extend farther from the origin, as expected for a more energetic oscillator.

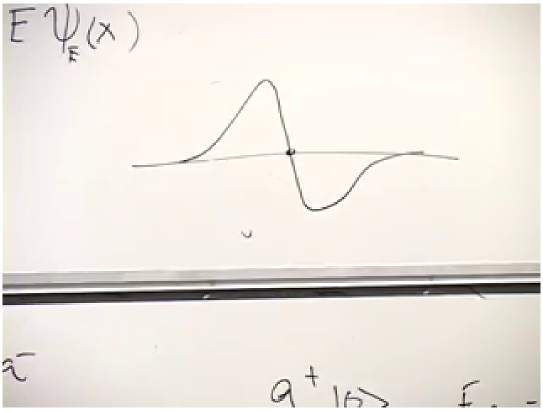

Figure 10.24: The first excited state is an odd Gaussian times x, with one node at the origin.

Lecture frame: 01:37:30.

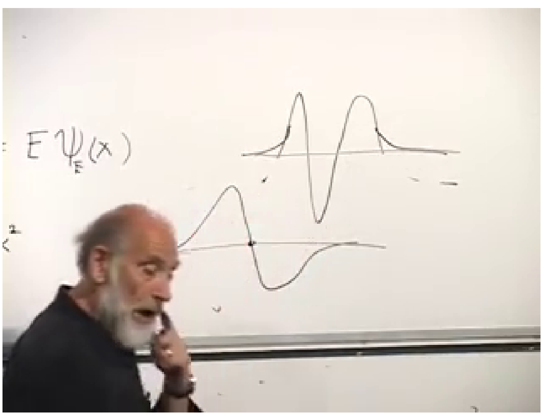

Figure 10.25: Successive oscillator states acquire additional nodes while retaining Gaussian decay at large distance.

Lecture frame: 01:39:12.

Question from the audience. Are the nodes of the Hermite-polynomial wave functions at real values of x, and is the probability exactly zero there? [a]

Response. Yes. The nth oscillator eigenfunction has n simple real zeros. At each node $\psi_n(x) = 0$, so the position probability density $|\psi_n(x)|^2$ also vanishes. The zeros arise from destructive interference within the stationary wave function.

[a] Lecture timestamp: 01:41:44–01:42:58.

Question from the audience. What is the physical reason for quantized energy rather than a continuum? [a]

Response. The mathematical and physical statements are the same boundary condition viewed from two sides. The oscillator is confined: its potential grows without bound, so a physical eigenfunction must decay at both infinities. Those two boundary requirements can be met only at discrete energies. A free particle can escape to infinity and therefore has a continuous momentum and energy spectrum. More general potentials may have discrete levels below an escape threshold and a continuum above it.

[a] Lecture timestamp: 01:42:59–01:44:35.

Question from the audience. Why are the exponentially growing solutions excluded rather than treated as unusual states? [a]

Response. They cannot be normalized to a probability distribution on the line. Relative to their divergent weight at infinity, every finite region would have zero probability, and the oscillator potential would give them divergent energy. They are mathematical solutions of the local differential equation, but not state vectors in the oscillator Hilbert space.

[a] Lecture timestamp: 01:44:38–01:46:24.

10.9 The Completed Oscillator Picture

The harmonic oscillator begins with the most ordinary classical motion: $\ddot{x} = -\omega^2 x$. Quantization turns its coordinate and momentum into noncommuting operators. That single change prevents zero energy, selects a normalizable Gaussian ground state, and leaves a half-quantum of irreducible energy.

Factorization then converts a differential-equation problem into an algebra problem:

$$[\hat{a}, \hat{a}^\dagger] = 1, \qquad \hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right). \qquad (10.62)$$

The creation operator climbs an equally spaced spectrum, the annihilation operator descends it, and the ground state terminates the descent. In position space the same ladder appears as Gaussian wave functions multiplied by successively higher Hermite polynomials.

Angular momentum has not been covered here; it is deferred rather than silently omitted. The oscillator itself supplies the bridge to the next subject, because every normal mode of a quantum field behaves as a quantum harmonic oscillator, and its raising operator creates one additional quantum of excitation.